

Fin 350 The History of Financial Crises

Course transcript with numbered paragraphs

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Highlighting key

key vocabulary terms and concepts distinctive to this course

key figures thinkers, characters, and figures central to the lectures

Paragraphs are numbered lecture.paragraph (e.g. 3.12 is the twelfth paragraph of Lecture 3). Each lecture opens with an "At a glance" overview, which is not part of the transcript.

1. The Anatomy of a Crisis

AT A GLANCE

Defines a financial crisis as a sudden decline in the value of an important asset, usually driven by a sudden repricing of risk, and asks why such costly events keep recurring. Contrasts the Minsky-Kindleberger view (irrational cycles of greed and fear) with historical particularism, and proposes a middle ground: a crisis taxonomy. Argues that crisis risk is adaptive because it is stapled to things societies want: political risk subsidies, geopolitical competition, learning about new technologies, privacy, and discretionary fiat money. Sets the motto "know thyself," previews the deep dives, analyzes the Roman panic of AD 33, and introduces the early modern world and John Law.

- 1.1 Welcome, everyone, to this course on the history of financial crises. It's a pleasure to be with you. I want to begin with a definition. We're going to be spending three days talking about financial crises. What do historians and economists mean by the term financial crisis? Well, I don't know what they mean, but I know what I mean, which is a sudden decline in the value of some asset that's important.
- 1.2 In other words, it could be land. It could be stock that you own in a company. It could be a sovereign debt, a bond, of a government. It could be a currency. Whatever it is, if it declines suddenly, and if that's an important asset in the economy, that's what we call a financial crisis. Usually, those declines of assets that happen suddenly are related to a sudden change in risk perception. That is, when an asset, which is a claim to future revenues in some sense, when that asset becomes viewed as riskier, its value falls.
- 1.3 Because effectively, the discounting, as we say in economics, of that asset, the required rate of return on the asset, rises. That causes, even for the same perception of cash flows that you're expecting from that asset, even if the expectation were the same, if the risk goes up, then the value falls. Just to start off with a very simple idea, asset declines happen usually because of a big change in perceptions of risk. It could also be, as I just noted, a change in the perception of the revenues, the expected revenues, the average forward-looking expected revenues.
- 1.4 But some combination of those two, in economic theory, is what drives declines in asset values. Has to be a change in perception of risk or a change in the expectation, the average expected future revenues. Usually in financial crises, it's both. But what I want to emphasize is, what we're really talking about is often a sudden change in the perception of risk. That's interesting. Why is it that perceptions of the uncertainty of the revenues that you're expecting from an asset, why would that change precipitously and suddenly? Why do we care about this topic?
- 1.5 The answer is, not just because we care about asset values, which decline, but also because those declines tend to be associated with major disruptions in economic activity, not just the value of the assets, but the total of current income and wages go down. What's interesting is

that when a **financial crisis** happens, it's very disruptive to the economy, by which I mean, the production of income currently, not just the value of assets.

- 1.6 How disruptive? Well, one study finds in an analysis of **banking crises** that when a **banking crisis** happens, real economic activity falls by about 6%. It's also true that, in recent era, that is, in the last 30 years, **banking crises** around the world have been associated with costs of distress for banks, meaning the amount that banks required from governments to be bailed out, to be about 16%—that's the median—16% of GDP.
- 1.7 So these are hugely costly events, both from the standpoint of how big the asset declines are, and in the case of the **banking crises** I just mentioned, how much the government had to give the banks in order to bail them out if they chose to bail them out, 16% of GDP. But also, while that's happening, it's not just the asset values that fall, but the economy declines in terms of what people are earning currently.
- 1.8 OK, so these are pretty disruptive events, these sudden changes in **risk perception** that are affecting asset values and also affecting current economic activity. That leads us to a puzzle. How come we keep doing this? If there were a single phenomenon, let's say, called a **banking crisis** or a **financial crisis** more generally, and if it's so costly, you would think that we might learn to avoid them. The key question there, of course, is, is there something to learn?
- 1.9 When you're in these pre-crisis periods, just before things collapse, is it possible that you can figure out that you're in a really risky time? And if so, do something to prevent the collapse, which may also mean, do something to prevent the wrong perception of low risk that preceded the big increase in the **perception of risk**? That's really the first, I think, most important question about **financial crises**, addressing this possibility that we shouldn't be having them still.
- 1.10 I'm going to begin with a discussion of a **financial crisis** that happened 2,000 years ago, and then we'll be talking about crises that happened more than 300 years ago. And so you get the picture pretty quickly that the history of **financial crises** is something that goes back a long way in time. How come we don't learn? Now, of course, one possibility is we're incapable of learning. And one of the most influential theories of **financial crises** by Hyman Minsky and Charles Kindleberger is that view. It's the view that human beings, behaviorally, in some mass psychotic waves, oscillate between massive greed and massive fear.
- 1.11 When massive greed is happening, assets get bid up because people have a low **perception of risk**. Oh, the risk is very low. They're very optimistic, and they're excessively optimistic. Then, almost naturally, through a behavioral cycle of mass hysteria, that then changes from excessive optimism to excessive pessimism. If you take that view, crises are all quite similar. They come from a common behavior oscillating between fear and greed, and there's no way to fix them because it's part of human nature. You may say it's irrational, but maybe that doesn't matter because humans are the way they are.
- 1.12 Now, historians don't necessarily agree with that view. Some historians would say, oh, all crises are different. There's really no way to learn how to alter your behavior. The **Minsky-**

Kindleberger view is wrong because it just lumps together all these very different events. It's really more that there's particularity to each of these crises, and so there's little that really can be learned, and we shouldn't jump to the conclusion that people are behaving irrationally through these cycles of greed, followed by fear.

- 1.13** Now, however, other historical opinions aren't of that same view. That is, it's not necessarily the case that all these crises are unique. In fact, there's a literature that's identified some commonalities. For example, when crises involve the collapse of banks, we tend to see that banks lend a lot in advance of the collapse. That is, the growth rate of loans tends to be unusually high. We also tend to find that banking systems that have been given a lot of observable government protection, for example, very generous **deposit insurance** of banks by the government, also tend to be much more likely to have a **banking crisis**.
- 1.14** There are observable commonalities, statistically, that tend to forecast **banking crises**. Similarly, with exchange rate collapses, where exchange rates depreciate massively and suddenly, we tend to find that unsustainable fiscal and monetary policies tend to be an important part of that story. Again, statistical evidence of commonality. In stock market collapses, statistical evidence tends to point to what's sometimes called momentum effects or returns extrapolation, which just means that, when returns are high, there is a tendency for them to be high in the adjacent next period.
- 1.15** That's sometimes viewed as a behavioral response that might be, in some sense, related to this **Minsky-Kindleberger** view of greed. That is, you see stock prices go up and you extrapolate out of optimism that if you get into the market, they will go up again, and that forces them up again. We have some disagreement, theoretically, and we have different views about crises. In addition to the statistical evidence about some common features among crises, there's also narrative evidence that shows that they do have some common features.
- 1.16** For example, we're going to talk about the Mexican crisis of 1994-1995 in this class. Those of us who were there at the time noticed a remarkable similarity between the Mexican experience of 1994 and the Chilean experience of 1983. One could also say that the Greek crisis of 2010 had a lot in common with the East Asian crisis, which we're going to study in some detail in the case of Korea in 1997.
- 1.17** Now, how do we sort between these views that, I guess we'd say they're two extreme views. All crises are the same—that's the **Minsky-Kindleberger** view, all driven by a common factor of human nature. Or the **historical particularist** view—all crises are different; you can't learn from one about another. Of course, in this course, we're going to find a middle ground there. That there are discernible similarities across crises, but that doesn't mean that all crises are the same.
- 1.18** Because we're going to point in that direction, that there is the possibility that you can learn from the experience of one crisis to another, then that raises the puzzle, why is it that we don't learn to avoid them? Of course, in the **Minsky** theory, it's just because we're incapable of learning as a society. We're driven by irrational **greed and fear**. We can't learn. That seems

strange to me as a historian and as an economist, because, of course, we have collective action through government, and we could learn through government policies and regulations to try to improve our situation.

- 1.19 We're going to take a different approach to this question of why learning is so limited. It's a very, you might even say, a surprising approach. What I'm going to argue is that crisis risk, in other words, allowing as a society for us to have the risk of a crisis, despite the costs of that crisis, might actually be **adaptive**. I mean that in this evolutionary sense. Why would it possibly be that having these explosions occasionally occur in **financial crises** might actually be **adaptive**? I want to point to five examples to give you a sense of what I mean by **adaptive**.
- 1.20 First of all, I'm not saying that the losses in the crisis, that the destruction, is a good thing. I'm not saying that. What I'm saying is that the bad thing of the crisis might be stapled to other things that we want. When I say we want, I don't mean that I can tell you are desirable. What I mean is, we may want some things as a society, despite the fact that they're not desirable. Let me give you an example. Political systems, our democracy here in the United States, are designed to achieve outcomes. Those outcomes aren't always perfect.
- 1.21 In fact, you could think about the way our system works as forming coalitions that win elections. Those coalitions may not be best for everyone. They may not choose policies that are the best for everyone. But that's not the way a democracy works. A democracy chooses things that are best for the winning coalition, maybe at the expense of the people who aren't part of the winning coalition. That's a successful democracy. That's the idea. Democracy isn't designed to produce the perfect situation for everyone on average. What it's designed to do is to allow majorities to win.
- 1.22 As part of their winning, they then get to make choices that are going to benefit those who are part of the winning coalition, maybe at the expense of those who aren't. It could be that that coalition wants to do what we in economics call **rent extraction**. That is, to create systems, to create regulations, to create ways of organizing ourselves that systematically take money from one group and give it to another. You could say, well, that sounds horrible. That's a failure. It's not a failure of democracy. Actually, democracy is often designed to do that. In fact, almost always does that.
- 1.23 Why would a crisis be related to that process? Because creating a subsidy for risk, which might lead to excessive risk-taking, might be a very effective way to create **rents**, to take from some people and give to others. It's seen from that perspective. When I, for example, subsidize **mortgage risk** by telling people, oh, you can take out a mortgage with no down payment—by the way, we do that for veterans in the Veterans Administration. Veterans are able to borrow and have a house with 100% mortgage at an interest rate that's artificially low because of the insurance coming from Veterans Administration Mortgage programs.
- 1.24 Now, does that make veterans more likely, all things equal, does that fact make them more likely to default on their mortgages? The answer is, well, of course. If I create a policy that

makes the mortgage riskier and that subsidizes taking risk, you will take more risk. If you take more risk, you'll be more likely to default, all things equal. But that choice of subsidizing that risk might be part of an overall social bargain that politically works for us. If it politically works, that might make it **adaptive**, even though it leads to some financial fragility.

- 1.25 The first ingredient of understanding what I mean by **adaptive** is that there are some politically **adaptive** ways to create **risk subsidies** that might serve the purposes of democracy by making the formation of political coalitions sustainable, despite the fact that things blow up and are economically costly. They may not be ideal in some abstract economic sense, but they might be **adaptive** politically.
- 1.26 A second example is **adaptive** geopolitics. Suppose that you're in a country, we're going to talk about two examples today from the early 18th century in early modern Europe, but we could also, later, we'll point to examples from **emerging markets** today, where countries are in an extreme competitive global situation, where countries are competing with each other, competing for trade, competing for market share. As part of that competition, they may decide to take risks because failing to take the risk might lead to an even worse outcome.
- 1.27 It's not that I want the risk per se, but I'm involved in a competitive situation where if I don't take the risk, something else happens that I don't like. Think about it this way. Suppose that I'm lagging behind as a competitor in the early 18th century, like France or Britain is lagging behind Spain, Portugal, the Netherlands. If they don't take some risks, maybe they lose even more territory in the global competition. By the way, that competition has both a military aspect and a mercantilist aspect. It's about competing for establishing markets.
- 1.28 The way you do that is, you invade different parts of the world; you take them over, then you establish trade markets. So if you're behind in that game, you're trying to figure out, how do I catch up quickly? Because I don't have forever to wait. If I don't catch up quickly, I'm gone. And then not only do I lose the ability to have an empire, but I might get invaded eventually myself. If you're France or England at the turn of the 18th century, you're not just thinking, what risks do I want to take from some sort of isolated economic policy standpoint? You're thinking, what risks can I take? What things must I do to be able to survive global competition? That can be, therefore, **adaptive** to you from a survival standpoint to take risks that might lead to **financial crises** because they produce something else.
- 1.29 A third example of an **adaptive** risk-taking strategy comes from the advantages of learning. There are risks that are brand new. We're going to talk about Florida, where we're filming this class, we're going to talk about Florida in the 1920s. New technologies allowed people to come to Florida and to think about developing it. There were dreams that it would be a paradise. By the way, those dreams were correct.
- 1.30 But they thought it would happen a little faster than it did. But the reason that it was so interesting, as we'll describe, is that nobody really knew how things were going to play out. If you didn't take the risk, you wouldn't learn. You wouldn't learn, what is the market for Florida

land? What is this land going to turn out to be? How much is it going to be worth? Where are things going to be developed? What are we going to do about all these swamps? By the way, the coastline was dramatically changed by technology. Florida was a complete puzzle. We didn't know in the 1920s what Florida was going to turn out to be.

- 1.31 How did we learn and make it happen? By taking huge risks, which in the short term, didn't turn out so well. Similarly, when we look at the stock market in the US in the 1920s, because we were such an innovative society, we had a boom in the stock market, and then we saw that that boom collapsed. Why exactly it collapsed, we'll see, is not so easy to explain. But what is clear is that it was something about learning about new technologies that involved that new risk, both in the 1920s Florida land market and in the stock market in the 1920s. You could say we should have avoided those risks, but to avoid those risks, you would have had to avoid learning. You would have had to avoid new technologies.
- 1.32 What about fraud? That's going to be my fourth example. You might say, well, there are a lot of episodes where fraud on a massive scale—Florida, it turns out to be one of them in the 1920s—substantially magnified the losses that were going on. You might say, isn't there a way for us to reorganize ourselves so that we make fraud less likely? The problem there is, as we'll see when we talk about some of the examples, including the land speculation in Florida in the 1920s and the frauds that are happening in the government, actually, as well as in the banking system at that time.
- 1.33 You might say, well, there should be a way to prevent these. But on the other hand, notice that hidden actions, the ability of people to keep their actions somewhat secret unto themselves, which is necessary for fraud to happen, has a lot of favorable qualities. It allows private information. It rewards entrepreneurship.
- 1.34 We don't want to have Big Brother monitoring us so closely and restricting us so closely and sharing all of our information about everything so that there's no possibility of investing in and creating private information. A market-based society actually depends on decentralized information and private information. So fraud is an undesirable consequence of something that is, in fact, highly desirable—privacy.
- 1.35 Finally, another example of something that creates a lot of risk but has a silver lining is fiat money. We're all used to living in a society where central banks create paper currency, and it's through the central bank's production of the paper currency over which they have a monopoly, the Federal Reserve determines the amount of dollars. We might say, well, there are going to be lots of examples in this class, in 1929, in the period 2002 to 2007, where we're going to see that monetary policy made big mistakes.
- 1.36 Monetary policy was too tight or too loose, and that there's a connection, which we'll explore almost right away today, there's a connection between those monetary policy errors and that sudden repricing of risk. So when monetary policy is too loose, it tends to, for some temporary period of time, make risk too low. In other words, people are underestimating risk when

monetary policy is weak. When monetary policy is tight, it has the opposite effect. It tends to cause risk perceptions to rise very quickly.

- 1.37 Another phenomenon, another source of this oscillation of perceptions of low risk changing to perceptions of high risk, economic research has shown, comes from central bank policies. You might say, let's get rid of those guys so that we don't have such oscillations. But then the reason we have, across the whole world, agreed to have fiat money regimes as opposed to, let's say, gold or other kinds of monetary systems that one could imagine, is because we think it's the best of available technologies.
- 1.38 That doesn't mean that there isn't room for reform of central banks. Many economists, myself included, believe that central banks, including the US central bank, could substantially improve its monetary policy to reduce these oscillations of risk. By the way, Milton Friedman, a famous economist in the 1970s, in the wake of the inflationary episode that happened in the US, made a lot of progress convincing people that there was big room for improvement. Some of that improvement happened, but some remains to be done.
- 1.39 I'm not saying that we live in an ideal world. I'm just saying that we have decided as a society that when people like me say, let's make monetary policy more predictable and systematic and remove these oscillations from it, that's not a popular view. You voters are not supporting people who are running for office who are advocating that. Whether you know it or not, you're supporting people who advocate what's called discretionary monetary policy, where central banks get to decide without much accountability while they're making the decision to change things in one direction or another.
- 1.40 We all just, in the US, lived through a major inflation. It's still happening, where the Fed has deviated from its target massively. No one doubts that the Federal Reserve made a huge mistake from 2021 to 2023. No one doubts that. That's not controversial. The Fed doesn't say, "We didn't make a mistake that caused that high inflation." The primary objective of the Fed is to not allow those kinds of inflations to happen.
- 1.41 The question is, well, you could say, well, A, maybe we should just get rid of the Federal Reserve and go back to a gold standard. No one's advocating that. B, you might say, well, let's change monetary policy to make it more systematic so there isn't so much discretion. That's my view, personally. That's not a popular view. The most popular view is, despite the fact that the Fed made a big mistake, we like the status quo of the way this is done. And politically, this is currently the winning operation.
- 1.42 Notice that all five of the things I talked about—domestic subsidies as a way to reward constituents in a political equilibrium, geopolitical competition, innovation and learning about new risks related to innovation, the fact that risk is hard to observe because we keep information private, and the fact that we have central banks that manage monetary policy through discretionary fiat money regimes—those are all things that seem to be, they may not be

economically optimal, but they seem to be part of what I would call an **adaptive political equilibrium**, that as countries and politicians that support these things, seem to succeed.

- 1.43 We might say, well, things aren't perfect. We're having **financial crises**. But I like to think about a little different. I like to think that the reason we're having **financial crises** is because we want them, not for themselves, but because they're stapled to things that we're very attached to. We're very attached to **rent-seeking** in our democracies as a way to allocate benefits. We're very attached to surviving global competition with other countries. We're very attached to innovation and learning about new risks, and to privacy, and to **monetary policy** through **fiat money** and discretion.
- 1.44 I'm not saying we live in the best of all possible worlds. What I'm saying is that we live in a world in which all of those things are politically chosen, and they seem to be successful. That is, politicians who would advocate differently from those wouldn't win. Our system is designed to actually promote those kinds of choices. It's not just our system. I think if you look around the world, you'll see these are common choices.
- 1.45 If you were just a biologist looking at this situation from the standpoint of what looks like it's being selected as part of a biological equilibrium, if you apply that same logic to social choice, you would say, these are choices that seem to be common around the world, and they seem to be part of an **adaptive** social-political system. Again, I'm not saying that means they're ideal economically. I'm just saying that we can understand them as choices that many countries are making, not just some failed political regime, but this seems to be a very successful political regime.
- 1.46 That's why the motto of this course, I think, is "**know thyself**." Instead of approaching **financial crises** as some sort of a foreign invasion by an alien being that we could somehow defeat or learn we don't want that; instead of tut-tutting about, oh, look how we keep repeating those same mistakes or blaming people on Wall Street; instead of doing any of that, what we should do is say, let's look in the mirror. Let's learn to know who we are and why we're making these choices repeatedly.
- 1.47 It's not, as I'll argue, I'll just come out right now and tell you, it's not an irrational cycle of **greed and fear** that's driving all this. It's not that at all. It is a conscious set of choices that seem to be **adaptive**, that seem to be part of who we want to be. To give you a few windows into how we as economists analyze these states of the world where crises happen more or less across time and space, I'll just point to some graphs. This is part of— simple graphs can lead to a lot of self-knowledge, can pose questions, make you ask questions about why your society that you're living in is different from other societies and why choices have been made.
- 1.48 This is a graph of the number of countries in the world that have some kind of **deposit insurance** system for their banks. One theme of this course will be **deposit insurance**. This is maybe the least controversial empirical statement in financial economics. **Deposit insurance** makes banking systems much riskier. Now, you didn't expect that, probably. You expected to hear that **deposit**

insurance, which removes the incentive of depositors to withdraw their funds because the funds are guaranteed by the government, that should make banks less risky because the banks don't have to worry about somebody getting, a massive number of people coming in and trying to withdraw their money all at once.

- 1.49 Well, that's an interesting theory. That theory is wrong. In fact, as you'll see, the most robust and agreed-upon empirical fact about **deposit insurance** is that it makes banking systems more likely to collapse. Very interesting. Of course, that's related to the fact that when you remove the discipline of people worrying about whether their deposits are safe, then the bankers end up taking a lot more risk. **Deposit insurance** is a **risk subsidy** for the banking system. When you **subsidize risk**, you get more of it.
- 1.50 Notice in this graph, what you see is that, as of about 1970, there are really only a handful of countries. The US is the first in 1933. Prior to 1933, no country has **deposit insurance**. We've been having depository banking in the world since about the 7th century BC in Athens. When did we first get **deposit insurance**? Not until the US in 1933. Then for another 40 years, only a very small number of countries followed the US example. But then all of a sudden, from the 1980s and the 1990s, we see **deposit insurance** spreads like wildfire throughout the world. That's a really interesting political choice.
- 1.51 Why does that happen? When and where it happens? And believe it or not, we will be talking about that toward the end of this class. But the point is, if I said to most people in the world who now live in a system with **deposit insurance**, "You know, **deposit insurance** is a bad idea. We can accomplish better the same objectives that you think we're achieving with **deposit insurance** with something else and without promoting a huge amount of risk in the banking system," if I said to most Americans, "I want to get rid of the **deposit insurance** you have," most of them wouldn't like that. That wouldn't be popular.
- 1.52 As a person, you can either decide, well, that means that people aren't being smart. They should listen to professors, right? When I tell them something, they should obey, do what I say. Or you could say, professors aren't smart, because professors aren't understanding that there's something **adaptive** about this **political equilibrium**. It's happening all over the world. We may think it's a bad idea economically—by the way, I do—but now we have to somehow respect it, too, even if we don't think it's making the world a better place economically. We have to try to understand, why is this happening?
- 1.53 It doesn't seem like it ever gets reversed. So if I think about evolution, biological evolution. Suppose I told you there's a time when, all over the world, organisms adopted a new trait, and that trait was never reversed. You'd say it was **adaptive**, right? We have to learn to think that way in social science, too, even if we, as economists, tend to think a little differently about what's optimal. We have to think about not just what's optimal economically, but what's happening.

- 1.54 This is an interesting part of our thinking, is to be able to come to grips with things that seem to be happening all over the world at the same time, not getting reversed, and being fairly popular as policies, despite the fact that they might produce **financial crises**. This is the prime example. Here's another example, and I will talk about these later in the course as the **two gorillas of financial crisis** in the world, and especially in the US, but all over the world. These two risks that are getting subsidized.
- 1.55 One is banking risk, **default risk**, coming through **deposit insurance**, and the other is **mortgage risk**, coming from programs all over the world that subsidize risk-taking by allowing people to borrow, to fund the acquisition of a house, at an artificially low rate by taking on a lot of risk that they don't have to pay for. I gave you the Veterans Administration as an example earlier. Here's a graph that shows that, at the end of World War II, around most of currently developed economies, so this is a list of about 17, I think, 18 countries, all of which are developed economies, their total ratio of mortgage loans to total loans in their banking system, as of 1950, mortgages were about 30%.
- 1.56 Today, they're more than double that. How did that happen? What's going on there? The answer, of course, is, **mortgage risk** is being subsidized. That's why you got more of it. Now, this housing collapses that we're living through, and we had one in the US in 2008, you might say, well, it seems like that's a bad idea. That was a pretty disastrous experience. Subsidizing housing risk leading to this excessive risk-taking was very costly. But I will tell you that I don't see any evidence that this trend that we're seeing is reversing. **Subsidizing mortgage risk** is now happening again in the US.
- 1.57 We are once again undertaking policies to allow people to make what you might call financially unsound investments in a house by borrowing because it's the tool that the government can most use, most politically, effectively and easily, to help promote borrowing for houses, and because that's politically popular. You might say, "That's stupid. We shouldn't be doing that."
- 1.58 By the way, it's also the case that subsidizing mortgages doesn't seem to actually increase the house ownership rate. I'll show you evidence about that. Because when you subsidize **mortgage risk**, it tends to bid up the price of houses so that you're chasing your own tail in some sense. But that doesn't mean that we don't want to do it politically. Just because it doesn't make sense, doesn't mean we don't want to do it.
- 1.59 A third example, this has to do with **monetary policy**. What you're looking at here is a measure of risk. There are two measures of risk. The orange line is measure of risk in terms of spreads in, I'm sorry, is the **VIX**, that is, a measure of the volatility of the stock market. The blue line is a measure of bond risk spread. A particular physical risk in the bond market, how much extra yield do you get for that risk? That spread, you can see, varies. In fact, notice that the **VIX** and the bond spreads tend to vary quite closely together, don't they?
- 1.60 You can see that, from 2002 to 2007, those risks declined dramatically. Remember when I said **financial crises** are often a time when there's a dramatic **reappraisal of risk**? Look at 2008. You

see the dramatic **reappraisal of risk**. But that **reappraisal of risk** was preceded by five years of dramatically declining **perceptions of risk**. The **pricing of risk** was very low and declining for five years, and then it suddenly went up. When we look at that, we know that we're not just talking about **mortgage risk**. We're not just talking about **mortgage risk** subsidization.

- 1.61 What's driving that bus, for the stock market and the bond market? Well, it's **monetary policy**. We will get back to that. We'll talk about that specific example later in the class. But the reason I'm pointing to it right now is, that's an example of **fiat money** regimes under discretion, the **Federal Reserve**, choosing to enact monetary policies that had predictable consequences for risk. This is a predictable regular phenomenon of **monetary policy**.
- 1.62 When **monetary policy** is excessively loose, and it was observably excessively loose from 2002 to 2006, as we'll see, it was unusual— it wasn't just looking back at that and saying, "Oh, it was loose." It was during the time we were living through it, it was obvious that they were loosening **monetary policy** excessively. That had big consequences for these markets. Yet, we haven't decided to stop the Fed from doing that. I don't think that we're going to do that anytime soon. These are examples.
- 1.63 When I say, "**Know thyself**," these are the United States that we're talking about or other developed economies or, in the first graph, it was worldwide. These are phenomenon that are happening all over the world. They're quite similar. It's not like there's somebody out there with some better idea that's very popular that's getting elected. Why? Why is this **adaptive**? Well, that's a question which we'll get into, in part.
- 1.64 But what I want to get you to think about, which is very different from the normal tut-tutting about those unwise people who take risks on in advance, I want you to think, no, no. A lot of these risks are predictable. Often, they're going to end badly. They're done on purpose. And now, the even more controversial statement, they're **adaptive**. The next politician will make the same choice because we choose it. He might say, "Well, we should get smart and not do that." I would agree with you. But that's not the question.
- 1.65 The question I'm here, as a positivist, as an economist and historian, trying to understand, why do these things happen? The answer is, they're chosen, and there doesn't seem to be much evidence that that's going to change. It seems like the **political equilibrium** likes them. Why? Well, because it promotes learning, that's good. Because it promotes private information, that's good. Because it promotes **fiat money**, I would say that's good. But that doesn't mean that it's the best we can do. I'm not saying that. It just means, we're doing it on purpose.
- 1.66 As we get through talking about these various crises, I want to suggest a middle ground between the **historical particularist** view. All crises are different. You can't learn anything from one, from looking at another. That's not true. But I also want to suggest that the opposite extreme is not true either. All crises are the same. That's not true. When we talk about these influences on crises—international competition, domestic **rent-seeking**, political subsidization, learning, **fiat money** policies—some of these influences are more important in some crises than in other.

- 1.67 What that suggests to me is that, just like I like making biological analogies, I think we want to think about constructing a **crisis taxonomy**. That is, where some crises are more similar to other crises than they are to others. They're more similar to some crises than they are to others. So there are these multiple dimensions that we've talked about that tend to influence crises, make them happen, make them more likely. But they're not always present. It's more like some are present in some events, some are present in others.
- 1.68 I'm writing a book right now with Matt Jaremski, and we've already published a preliminary article looking at 10 crises this way, which are the ones we're going to talk about. We have, in our book, 35. What we find is, it's extremely useful to ask a common set of questions that help us organize our thinking about these crises along the lines I've just developed with you. But then notice how some crises are really driven by one of these or two of these phenomena and others by very different ones.
- 1.69 The middle ground is, crises aren't all unique, but neither are they all the same. Just like we think about organisms, that we can usefully divide the world using some **taxonomy** that's reflected in zoology, we're going to think we're going to be able to do the same thing for **financial crises**. This table, which I know you can't read, the point is not for you to read it, but just to get the ideas I'm telling you about it, which is, so what we do is, we look at each crisis and we ask a set of questions. On the left-hand extreme column, we have a list of questions we're asking about each crisis.
- 1.70 Was the collapse predictable? And when I mean that, when I say, "Was it predictable?", I mean predictable using some kind of **ex ante** tool. We're going to talk about the Korean crisis, which happens in, toward the end of 1997. But in March and April of 1997, both the Financial Times and the Economist predicted a crisis in Korea, six, seven months before it happened. And so did I, and so did many economists, because we saw common features. That's what I mean. Predictable based on some **ex ante** criteria that we can also see reflected in news reports long before the crisis happened.
- 1.71 The Mexican crisis of 1994 was predicted by lots of economists two years ahead of the crisis happening. So is the crisis predictable? But there are some for which it wasn't. Secondly, can we point to a price boom just before the bust? In some cases, we can. I mentioned the housing crisis of 2008. As you could see from that graph where we looked at **monetary policy** and its effect on bond markets and stock markets, all asset prices were elevated by the very low **perceptions of risk** from 2002 to 2007. Then asset prices declined precipitously when those risk perceptions changed.
- 1.72 There was an excess price boom in the case of 2008. But when we talk about Korea or some other examples, but Korea in 1997, there was no excess price boom. There isn't just one phenomenon. In most cases, we're going to see examples of political **risk subsidies** of one kind or another. But there are examples where there isn't a political **risk subsidy**, the **Florida land boom**, for example, or the **stock market crash**, boom and crash, of 1929.

- 1.73 What's interesting is, by asking these narrative questions that each crisis has to answer, we can then create a data set of what makes crises different or similar. What we see is, there isn't just one phenomenon, but there are some clusters. There are some things that go together. For example, political **risk subsidies** tend to be highly positively covarying with the predictability of the excessive pricing. That's not surprising. If you're subsidizing something politically, it's not surprising that the price goes up a lot.
- 1.74 That's our approach. Our conclusions are that this taxonomic approach really leads us to some useful insights. What's our plan for this class? We're going to do **deep dives** because so far, I've just told you a lot of things in a very general way. I've just given you the introduction to how I'm thinking about presenting this material to you and what I think we can learn from thinking about **financial crises**. But to really show you what I'm talking about, and for you to actually learn it, you have to do what I call a **deep dive**. A **deep dive** means that you're going to have to get into the sordid details of each of these crises.
- 1.75 Some of them are quite arcane and complicated. But without that, it's just vague talk. We're going to talk about the banking and land market collapse in Rome in the year 33 AD. We're going to talk about the **Mississippi bubble** in France in 1720, the **South Sea bubble** happening immediately thereafter in England. We're going to talk more generally about the situation of bank chartering and crises with two different comparisons, one between England and Scotland, where England is the relatively highly crisis-prone banking system in the 18th and 19th century, early 19th century, and Scotland much less so.
- 1.76 Then we'll do another similar comparison between the crisis-prone banking system of the US in the 19th and 20th century and the Canadian system, which we're still waiting to have a **banking crisis**. Canada, since it starts chartering banks in the 18-teens, has not had a **banking crisis** yet. What's going on? These two very similar countries in so many respects have completely different **banking crisis** histories. By making those comparisons between England and Scotland and between the US and Canada, we're going to learn some things.
- 1.77 We're going to learn that crises are caused by political choices, and that's why they can be different across countries, even countries that are very similar and neighbors like England versus Scotland or the US versus Canada. We're also going to learn how to think about why learning doesn't happen. It's not that the US people weren't looking at Canada. It's not that the English people weren't looking at Scotland and noticing that their banking system wasn't exploding. "Why can't we have a banking system like the Scots?" say many people living in England, let's say circa 1820 or even before then, 1800.
- 1.78 Similarly, many people in the US are wondering, "Why can't we have a banking system that's as stable as Canada's?" And yet, they're asking those questions for many decades, and there is no imitation. It's a great example of this phenomenon of the puzzling persistence that we started with. When you see adjacent countries with drastically different experience, where there isn't learning going on, I can't learn from your history because, guess what? We're in a different **political equilibrium** from your **political equilibrium**.

- 1.79 Then we're going to talk in some more depth about those. The US crisis is very crisis-prone, but we're going to do some nuanced discussion there about how the shapes of those crises are different over time. We're going to compare the era from 1873 to 1907, the so-called **national banking era** crises, of which there are six. We're going to compare that to the **Great Depression**, which was a much more severe event for US bank failures. Then we're going to move on to discuss the **Florida land boom** and bust of the 1920s and the stock market boom and bust that happens also in the 1920s in the US.
- 1.80 We're going to then talk about **deposit insurance** because it turns out to be such an important topic. We're going to start with an analysis of the growth of **deposit insurance** in the 1920s in the US. Here's another economic puzzle. All of the states that adopt it, not the country as a whole, but the states that adopted it, ended up seeing complete collapse. So that by 1930, it was absolutely clear that every state that had chosen **deposit insurance**, their banking system collapsed. It was a complete disaster. Then in 1933, the federal government adopted federal **deposit insurance**. That seems kind of strange.
- 1.81 Just to add to the puzzle, and I don't want to tell you everything at the introduction, in the 1932 election, **FDR**, who ended up winning the election, said there was definitely one thing he would not do when he was President, and that is to support **deposit insurance**, because it was so obvious what a catastrophe it had been for the states. He went on to—**Roosevelt** was actually fairly knowledgeable about banking. He went on to talk about how **deposit insurance**, by removing **market discipline**, means that depositors aren't concerned enough about their money, which makes bankers act excessively risky.
- 1.82 He not only was opposed to it, but explained his opposition, clearly showing that he had been paying attention to all the collapses that had happened in the eight states in the 1920s. Nevertheless, he's the President who signs federal **deposit insurance** into being. Is that because his mind changed about the economics of **deposit insurance**? No, it is not. That's really, I think, a great lesson for us as we're thinking about how learning takes place, what it means for something to be **adaptive** versus economically optimal.
- 1.83 Then we're going to move on to talk about other **risk subsidies** that are consciously chosen. The **farm credit system**, it starts in the first decade of the 20th century, and the residential mortgage subsidies that begin in the **New Deal**. So 1934 is an amazing year for federal government decisions to **subsidize risk** in the residential mortgage area. That was a landmark change, and it culminated quite predictably in the **thrift** crisis in the 1980s, just as the **farm credit system** culminated in the agricultural bank and land crisis of the 1980s.
- 1.84 What's really interesting is, when we create these political subsidies for agricultural borrowing against land or for residential borrowing against land, they were quite conscious. They weren't done because people were making some economic choice, rather because of a political choice. They resulted ultimately in greater crisis risk, and the crises happened. And by the way, we haven't gotten rid of those. We didn't learn to get rid of those. And then we end up, finally,

talking about that graph that I showed you, where **deposit insurance** is rising throughout the world in the 1980s and 1990s.

- 1.85 And so what I'll do is, I'll explain to you why that happened, and then also show you that the consequences of that are well-known to be making **banking crises** much greater and more severe around the world. This period, from 1980 to 2010, was the most severe period all around the world for the **banking crises**. That is, we had more of them, and they were of greater severity than any other time in human history. That was the result of a conscious choice that we have not reversed.
- 1.86 We have done some reforms to try to mitigate those risks, and especially in **emerging market** countries. But what's interesting is, those choices were made despite the fact that the economic research was telling us that these risks were going to happen. There was nothing surprising, in a sense, about this period. As I said, our approach is going to be **deep dive**. The other thing about our approach is, we're going to avoid **ex post** thinking. That is, we're not going to be looking at this— if you read the books by John Kenneth Galbraith's terrible book about the **stock market crash** of 1929, it's this exercise in superiority thinking.
- 1.87 He's making fun of everyone. Oh, how silly, how stupid. By the way, he wasn't much of a historian about the 1920s, and most of the views that he expressed in that book, I would say, have been shown to be false. A lot of times, people who tell you they're analyzing **financial crises**, what they're really doing is looking back and making fun of the people who took risks that didn't work out well, pretending that they're much smarter than those people were.
- 1.88 That's not the approach we're going to take. We're going to take the **ex ante** approach. When I say that the risk was predictably too high, it's because, based on evidence that you could have prior to the collapse, you would have known so. And surprisingly, there are some events where that's true. So we're going to keep ourselves with **ex ante**, not **ex post** thinking. We're going to look at how people actually were gauging risk. We're going to try to ask what they were thinking, what information they had, and how reasonable their actions were on an **ex ante** basis, not an **ex post** basis.
- 1.89 OK, let's begin with the Roman banking **panic** of AD 33. This is the oldest crisis that we're going to talk about. But the Roman world, despite the fact that it's happening, this is happening 2,000 years ago, it seems remarkably familiar. There are two kinds of banks or lenders, I would say, in Rome at this time. One are deposit banks, maybe more familiar to us. The other were money lenders who were really members of the political elite who just ran their own lending operations as individuals.
- 1.90 Lending of both groups was regulated, as it is today, for political purposes. I would say that most of the literature by scholars of Rome would say that the goal of this regulation was primarily to sustain the empire and to sustain the elite who controlled the empire. The people who were running the empire, the Roman senators, were wealthy landowners. The regulations, we'll see

very clearly in the example we're going to talk about, the regulations were geared, of course, to help the members of the elite because they're in control.

- 1.91 But also to promote the interests of the Empire, a way to coordinate a good future for all of their investments because they all cared about promoting the future of the Empire. The Empire, at the time this is happening, has been expanded recently in the recent decades, as you all know, if you know Roman history. But with that expansion came new concerns about the stability of the Empire, about the ability of the Empire to hold itself together as a political entity spanning the whole Mediterranean area and beyond. The banking regulations reflected that.
- 1.92 There was also a maybe unwitting macroeconomic variation. At the time of Julius Caesar, generally speaking, the Roman policies resulted in making, through lots of government expenditures, interest rates very low, and credit and money were quite abundant. But under Tiberius, just prior to this crisis, that had been reversed. Interest rates were rising, which made, as we'll see, one of the regulations, the usury ceiling, become binding.
- 1.93 We've got some background. I just provided you the background. One is the empire run by the elite and for the elite. But also what that means is wanting to preserve the stability of the newly expanded empire. There are two main bank regulations. One of them is limits on the interest rate that you can charge. This is called the usury ceiling. Now, that had not been enforced, partly because interest rates were generally pretty low. The purpose of a usury ceiling in most of the finance literature is seen as favoring the elite, because if you are a member of the elite, your risk level tends to be lower than some new entrant.
- 1.94 By keeping interest rates low, by saying you can't charge high interest rates, you implicitly are forcing lenders to lend to members of the elite. Now, that might be justified. Usury laws might be justified by politicians for a lot of reasons. But as economists and as economic historians, we tend to find that constituent interests tend to explain regulations pretty well. In this case, the constituent interest, this is true also of the US in the 19th century, of studies of usury laws there, that what we find is that the best explanation is that usury laws favored low-risk borrowers, and in this case, members of the elite.
- 1.95 The other regulation was requiring that when you lent money, you had to make sure that a large amount of your own wealth as a lender, as a bank, was actually in the form of Italian land. It limited the amount of loans you could make because you had to make loans in proportion to your holdings of Italian land. Why were they so interested in making sure that lenders, the wealthy people in the economy, held Italian land? I think one interpretation is fairly obvious, and the historians have made, is that Italian land is at the center of the Roman Empire.
- 1.96 And so as the empire is scattering, we want to make sure that people who are part of the key governance are, as much as they may be drawn to distant places, are still investing locally, are still part of a critical mass of their wealth being near to the center of the Empire. Those were the two regulations. All right, now here's more of the background.

- 1.97 What happens is, as under Tiberius, as I said, interest rates start rising. Some of the people who are borrowers, who are also in the Senate, start complaining that the interest rates are rising; oh, and by the way, there's this dormant law out there that says interest rates can't rise above a certain point. And so they complain, and then they enforce the law. When they enforce the law, the amount of credit declines in the economy. I'll show all this to you on a graph. We'll review it together. The amount of credit declines.
- 1.98 When the credit declines, another action is taken. Because when the credit declines, land also declines. What's happening is, as credit's declining, that also means that credit available for buying land is also getting more scarce. What happens to land prices? They fall. When the usury ceiling is initially enforced, it leads to a scarcity of credit, which tends to also be reflected in a decline in land values.
- 1.99 But then the government's reaction to this is, well, we don't want land values in Italy to decline. So what we're going to do is increase the regulation, increase the proportion that money lenders have to hold in land. But then what happens when they do that? Well, that means that the money lenders have to lend out even less. What that does is, it makes land prices fall even more. I'll show this to you in a graph. The point is, these were bad policies in the sense that they led to an undesirable contraction of credit.
- 1.100 Then when they tried to fix the effect of that on land prices by using the landholding requirement, by boosting the landholding requirement, they made the supply of credit fall even more, causing land prices to collapse. At this point, the banks and everyone in the economy is in chaos. The Emperor resolves the problem by becoming the first-known lender of last resort, making a three-year interest-free loan to all the lenders from the Roman Treasury to bring an end to the crisis.
- 1.101 It's a pretty interesting example of a crisis that's clearly related to government policies that have clear political constituencies. Here's a graph that illustrates this. The initial equilibrium, pre-crisis, is the intersection of the two blue lines. So pre-crisis, you can see the interest rate is above the legal usury ceiling, which is the flat orange line. Then you can see the first thing that happens when they impose the usury ceiling, the new equilibrium. We can understand this evolution through this graph.
- 1.102 If we go over here, you can see, before the crisis, these two blue lines— this is the supply of loans, this is the demand for loans—pre-crisis are intersecting at this interest rate, which is above the usury ceiling interest rate that's not being enforced at that time. Then when the usury ceiling is enforced, well, loan demand stays the same, but now the intersection of loan demand and loan supply is here. You can see a huge decline in credit equal to this distance.
- 1.103 That means that land prices are declining initially because credit is more scarce, because we see why credit is more scarce, because interest rates are being kept too low. But then to fix it, the Senate says, well, we're going to force lenders to hold more land, which we hope will drive up the price of land. But that means, if you're forcing them to hold more land, they have to hold

less loans. That shifts the loan supply curve inward, restricting the amount of credit, which means that the new equilibrium over here is even worse.

- 1.104** So, the first policy of the imposing the **usury law** shifts credit negatively by this much. The next policy, by boosting the land ownership requirement, reduces credit by this additional amount. What this analysis shows is the two regulations that we can completely understand from a political perspective, **usury laws** and limitations on wealth holding, they tried to make sure that wealthy people are holding a portion of their wealth in politically important assets. We can make perfect sense of those.
- 1.105** By the way, those regulations weren't repealed, and we're still living with them in most of the world—regulations that are doing one or the other of those things, despite the fact that those regulations were very costly. That leads us then to ask, well, should we have learned differently? Of course, there was no repeat of the specific circumstances. But the same political choices, the same regulatory choices, are reflected in capital control regulations and **usury laws** that are still used as political tools today.
- 1.106** By the way, economists tend to complain about them, myself included, saying these are generally not good for economic efficiency. There are better ways, even if you want to do redistribution, there are economically better ways to achieve those. And yet, what we see is, just as these tools were used in ancient Rome, they're used by economies today. The reason, I would say, is that there's some kind of political purpose that they tend to serve, so they continued to be useful.
- 1.107** It wasn't that the Roman Empire had these regulations in place because they were trying to optimize some economic achievement of growth. It was more that they were trying to maintain a **political equilibrium** that rewarded members of the political elite and that made sure that the members acted cohesively to have a critical mass of their wealth invested in the center of the Roman Empire. You can make sense of that politically. It isn't about economic growth of the Empire. It's about trying to create a stable political coalition.
- 1.108** If we were going to talk about, for example, Mexico, around the year 1900, we would be talking about the dictator **Porfirio Díaz** and what was called the **Porfiriato**. You could say, well, that was a very inhospitable regime. It wasn't pro-growth, but it was very much pro members of the elite that preserved the stability of the country until 1912, when it didn't preserve the stability of the country anymore. That choice, though, actually made economic success in Mexico much greater than it could have been probably with any other system.
- 1.109** When you're looking at the question of, "Does this promote economic growth?", don't compare it to a vacuum. Compared it to what are all reasonable alternatives. Because you can't have economic growth or a banking system occurring in a vacuum. For example, after 1912, there was no banking system in Mexico for decades. Because if you have a complete chaotic situation of bandits running throughout the country and no centralized government in control of the peace, you can't have a banking system.

- 1.110 It's like an invitation. Put up a sign, "Come steal my money." Banks disappear. When you're asking the question, "Did this promote the growth of the Roman Empire?", which is a very good question, it's not so obvious how to answer that. Compared to what? Because you can't have an Empire if it's not stable. If the purpose of these regulations was to create a stable coalition to run the Empire, well, that's a precondition for even being able to talk about economic growth. Economic policies can't be understood in isolation without thinking about how they affect the ongoing **political equilibrium**.
- 1.111 Now we want to pivot to talk about the rise of the modern world. We're going to skip a lot of history. The whole Middle Ages just got done away with here. But we're going to talk now about the rise of the modern world. The modern world is the world that we still live in that starts, let's say, around 1600. What defines this modern world are changes in technology for weapons, shipping, navigation, which centralized power. Instead of a medieval world where power is very fragmented, think about all those beautiful castles that you've seen throughout Europe. That's what decentralized power looks like.
- 1.112 Somebody can sit in a castle out in the middle of nowhere and be pretty much in charge of that local area. But if you have a cannon, castle is kind of useless. That's a nice way to think about the transition, from a decentralized power in Europe to a centralized power, is the creation of a cannon, which makes the king not just the head of a coalition of local fiefdoms, but now the king is, if the king can raise enough money to fund an army and the weapons and the navigation and the shipping to now exert power in a much more centralized way, not just within the country, but now across the whole world.
- 1.113 That's what defines modernity as we use it in economic history and political history. It's a change of power that makes the center of power more centralized and it makes the breadth of the power more sweeping. It's fundamentally a technological change. Sometimes this era, the early modern era, is referred to as the era of **mercantilism-imperialism**.
- 1.114 Because there's a coalition between these new, powerful, centralized leaders, in Spain, in Portugal, in the Netherlands, later in Britain and France, a coalition between them and the merchants who are going to execute, not just the conquest, that's for soldiers, but they're going to set up trade routes so that the consequence of conquest can be the increase in trade. This is also physically associated with a complete shifting of the center of gravity of the world from the Mediterranean to the Atlantic Ocean. I'm not saying other oceans weren't important, too, but the Atlantic Ocean was really the key.
- 1.115 Now, in addition to the new technologies, there were new institutional tools of **mercantilism-imperialism**, which were fundamental and extremely novel. What were these? The granting by sovereigns, to particular groups of people, **monopoly rights** over some kind of activity. It could be a monopoly right to form a bank, like the **Bank of England**, or it could be a monopoly right to organize trade with a certain region, like the East India Company, or it could be other kinds of **monopoly rights**, to run a lottery, for example. So grantee of **monopoly rights**, that was

something new, as part of the strategy, new partnerships between particular individuals or groups or merchants and the sovereign.

- 1.116 Secondly, the chartering of companies, initially called **joint stock companies**, basically what we would call corporations, which allow the pooling of funding by a wide variety of investors. Again, it's all part of this need for large sums to help to fund centralized power. The issuance of new types of debt. We'll see that a lot of these sovereigns are issuing lots of different debt, like IOUs here, short-term this, a little bit of that. All of a sudden, we see big advantages to creating much more homogeneous types of debt, long-term bonds, what eventually became in Britain known as **consols**.
- 1.117 We'll see that there are big advantages to the sovereign of creating a single class of homogeneous debt, mainly because it raises the price by making the debt more liquid. The chartering, some of those corporations that are now being created are banks. Because a major idea that's emerging in the modern era is that there's a big advantage to the sovereign of chartering banks that have a mission, from the sovereign's perspective, of making the Empire great; and from the banker's perspective, having the sovereign as your ally is a very helpful thing.
- 1.118 When you're thinking about these new tools, don't just think technological tools, but new innovations of institutions. Again, what's the list? Monopoly privileges, **joint stock companies** or corporations with wide ranges of investors, new types of sovereign debt, what will emerge in Britain, the **consol**, and the chartering of privileged banks. Now, of course, Europe already had banks, but they were just privately created things, the Medici. This is not that. This is where the state says, "We're creating a bank, and it has special privileges."
- 1.119 Now, we could decide to charter lots of them. Some countries do that, or we could decide to charter only one. We're going to talk about that distinction, too. OK, so the point is, the modern era is all about, initially, technologies and later institutional innovations that are promoting this new centralized government with wide-ranging imperial ambitions. When we're thinking about **financial crises** in the early modern era, it's all about sovereign risk. It's all about the competition among these countries because that's what's defining the risk of the era.
- 1.120 OK, there are individual companies, they are issuing stock on the London Exchange. Sometimes it goes up and goes down. But the thing that really defines the crises we're going to talk about, the risks that make systemic problems, are the sovereign risks. The first two examples we're going to focus on, the **Mississippi bubble** in France and the **South Sea bubble**, are wonderful examples of how backward sovereigns, sovereigns that are late at the **mercantilism-imperialism** game, Britain and France, are desperately trying to use these innovations, these new institutional innovations, to catch up with the incumbents who were running the show at the time.
- 1.121 The thing that makes the **Mississippi bubble** so much fun to study is the character who's at the center of it, who was, perhaps still to this day, the most colorful character in the history of

finance, a Scot named John Law. John Law had been a schemer for a long time prior to this. But I think what might surprise you, if you've only read superficially about the Mississippi bubble and about John Law, is that financial historians that are studying what John Law did, and economists, do not characterize John Law as some insane person who managed to inexplicably drag the whole French Empire down through his cockamamie schemes.

- 1.122 No, no, that is not what I'm going to teach you, and that is not the received wisdom. The received wisdom is, John Law actually was an extremely creative person who understood all of the possibilities of these new institutional innovations of modernity—the corporations, the banks, the funding schemes— and who saw all of the mechanisms that others were also using at the time, understood them, and wanted to invent them in a new way. His particular ideas, qualitatively, all made sense.
- 1.123 I think now the consensus view is, he just took it a little too far. In particular, the thing that made that a little too far was when he thought that he could also fix the market prices of things. He had his scheme for his company, and he also was running a bank. And so then he said, "Well, I don't like the current price that the shares of the company are getting, so why don't I just print a lot of notes in the bank and set the price higher than what the market wants to set it at?"
- 1.124 And that was really, that idea or doing other kinds of behind-the-scenes, you might even say, quasi-fraudulent in the sense that, if you print up a lot of money, it's obvious that you're probably going to fuel inflation. He knew that. He was smart enough to know that. But people sometimes get stuck wanting to achieve a short-term objective and convincing themselves that, "Oh, all I have to do is just help the market a little bit see the wisdom of that price, and I won't have to flood the market with notes to achieve that." It turned out that he was wrong about that.
- 1.125 The lack of market discipline, the fact that one person is in charge of the whole country financially and can make, through centralized decision-making, can make whatever decision he wants, and can decide to try some scheme that's going to say, "That price, I want to keep that price high. I have all the tools I need to do that. I can print money like crazy." But of course, when you overdo it, then it doesn't work because, ultimately, you can't fix the stock price by just issuing money, at least not in real terms.
- 1.126 And so, as he learns the hard lesson of, there's also a market out there that decides prices, just because you have all the powers of France doesn't mean that you can set the price of something—that's a very important lesson. But the fact that he didn't get that lesson means that the whole thing came tumbling down. He is driven out of France, and for, I think it's more than a century, France doesn't even use the word bank in any chartered company because it's such a disreputable term. So it was a very major financial crisis and disaster.
- 1.127 But what I want to, when we come back to talk about it, what I want to emphasize is that it was not a crazy idea by some lunatic who randomly got control. We understand why he got control. We understand that his ideas were quite sound, were related to other things that people were

doing throughout Europe, that they were original, too. They were probably quite smart. If he hadn't taken them a little too far, they might have been very successful. In fact, they might have put France on a much better historical trajectory in terms of its global competition. Let's take a break and we'll come back to pick up where we left off.

2. Shaping Financial Systems

AT A GLANCE

Follows John Law from his Scottish land-bank proposal to the Mississippi bubble, showing that his ideas about credit, sovereign partnership, and installment shares were sound but pushed to excess once he abandoned market discipline and printed money to peg the share price. Turns to the South Sea bubble in Britain, explaining the equity-for-debt swaps of the Bank of England and South Sea Company, how they raised sovereign debt prices through liquidity and reduced default risk, and the insider-outsider dynamics documented by Temin and Voth. Compares Scotland's free-entry banking with England's Bank of England monopoly, closing with Calomiris' law of banking.

- 2.1 Welcome back. I want to talk a little bit about John Law before going into his adventures in France and the Mississippi bubble. John Law is a Scot. Why is that important? Where did he start his scheming? Scotland was a poor country in 1705, when John Law wrote his first scheme for Scotland, which was the creation of a land bank. As we'll see later, what's interesting about Scotland at this time is it's the relatively poor stepchild within the British Empire, as of 1705.
- 2.2 But by about 40 or 50 years later, it would become very wealthy and very successful and having a rate of growth and economic development over the next 40 or 50 years that was really almost miraculous. That miraculous growth rate, historians will tell you, was directly related to its ability to charter banks. There are two interesting things about that. One of them is, remember, we talked about innovations of institutions relating to the chartering of companies.
- 2.3 Scotland is perhaps one of the great examples of the use of some privileged chartering, three banks that had privileged charters, but then other banks that were allowed to have free entry and the creation of charters that weren't privileged. That is, you could get a charter simply because you could put together the capital to charter a bank. What's interesting about this is, Scotland is a poor country, first of all, that, as we'll see later today, in fact pretty soon, manages to propel itself to success with a very stable and innovative chartered banking system.
- 2.4 That's strange in the sense that it's happening within the same sovereign power as England and Wales. England and Wales, same king, different policy. What we'll see when we compare England and Wales to Scotland is that there's only one chartered bank in England and Wales during this whole period from 1694 until the 1820s, and that's the Bank of England. So first thing to notice here is, sometimes the differences in political equilibrium are happening on the same island with the same king, two different regions, completely different chartering policies. That's interesting.
- 2.5 Scotland had the power to make, on a decentralized basis, to make decisions in 1705 about creating banks. Of course, it made use of that power. And Law's first idea for this poor country of his was to use a land bank as a way to propel growth. And specifically, he proposed the creation of a land bank in which the bank would be having, as its assets, loans related to land,

and as its liabilities, paper money. This idea was rejected in Scotland. They didn't create John Law's **land bank**. But it was quite similar to ideas that ended up taking hold, very much so, in colonial America.

- 2.6 One of the great advocates for **land banks** on similar but not identical logic to John Law's logic in 1705 was Benjamin Franklin for Pennsylvania. Interestingly, Franklin's argument also was an economic development argument, saying that a **land bank** makes a lot of sense, and paper money, liabilities of a **land bank**, makes a lot of sense, especially if you're a poor place who's competing for immigrants called Pennsylvania. We want people to come to Pennsylvania rather than New York or Massachusetts or other places. One of the ways we can do that is to create institutions that make the creation of economic success easier.
- 2.7 Therefore, he promoted a **land bank**. It was almost identical to what Law was doing. I want to emphasize, Law wasn't just coming up with crazy ideas that nobody else believed made sense. In fact, the basic idea of the **land bank** was quite sensible, although maybe not perfect. But it was quite sensible to argue, we're poor, but we've got a lot of land. The land's value is only going to go up. Well, of course, that isn't always true, but it's going to trend up. We have an asset here that we can reliably lend against, at least to some extent.
- 2.8 We have no existing credit system. In 1705, it was a very limited credit system. It was growing. It changed very dramatically, very soon thereafter. But we don't have enough credit. The other problem we have, and a lot of Europeans had this problem, was that money that was based on gold or silver often got debased, either by the fact that the supply of gold and silver increased a lot, or by people imitating the coins or chipping the coins or whatever. And so, an idea that, again, was not crazy, and in fact, the **Wisselbank** in Amsterdam had proven it to be quite true, was that by having paper money based on a stable source of value, you'd have a better transacting medium that would preserve its value more.
- 2.9 So we were saying, let's have a **land bank**. Why? Because we can have more credit. We're a poor country. We can bootstrap ourselves by using the one thing we have, which is the land that we believe through a development process will grow in value, as the basis of the asset that we can lend against to create credit that will allow people to make more productive use of the land. Then let's have the liabilities of that be in paper money, which will hold its value better.
- 2.10 Ben Franklin argued that would also save us from having to import the silver and gold. We could avoid the expense of importing the silver and gold. We could import more hoes and other kinds of implements instead. This is not a crazy idea at all. In fact, you might say it makes perfect sense. It shows the creative potential of John Law from a very early point. He also understood another thing that turned out to be a general principle of modern finance in the early modern era, which was that there's a big benefit to having the government connected to a corporation.
- 2.11 The benefit was that the corporation had less negatives from the government as a result and more positives. The big negative of any bank, of course, in the Middle Ages, was that the sovereign defaulting on it would often be the thing that made the bank go bust. But if the

sovereign owns an interest in the bank and is in partnership with the bank to achieve mutually desirable objectives as part of a charter that defines the mission of the corporation as being something that's friendly to the sovereign, to lend to the sovereign, to promote the sovereign's interests in trade, to do other kinds of things, that's something that the sovereign is not going to want to destroy.

- 2.12** Furthermore, if the government is willing to use its tax revenues as a part support for the liabilities issued by the bank, and this was something that Adam Smith talked about, too, but John Law talked about it quite a bit; if the bank that's chartered by the government issues liabilities, generally, bank's liabilities, like the notes that they issue, are backed by the assets of the bank. But what if the sovereign also connects the bank to its own tax revenues as a further backing and says, well, we, the sovereign, are willing to insure you during difficult times as part of our partnership.
- 2.13** Law recognized that this could be mutually advantageous. This could create long-term stability for the banking institution. I want to emphasize, there are some risks and problems associated with these ideas of partnering between sovereigns and banks. But his emphasis was on the developmental aspect of, how do we get from being very poor and what do we have to go with? We've got land, which is mainly valuable because of future things that are going to happen here. We've got our sovereign, whose debt is mainly valuable because of future taxes that the sovereign is going to do.
- 2.14** So we are thinking about our future world, and we're bootstrapping ourselves by saying, "What have we got going for us that's most reliable, and how can we build our development from there?" This idea that sovereigns can reduce risk by giving backing to corporations and allow those corporations to develop faster, and that you should go with the things that you have to try to build up the credit and the money that will drive your growth further, those are not crazy ideas.
- 2.15** Now, I want to emphasize, though, not all of his ideas were right. He assumed that land values would be inherently more stable than silver or gold. And of course, that's not true. Land values, as you all, I'm sure, have experienced, from being around for the last 20 years, can go up and down. So he exaggerated that case. But his basic idea about using credit to promote growth and maybe using land, that was a very good idea. It was rejected.
- 2.16** So he goes to France. And his proposals don't immediately catch on in France. He's coming in and he's trying to, he ends up engineering a radical restructuring of French finance. But you might wonder, how does a Scot who lands in France get there? We'll talk about it. This is radical in the sense that it involves a complex network of institutions that he creates, issuing shares in a joint stock company that's going to end up owning land rights in Louisiana, trading rights for beavers in Canada, all of the tax collection rights within France, a new bank that's going to be chartered as a monopolist bank that's also going to issue legal tender paper money.

- 2.17 That company that he's going to establish is also going to restructure entirely all of the existing debt of the French government into a new form of debt. It's going to exchange the outstanding debts that are held, there are many different kinds of debts, for one homogeneous debt, and then the main holder of that debt is going to become that company itself. This is a fairly ambitious system. It's going to take all of the trading rights of France, all of the land rights of the new colonies that it's expanding into, the bank chartering rights, the paper money creation rights, as well as restructuring all of the debt, as I mentioned, also with all the tax collection rights, and it's going to put those into one company.
- 2.18 Now, he recognizes that he needs investors in this company. One of the tricky things is that there's not a lot of liquid wealth in France to invest in his company. So he also comes up with a really important idea of **installment plans**, so that you can buy, let's say, 5 or 10% of the shares now. And then if you don't pay the installments in the future, you lose the shares. But it allowed people to invest in something even though they had very little money up front. Again, a very creative and interesting idea.
- 2.19 There are some problems with it because, of course, it can mean that your stock isn't really invested yet. That is, people are investing in your company, but then if the share price falls, that stock isn't forthcoming. Of course, that ended up happening to him. But it was a part of a scheme, again, to try to boost a country that was just developing. The creation of the bank is a particularly interesting part of it. He wanted to try to, as part of this mutually enforcing system, to make the paper valuable, partly by making taxes collected only in the paper. So saying, "You wonder whether this paper is valuable? It's the only way you can pay your taxes."
- 2.20 Then, of course, on the other side, since the bank is owned by the government, the paper is fully backed by future taxes. He did a lot of other things to try to make each element of his system, including the bank, accepted and considered reliable. One thing that's a little unusual is that along the way, in May 1718, he nationalized the bank. You might wonder, why would he do that? The answer is that he wanted to eliminate any other voice in making decisions. If you have shareholders, it means shareholders get to vote.
- 2.21 What he was all about was **complementarities** and synergies, which he believed he could achieve—I'll show you a graph next page—which he believed he could achieve by doing all these things together. He didn't want any dissenting voices. In fact, this company ended up owning everything, all of the trading rights, all of the land rights, all of the rights to collect taxes, all of the rights to the bank. And held most of the debt of France, too. It had all of those things. He was the head of that company, as well as the finance minister of France.
- 2.22 It was a complete centralization of power, where the whole idea was, all of these **complementarities** that all the other people are theorizing about, and that he is also theorizing about, he's going to realize those **complementarities** with no dissenting voices and no reasons to have to wait, do things really quick, and get to where he wants to go. That is part of the story of the nationalization of the bank. Because if he nationalizes it, well, he's the finance minister.

- 2.23 Even though you might say, "Why would he give up being the head of the bank?", he didn't give up being the head of the bank. He was the head of the bank either way. He also got rid of the redemption requirements for bank notes so that the notes, initially when the notes were issued, he made them receivable for taxes to boost their value. But then he decided he didn't want to have to deal with the inconvenience of having to maintain parity of the notes in specie.
- 2.24 This is the beginning of the weak side. The weak side is, "I can be in control, I can get all these complementarities, but I'm going to not have any market discipline. I'm not going to have anything holding me to the fire, making me have to make good on a commitment to maintain the value of things." You can see that fairly early by 1719. This is a graph that I constructed to try to give you a sense of what these complementarities look like. I'm not going to go through each element of this, but what I want you to see is, the point of each of these arrows is, this causes something to increase from beyond what it would otherwise be.
- 2.25 So the point is, by having control over each of these elements, you make each of them bigger because they're mutually enforcing, complementary, in their structure. And as I said before, each of these arguments makes sense. If I'm the person that holds the debt, well, obviously, I'm going to be the most aggressive one to do tax collection, right? So making me both the debt holder and the tax collector, there's a clear incentive complementarity there. He made these arguments along the way about each of these arrows, each of these elements.
- 2.26 What I want to emphasize is, a lot of these arguments made sense. I've mentioned the alignment of interest. If the sovereign and you are linked, the sovereign is not going to default on you as frequently. In fact, the sovereign is more likely to help you through difficult times, knowing that you're also there and, as part of your charter required, to help the sovereign through difficult times. We're going to get a lot of good things out of this, but what are we missing? What are we losing? Decentralized decision-making and market discipline.
- 2.27 So autocracy has its limits, right, in terms of wisdom. So even very good ideas maybe aren't good enough. And that's what I want to emphasize as the major governance flaw. And it comes up when he believes that he can also use this new power to create currency and to have made it a legal tender, so it has to be used as the sole currency to control everything, and then secretly printing more money to try to boost the stock price when he wants to boost it.
- 2.28 Why is he driven to boost the stock price? Because once he's gotten these people to invest in installments of the stock, to get them to complete their investment, he has to keep the price up. So he doesn't want the price to ever fall because then some people might decide not to keep investing. So he's balancing these things, trying to get everything to work. It's a little bit too much centralized control, of course, to be able to pull it off. Of course, his ambitions are quite great.
- 2.29 Here's a graph that gives you a sense of how the price of this grand company goes up and down. I'm going to be talking about these prices a little bit. So you can see it reaches almost 10,000 a share, and he starts pegging it. You could see, let me come over here. Right here, he starts trying

to peg the price as it weakens to sustain it, so he can get all those installments to be realized. And though as he does that, once it becomes clear that he's just printing money, then we have a collapse of the whole scheme. He's forced to flee the country.

- 2.30 But when you look at this, I'd say some of the early reactions were, "Oh, this whole scheme was ill-advised from the beginning." And that's not what I would say financial historians generally believe. What they believe is that it probably was sustainable. The price of this probably could have been sustained somewhere in this middle range. It was this last bit that was an excess. If he hadn't been so confident of his ability to prop that price up by the additional amount, this scheme might have lasted and done quite well.
- 2.31 I'll show you, what would it take to be able to maintain, for example, about a 5,000 price? François Velde did a bunch of valuations, and they have different notions of what's the sustainable price. But what I want to do is, I want to focus on this one, which I think is quite reasonable. Under these assumptions, the way we work in finance is, we make different assumptions and we see what would the price be under these different assumptions. I regard this as a fairly reasonable assumption.
- 2.32 If you had a 6% discount rate in valuing the shares, you assume about a 3% dividend growth, and with certain assumptions about which were real shares and which weren't real shares, you could maintain a price of about 5,000. You can see what I was saying earlier, that you can show that these prices are sustainable pretty much in the middle range there if he hadn't pushed it to the end. But you might say, well, why was he driven to push to that high price?
- 2.33 Because in the beginning, he was operating in a very illiquid market where he had to use this installment plan, which made him need to keep propping up prices to make sure people saw continuing incentives to continue their investments. This gives you a little bit of a sense of how much money creation came about in that 1719 to 1720 period.
- 2.34 Once he started trying to use the money supply to prop up the stock price, you can see that he used it in excess, and of course, it didn't work, ultimately. This is a graph of the money supply, which is the solid line, and the price level, which is the boxes. You can see quite a big effect on prices, which made this very unpopular. This scheme was very unpopular because it pretty much doubled the price level over a very short period of time.
- 2.35 I want to move on now to talk about the next example of this. But I guess one thing in summary, notice that this financial crisis was created by a sovereign. It was created by France. The whole scheme, all of the risks that were created, were in pursuit of the expansion of sovereign power. The logic of the creation of this grand scheme and all these financial tools actually made sense. The thing that didn't make sense was abandoning the discipline of the market in deciding what the price of things was, and thinking that you could take your sovereign autocratic power to the point of even deciding the market price.
- 2.36 If he had been a little less ambitious, gone a little slower with his installment plan, making himself less dependent on the constant increases in price to drive interest in the stock, he

wouldn't have been pushed to use the inflationary tool, and he might have succeeded quite well. Let's go to another example. It shares a lot with this, which is the **South Sea bubble**. Like France, Britain was relatively behind in the international competition as a nation in the 17th and early 18th centuries.

- 2.37 A little bit of political background to Britain. They undergo, in 1688, something called the **Glorious Revolution**. This is a change in monarchs where they bring in **William and Mary** from Netherlands, **William of Orange** comes in. It's not just a change in ruler, it's also a change in regime toward a more liberal set of policies, more tolerant of religious differences, more internationalist in its ambitions, and very anti-French and anti-Catholic in terms of international competition.
- 2.38 This is also the beginning of having war between Britain and France that's going to last from 1688 until 1815. It's going to be the defining feature of British public finances and financial system for this whole period. How can we beat France in this continuation of war? You can't understand anything I'm going to tell you from now until we finish this topic on the rise of London, the shape of the English and Scottish banking systems, the nature of the public finances, the chartering of the **Bank of England**.
- 2.39 All of these topics are fundamentally driven by this single fact of the war between Britain and France, which makes Britain desperately in need of financial resources. Now, from the standpoint of the **South Sea bubble**, that has to do with the share price of a particular company called the **South Sea Company**. This **South Sea Company** was chartered in, I think it was after 1710. We'll find out. It's in the next slide. Oh, there it is, 1711.
- 2.40 Initially, this company was given two kinds of activities to do. There were lots of these. Companies would be chartered by particular groups at particular times with particular activities. We'll see a long list of them. The **Bank of England** was chartered in 1694. What was its goal? It was chartered to do an **equity for debt swap** to help sovereign funding. You might think, oh, the **Bank of England**, I think of it as now a central bank or a bank of issue at least, whose job is to create paper money or something like that, or to create credit for other banks. Those are jobs it got much, much later.
- 2.41 Its job was to help expand the sovereign's ability to raise funds by doing an **equity for debt swap**. So I'll talk about that. But it's worth mentioning that now, though, because you'll understand, the **South Sea Company** had the same mission. What's the difference? Well, the **Bank of England** was chartered by a group that requested a charter when the **Whigs** were in control, and the **South Sea Company** is when the **Tories** are in control.
- 2.42 So what's also interesting is, these companies that are being created as partners with the sovereign are private enterprises that are being chartered by governments depending on whether the government wants to ally itself with a particular political group at a particular time. These companies, so the **South Sea Company** had certain trade privileges and it had the ability

to enter into contracts with the Crown for debt swapping. I'll talk about that. The **Bank of England** only started off doing the second of those.

- 2.43 But the **South Sea Company**, its **monopoly rights** on trade with Spanish colonies in America became almost worthless as a result of certain Spanish victories that were happening in the late 17th and early 18th centuries. So the South Sea Company's business strategy was pretty much driven by converting sovereign debt. To understand the **South Sea bubble**, we need to understand the **South Sea Company**. To understand the **South Sea Company**, we have to understand what these **debt swaps** were that it was engaged in, just as the **Bank of England** had been earlier engaged in.
- 2.44 I also want to mention as background that what's happening here is, as all these companies are being created, we're also seeing the rise of London as a financial center. This is a big surprise. It's good to keep this in the back of your mind. Also, we'll try to understand how much of the rise of this is due to these companies, in particular, some influence they're exerting on the markets. That's going to be a subtle question to answer, but we're going to come back to this and argue, well, not as much as you might think.
- 2.45 But this is a quite surprising fact that's happening in parallel, that London's actually displacing Amsterdam as the key financial capital of Europe by, let's say, the middle of the 18th century. That is not something that you would have guessed would happen. It's not because England imitated the Dutch; they didn't. The **Wisselbank** in Amsterdam and the **Bank of England** weren't the same at all. The **Wisselbank** was a **clearinghouse** for bills of exchange of Dutch merchants and other countries' merchants, too, all happening in this one place. It was not a tool for lending to the sovereign. That's not why it was chartered.
- 2.46 The **Bank of England** had no role in exchanging the bills in the private market because nobody trusted it, because it was a political entity. But it had a central and initial role of just serving the funding needs of the sovereign through this weird new mechanism of debt exchange for consolidating debt and having that debt basically be held by this new **joint stock company**. This is the same idea that **John Law** is learning from, the Bank of England's idea. And the **South Sea Company** is also learning from it.
- 2.47 So we have to really understand. I just want to be clear here. I'm not making the argument that these sovereign debt schemes made London a financial center. I'm saying that it's an interesting question. I'm going to point in a different direction for that. But I am going to say it's interesting that these two things are happening in parallel, that we're getting the creation of these interesting debt exchange schemes. The **Bank of England**, later the **South Sea Company**, notice that **John Law** is doing the same idea. I haven't explained it fully yet of why this makes sense.
- 2.48 Then in parallel with that, there's the rise of London. My view is, just so you don't get confused, these two outcomes are both coming from, the successful public finances and the successful private finances are both coming from the relative strength of Britain from a governance perspective, the Parliament and the nature of the British sovereign after the **Glorious**

Revolution. It's not that one is causing the other. It's not that the funding schemes are fixing all of the public finances. They are helping, though, as we'll see, but they're not driving the bus.

- 2.49 Neither are the public finances driving the bus of the expansion of the private finances. But neither is the collapse of the South Sea Company's stock in 1720-21. Neither is that causing a big problem for the continuation of private development. These are happening on somewhat separate tracks, unlike in France. The driver that's favorable to both of them, I would say, is the governance structure of Britain. But they are happening in parallel, and that's interesting to keep in mind.
- 2.50 OK, it's useful to put into context all of, how desperate the public finances of the British sovereigns were and how much experimentation was going on. So the experimentation had to do with creating monopolies over life insurance called **tontines** at that time, which is a particular kind of life insurance policy, to allow the issuance of annuities to try to raise money for the state. Similarly, the million adventure lottery of 1694. So the state is pretty desperate for money. It's creating a life insurance company that it runs to try to find a way to attract funds.
- 2.51 Remember, it's a monopoly. It's creating a lottery. It's getting the **Bank of England** to do its debt swapping. It does another lottery in 1697. It's chartering a lot of companies, and all of these companies have some mission of helping to expand its funding. Maybe the hardest one to understand is this **debt swap** that the **Bank of England** really originates and the **South Sea Company** tries to build on. So, there's this constant struggle among political interests to have control over these swaps.
- 2.52 The **Bank of England** gets to do it first with the **Whigs** who are in control of that. The **Tories** enter with the **South Sea Company**. They want to do it. This gives you a sense of how successful the **South Sea Company** was in becoming a holder of this consolidated government debt. If you look at how much of the government's debt is the **Bank of England** from its creation is able to hold? It's holding a third of all the outstanding government debt right after it's created.
- 2.53 Then you can see, when you put the **Bank of England**, the East India Company, which also had as part of its mission to hold some sovereign debt, and you put the **South Sea Company** together, and you can see right here how, from its founding, it becomes even more important than the **Bank of England** as a holder of sovereign debt. You could see that a major purpose of these companies was to consolidate government debt holding into these monopoly corporations that were created by the government.
- 2.54 That's a question, right? That should be a question. This sounds like alchemy. Why would you spend so much trouble creating a corporation whose mission it is to hold government debt? Why not just issue the debt into the market? Because obviously, the whole story, we've got two stories going on in London at this time. One, the development of the private finances, the other, the development of the public finances. We're focusing first on the public finances. What's driving this bus?

- 2.55 Why is it that these **equity for debt swaps** that we'll describe happening at the **Bank of England** and the **South Sea Company**, why is it that these are useful at all? If they're going to be useful, what's going to be the measure of their usefulness to the sovereign? What is it? It's going to be raising the price of sovereign debt. Here's the question. How could it possibly be that creating companies whose mission it is to hold sovereign debt raises the price of sovereign debt?
- 2.56 In a frictionless finance theory, if you ever went to business school and you took finance classes, people would scoff at this. Well, how can this, having somebody whose charter is just to hold the debt doesn't affect the price of the debt. The debt is priced at whatever the debt should be priced at. But that's a very simple-minded view. So let's think a little bit about what affects the price of debt. Two things affect the price of any debt instrument.
- 2.57 One is its **liquidity**. If you have fragmented debt and you have an IOU that matures in 90 days, you have a different IOU that matures in two years, they're both promises to you two individually, they're small denomination—nobody's trading these, right? I don't know about you, but I don't get to trade my IOUs even if they're from the government, right? Short-term **instruments** that are idiosyncratic and customized to particular transactions don't get traded in markets.
- 2.58 That means that they're not very liquid. That means that, for a given sovereign **default risk**, their prices are lower than they would be if they were liquid. What makes a market liquid? Homogeneity. If we can create a single debt instrument, a single long-term debt, and we can take all of the existing small short-term **instruments** that are customized, and convert them into a single kind of instrument, that instrument is going to be a higher price just by virtue of the fact that it's more liquid.
- 2.59 But there's more. We're also doing something to the **default risk**. We're reducing the **default risk** of the instrument. How? Because we're putting so much debt into **instruments** that are owned by the most powerful political players in the country, in both the **Whig** party and the **Tories**. And by the way, they're also owned by the sovereign in large part. And beyond ownership, they have joint interests of a mission to complement each other and to support each other.
- 2.60 Now, the **Bank of England** notes aren't initially supported the way the Banque Générale's notes were by the taxes of the sovereign, but that changes; by the 1830s, they are. You might even say implicitly they were before. I'll talk a little bit about the Napoleonic Wars, where we see this complementarity between the **Bank of England** and the sovereign really coming out. When the sovereign's in stress during the war, the **Bank of England** basically has to do certain things and is required by the sovereign to do them.
- 2.61 What I'm trying to tell you is, this partnership is doing two things, this **debt swap** partnership. It's first creating greater **liquidity** of the sovereign **instruments**, which is raising their value, and secondly, making them less likely to default. Now, there's another thing, though, going on here that I want to emphasize, which is tax reforms that are happening separate from this whole swap process. I'm not saying, and it would be a ridiculous thing to say, that all the improvement

I'm going to show you in a minute in sovereign funding opportunities in Britain is driven by these swaps. That's not the case.

- 2.62 It's mainly tax reforms that are driving the bus here. But the swaps are contributing substantially. How big is the effect on the price from the swaps? Well, we know the answer to this because the swaps were a contract. The way it worked was, let's say you're the **South Sea Company**, you enter into a contract with the government that says, OK, if you give us a million pounds of all these different heterogeneous debts, we'll give you £900,000 of consolidated debt.
- 2.63 In other words, the government's making a big profit on this transaction, and the company is entering into a contract with the government saying, "Here's what we're going to do. We're going to go get all of these debts and give a million to you, and you only have to give us a million less 5% or a million less 10%." That's the contract. So it's clear that the government's funding capacity is substantially improving as a result of these contracts.
- 2.64 We can't say how much of it is coming from **liquidity** and how much of it is coming from reduced **default risk** because of the partnership. But what we can say is, this isn't alchemy. This isn't some fake chemical process. This is real. This is called real financial design, real intermediation. It's quite important to recognize that this is a major contributor to the sovereign debt capacity. It's very important to bear in mind that there had been a lot of experience with sovereign default, including in England in the 17th century.
- 2.65 Creating these companies, whether the Whig-run **Bank of England** or the Tory-run **South Sea Company**, that had clear political power-sharing relationships, was very important because it meant that you could rely on the fact that there wasn't going to be a continuation of that default experience. By the way, another thing to say that's very relevant is, bribes were legal. When they're selling these, the way these share swaps work is, the **South Sea Company**, let's say, enters into a negotiation with the sovereign. They enter into an agreement where the sovereign says, "You give us these amount of debts that you buy, and we will give you that minus 10% of consolidated homogeneous debt."
- 2.66 To get that to work, how are they funding that? They're issuing shares. So the point is, I issue shares in the **South Sea Company**, and then you can pay for these shares by giving me government debt. You don't just have to give me cash. You can give me the government debt. And then I bundle that government debt and convert it. So notice that the people buying the shares in the **South Sea Company** are implicitly understanding this alchemy. They're implicitly understanding that what the **South Sea Company** is buying is a lesser amount of a better government debt.
- 2.67 It's the same nominal amount, but it's going to be more liquid and more reliable, less **default risk**, because it's going to be held by the **South Sea Company**. It's a really interesting transaction where the shares are being actually paid for by government debt, and then they're converting that into a lesser amount of government debt. You wonder, how can anyone want to be part of this arrangement? It seems like it's a losing proposition.

- 2.68 I'm giving \$10 of government debt to get a share that entitles me to \$9 of government debt, plus whatever franchise value there is in the company, which is small at this point probably, because there's not a lot of trading rights having to do with the Spanish colonies in America. I think you get the point, right? It's a surprising transaction, and it's, what's so interesting is that it works. But a part of the reason it works, too, is bribes were not illegal. So you could go to influential government officials and bribe them to help encourage participation in the trade and actually go out and advertise for you.
- 2.69 And of course, why is that useful? Because it shows people that there's a commitment of powerful people to not default on this debt. So this is a different world from the world that we're used to. It's a little bit different. The way partnerships between government and the private sector works is much more fluid. There's ownership interest by the government in private companies. There's private companies relying on the government's commitment to their partnership not to default on them. There's bribes being paid to express support for the company.
- 2.70 You can see that some of this **default risk**, that what we're seeing from these effects is that, as I said, it's affecting the value of the government debt by 5% to 10%. It's not the only thing causing government debt to improve. Taxes, reforms are a big part of it. But 5 or 10% is no small amount. Now, one of the things that's a little bit different or interesting about the **South Sea Company** is that we have a lot of data about the actual trading that occurs by different entities in the shares of the **South Sea Company**, when the price run-up is at its maximum. I just want to show you a few more graphs about this.
- 2.71 One is, you can see how much borrowing is increasing during this period, that the sovereign is getting a huge benefit from being able to borrow. You can see also that we switched from a situation here, where almost all of the borrowing is in short-term form. As that debt is growing, almost all of it is in long-term form because that's the homogeneous converted debt that the **Bank of England** and the **South Sea Company** are creating. That debt is growing pretty remarkably. You might think that with that growth is an increase in interest rates. But you'd be wrong.
- 2.72 What's interesting is, the combination of these financial innovations in the **Bank of England** and the **South Sea Company**, as well as the increases in tax policy, are causing a major improvement in sovereign debt creditworthiness. You can see this in the form of the lowering of yields that's happening, especially during the 17-teens and the 1720s. That's not just true for London, for British markets, but it's also British markets relative to Amsterdam, so that there's a big improvement that's happening here.
- 2.73 Part of this has to do with, as I said, the political structure that wants, as this debt is very broadly held, that wants to see tax policies enacted that makes the debt paid. As this debt becomes more widely held, beyond just the South Sea Company's holdings, the Bank of England's holdings, **Antoin Murphy** has argued that that also created through Parliament a constituency

for raising taxes to make sure the debt was paid. But it's this combination of things that's really fixing the public funding problem.

- 2.74 Then on the private side, what's happening is, again, because of this kind of, I would say, pluralistic structure of England after the **Glorious Revolution**, you're getting a lot more companies being chartered. It's not just the **South Sea Company**. That's what's different about the Mississippi experience. **South Sea Company** was just one of many companies being chartered. And yes, its share price went up and down and wasn't sustainable and was a **bubble**, but it didn't fundamentally sidetrack the private development or the continuing public improvement.
- 2.75 So I would say the **South Sea bubble** is not really a very important event in the development of either the public funding of the sovereign or the development of the private markets. Now, having said that, I do want to still talk about the **bubble**. It's not—very unlike the **Mississippi bubble**, which is a defining experience in the history of France, and a very negative one; the **South Sea bubble** was not really as important.
- 2.76 The other thing that we can say about the **South Sea bubble** is that it was perceived as, by insiders, by knowledgeable people, it was perceived as a **bubble** while it was happening. And that made it a little bit different. So this is a case where, when we do a valuation of the **South Sea Company**, we end up seeing that, if you were a reasonably informed person who understood all of the finance logic that I've just been explaining here, you would have seen that the implicit value of the franchise—that is not the debt itself, but this other stuff that the **South Sea Company** was not doing.
- 2.77 It didn't have trade rights. It wasn't able to really exercise them. That implicitly, the market price was putting a huge value on that. And what's our best explanation for this? Well, early on, we know that insiders saw an opportunity because they had lived through the **Mississippi bubble**. It was even the same people who had made money from the **Mississippi bubble** but gotten out before the crash who, because they had money to invest, were now investing in the early part of the **South Sea bubble**.
- 2.78 So, some of the impetus for this was bribed politicians that were touting the South Sea stock. Some of it was investors coming from elsewhere in Europe, initially. That helped to propel it upward. But then what we also see is that, as the price got propelled upward, if you understood how to value the company, you'd realize that there was an implicit value being placed on the company's franchise. That is not just the assets that it was holding, but some kind of intangible that the company would be able to deliver on from trade rights or something else that you couldn't really believe made sense.
- 2.79 Then what you start seeing is informed parties withdrawing from the market. Because we still have some data about very informed traders—there's a bank in particular that I'm going to show you an article about—we can see them switching from buy orders during the initial period where they're making buy orders that are profitable. Why? Well, partly because the basic idea

of this wasn't bad, and also because there was a lot of money coming in from the rest of Europe right after the **Mississippi bubble**, looking for something to invest in.

- 2.80 So insiders were on the buying side when the market was going up. And then once it got to a certain point, insiders switched. And so this is a great example of one where, it's the first example I can think of where uninformed traders who didn't really understand what they were doing got hugely invested in the market and had big losses. And so, the importance of the **South Sea bubble** is not so much its consequences for the financial system, in contrast to the **Mississippi bubble**, but more as an example of the first time that outsiders versus insiders had very different experiences in the market.
- 2.81 Because, I guess now I'll give you my interpretation, the outsiders were simply extrapolating from the returns and thinking, wow, this is going great. I want to do everything I can. So I'll show you a poem in a couple of slides about ladies who are part of the elite who are selling all their jewelry to buy shares in this, and they've never done any financial analysis. This is something completely new. You'll see that **Newton** is saying he's puzzled. He doesn't understand why people think that the value of the company is as high as it is.
- 2.82 There's a lot of, and you'll see that **Hoare's Bank**, who's an insider involved in trading in these markets, a very experienced trading house, goes from being on the buy side to the sell side. We have this really interesting example, I would say, of the kind of thing that **Minsky** and **Kindleberger** talked about. I'm not saying that their point is never useful. This is an example of where the people who understand how to do valuation switch from being buyers to sellers, and the people who don't understand how to do valuation are riding the momentum all the way to losing their shirts.
- 2.83 Let's just look at that a little bit. I want to emphasize, it's not like the **South Sea Company** shares were worthless. It's just that they went from being shares whose value made sense, here they are, made sense, maybe even made sense at prices around here. But then it gets blown overboard, and then it comes back down to levels around 200. I don't want to go into all of the details of the valuation. There are some really interesting biographies of different traders who are involved in this.
- 2.84 But the study I want to focus on is **Temin** and Voth's 2004 study that looks at **Hoare's Bank**, a Fleet Street bank that's still in existence, at least as of 2004, that had been in operation since 1690. So they had their trading books. And what they were able to see was, you see the circles are when they're selling and the triangles are when they're buying. And so they're operating as a broker. They're on both sides of the market. And so they're able to observe when they're buying and when they're selling at different times throughout the whole **bubble** experience.
- 2.85 But then they do a statistical analysis. When they buy, is the market more likely to go up after or to go down after? When they sell, is the market more likely to go up after or down? What they find is, of course, on average, the market goes either way. But for the insider, when they're

selling, the market goes down afterwards, and when they're buying, the market goes up afterwards. That's what this table shows.

- 2.86 I mentioned Sir Isaac Newton who said that he cannot calculate the madness of people. This is an example of where it seemed like both the people who were coming in from abroad saw the bubble coming and got out in time. The local specialist investors got out in time. People who were calculating investors like Newton seemed to see it coming. Then this is a poem that gives you at least a sense of the point of unsophisticated investors being drawn in to the market, where it's focusing in particular on the wives of the elite.
- 2.87 Of course, probably some of them, their bribed husbands were out there jobbing the stocks and telling everyone why they should buy them, and maybe the wives listened to their husbands. That would be an unusual event, but we can't rule it out. I want to just emphasize, at the end of this whole discussion, thinking about the South Sea bubble and the Mississippi bubble, it's just so important to recognize that they're both about state schemes, and they're both driven by the funding needs of states that are involved in imperial expansion and wars, in fact, with each other, that defined their history from France and England from 1688 to 1815.
- 2.88 It would be hard to think that the lesson of these bubbles is about individual stock behavior, some constant behavior by individuals, because they didn't have constant behavior. They behaved very differently in the Mississippi and the South Sea bubbles, first of all. But also, this is mainly, despite those differences, this is mainly about the common need of the two sovereigns to engage in extremely creative and ultimately risky schemes to try to increase their revenues.
- 2.89 Larry Neil believes that the main problem with the South Sea Company's stock price was that in their negotiation with the British sovereign—remember, the beginning is, they negotiate what the exchange is going to be, how much they're going to have to give over what the sovereign gives them back. They were involved in bidding for that deal against the Bank of England. So the Bank of England's bidding and the South Sea Company's bidding for the same swap deal. The Bank of England puts up a price. The South Sea Company has to bid even higher.
- 2.90 And then, of course, if they hadn't been able to feel like they could enlist the political help of influential people to try to tout their shares, maybe they wouldn't have been as interested. And so Neil actually says that the government drove the South Sea Company to an excess bid by insisting on an unrealistically high amount of profit that they were going to get from the deal. If the government hadn't insisted on making whatever it was, 10% on the deal, if they had gone for 5%, there would have been a very different outcome.
- 2.91 So not just in a general sense, but in a very specific sense, these bubbles and these problems were absolutely driven by competition between these two sovereigns who were the two latecomers looking for some way that they could make a big splash in international markets. And another meta-thinking about these two crises is, the ideas made sense from the standpoint of aligning incentives to reduce default risk, creating more liquidity in markets. These ideas made sense, just as many of John Law's other ideas made sense. The point was that, because of

the excessive pressures on both of the states, the ideas were driven to a point of excess that made them untenable.

- 2.92 We also see how investors who are smart can do fundamental analysis and profit from it in a very innovative environment like this. What can we say about the repetition of crises? Do we ever see anything like this again? Well, in a sense, no. We don't see any repeat of the specifics. These debt for equity swaps by the Bank of England, the South Sea Company, the Mississippi Company, we don't see that. But what we do see is the LDC crises of the 1980s and the emerging market debt crisis in the 1990s and 2000s.
- 2.93 And what we're seeing is, I would say, reflecting relatively similar pressures to innovate, to do institutionally innovative but somewhat risky sovereign strategies. Emerging markets is all about countries deciding to fundamentally change many of their institutions all at once to try to catch up with other countries. It was the success of, in particular, countries like Chile and Korea by the 1980s that showed a lot of the advantages of economic liberalization that drove other developing countries to try to also become emerging market countries through a particular set of institutional reforms.
- 2.94 It's very risky to try to change so much in your economy all at once. I would say, even though those were good ideas and they've stood the test of time, and they're not even controversial anymore, that emergence was a successful strategy. There were lots of growing pains. When we talk Mexico in 1994 and South Korea in 1997, we'll see what I mean by those growing pains. But I think in some sense, that reflects a similar international competition among countries for global capital and for global markets more generally. I would say that, not only have we learned not to do this, but we've learned that doing this is probably a pretty good idea, as long as you don't do it to great excess.
- 2.95 I want to now turn back to something I mentioned earlier, which is this difference between Scotland and England in their banking systems. This is really a remarkable history. Try to imagine, in the United States, if we had our Congress, had one set of relationships with Western states and a different set of relationships with Eastern states in terms of fundamental things like banking laws or even parliamentary representation. There's a lot going on here. In 1707, Scotland gives up its independent parliament.
- 2.96 So let's just think about this a little bit. We have the same sovereign of England and Wales and also of Scotland. In England, only one bank is allowed to be chartered, the Bank of England. It's given the monopoly right to be the only chartered bank, the only one to issue notes, as we'll see, and to be the only chartered bank. All other banks must be partnerships of a small number of individuals. They have to be small, which means they can't issue notes, they can't issue paper money, they have no connection to the government per se, they aren't chartered, and they have to stay small.
- 2.97 In Scotland, there are three chartered banks, not one, that have special privileges that are granted by Scotland. But then in addition, they allow as many as want to come to be chartered

banks. So instead of limiting you to say, well, if you want to be a bank, you can't be chartered and you can't be large, the non-three privileged chartered banks, the other banks in Scotland, some of them got to be quite large.

- 2.98** Scotland develops its banking system much faster and much better in terms of credit supply, and also because it's grounded in large, diverse banks operating throughout Scotland, which is a fairly diverse economy, ultimately. Those banks are much more stable. If you look at England and Wales, they're experiencing credit scarcity and banking instability, coming from the fact that they're not allowed to have banks that issue notes that are operating throughout England and Wales that are big.
- 2.99** What's the purpose of all this? Why is the sovereign doing this? Why does it have one flavor in England and Wales and a different flavor in Scotland? OK, well, let me give you a theory. I need somebody that I can rely on. I need sufficient help and funding to be able to manage my public debt. We've just seen that the combination of the **Bank of England** and the **South Sea Company** provided huge support. Debt was able to grow dramatically, and yields were able to decline.
- 2.100** From that standpoint, you might say the first ingredient of the explanation is that the British sovereign didn't need to also have a monopoly partner in Scotland to deal with its public finances. It could do well enough with the **Bank of England** and the **South Sea Company**. Secondly, it seems to have wanted to placate the Scots. Notice Scotland's far from London. I'm having war with the Catholics. There are lots of Catholics in Scotland. And so what I want is a way to create a triangulation, where I give the Scots more freedom to charter banks, create credit, and develop their economies, which they do on a grand scale, in exchange for making sure I have their loyalty.
- 2.101** So another piece of this agreement to triangulate is to get rid of the Scottish Parliament in 1707. Scotland is still represented in the British Parliament, but they used to have a separate Scottish Parliament. And this is, I would describe it as unilateral disarmament. Kind of saying, we're not relying on you to fund the British government. We're not relying on bank chartering policies in Scotland to fund the British government. We're going to let you have your own autonomy in deciding your banking policies to promote your own local growth. You don't have to have this **Bank of England** thing going on there with a single monopoly and no ability to grow your credit.
- 2.102** But why don't you do something nice for us? Like, make sure that your political organizing only occurs in the context of the overall parliament, not in a separate parliament, too. I think that's quite interesting. Can you possibly understand then, if that theory is right, the difference between England and Scotland without appealing to a **political equilibrium** having to do with the driving force of war affecting the incentives of the sovereign to treat two different groups within a single island so differently? I think it's quite interesting.
- 2.103** When we look at this—I'm now just going to review some evidence for you about how we see that relationship being different. The Scottish banks are not doing any of the funding of the British Empire, not through these special exchange deals and not generally. They're mainly,

they're assets or just loans to enterprises in Scotland, many different enterprises. It's also extremely competitive. As I said, we have three different types of banks, the three specially chartered banks, but then all of the provincial banks that are allowed to enter and compete.

- 2.104** It's also extremely innovative. They develop pretty much in Scotland all of the major innovations of commercial banking by the middle of the 18th century. What are those innovations? The clearing of notes, on a centralized basis; paying interest on deposits; establishing and giving people lines of credit, not just making a loan upfront, but actually giving them the ability to draw on a loan. I could go on. Registering collateral interests so that you could be sure that when you lent against something, you could seize it if you needed to.
- 2.105** If you were going to describe, what are the foundational elements of what make commercial bank lending work, the place that you would point to as having innovated them, still today, if you were going to say, what's the list today? You would go back to Scotland, and all of them were in place by the middle of the 18th century. This propelled growth in Scotland in an unprecedented way. Now, of course, the English and the Welsh noticed this, and they weren't very happy about it. But somebody had to pay the price of the funding needs. Of course, the Napoleonic War really showed the advantages of having a **Bank of England** on your side.
- 2.106** Until 1797, everything was operating normal. But as the Crown's credit needs got worse as a result of the Napoleonic War, what did they do? They suspended **convertibility** of the Bank of England's notes into specie, and they did this over the objections of the **Bank of England**. The **Bank of England** didn't want to suspend **convertibility**. Why did the **Bank of England** suspend **convertibility**? Because the Privy Council told them to. Why did the Privy Council tell them to? Because the Privy Council wanted to remove the discipline, the limit, on what the **Bank of England** could lend it.
- 2.107** It wanted the **Bank of England** to be able to take a lot of new risk by lending it a lot of money. It saw that the only way to do that was to let the **Bank of England** notes fall below their par value in specie. In other words, they basically temporarily trashed the equity of the **Bank of England** by making the **bank of England** lend excessively to them. They realized that, as part of the deal, they'd have to suspend **convertibility** of the Bank of England's notes. But that's how you know that, in extremis, the **Bank of England** was there for the Crown. Because when the Crown needed the **Bank of England**, it basically took what it needed. They don't go back to converting until 1821.
- 2.108** Now, the Scots are not involved in that. They have to go off into a paper regime, too, but they're not converting the Bank of England's currency. They're also in a suspension, but they're not involved in lending. Even during this period of the Napoleonic War, they're not involved in lending to the sovereign. We really see, during this period from 1797 on, this dramatic difference between the two. Of course, as the **Bank of England** is expanding its note issues in order to buy the loans and the debt of the sovereign, during this period, prices are going up.

- 2.109 So wholesale prices are rising from 1797, you see from about 110 all the way up to 164. It's not until 1821 that they restore **convertibility**. And you can see the reason for that is because it's not until then that prices are coming down toward where they were before. What's driving the prices down? What's allowing the **Bank of England** to be able to confidently reestablish **convertibility**? The change in government deficits. The point of this table, and I won't spend a lot of time on it, is just to show you, as the government's deficits, which are the far left column, are rising to their height, that suspension is maintained. But by 1821, government deficits are contained.
- 2.110 **Bank of England** notes are being soaked up by the **Bank of England**, and the **Bank of England** is able to reestablish a discipline balance sheet. We didn't have a sovereign default because, after all, Britain won the war. But if Britain hadn't won the war, the **Bank of England** would have been in trouble along with the sovereign, and that would have been solved simply with an inflation, a permanent inflation. As it worked out, it wasn't a permanent inflation. After victory happens, prices came down, and the **Bank of England** was able to reestablish **convertibility** and open.
- 2.111 But during the war, the same risk of default that the sovereign was bearing and all the debt holders were bearing were being borne even more by the **Bank of England** and its shareholders because they had to lend to the sovereign in extremis. After 1821, guess what happens? All the people in England and Wales have been complaining all along about, why should it be? Why should the Scots be able to have all the fun coming from a competitive and ample supply of credit? And so what you see happening is, after 1821, finally, all the protests—and you have to keep in mind, people are protesting the monopoly of the **Bank of England** in England and Wales a lot.
- 2.112 They're specifically pointing to the contrast of the Scottish system as a reason to allow competitive banks to be chartered. But the sovereign doesn't listen to any of that. But then in 1826, once the suspension is over and once the war with France has been resolved, then what happens? They change the law in England and Wales, too, because they no longer had that same driver of the conflict with France, driving the policies that made Scotland different from England and Wales. It's a very interesting history. But again, we can't understand any of this as some economic discussion in isolation.
- 2.113 Now, this table is really just meant to illustrate the variety of the sizes of the banks in Scotland. The point is, in addition to the three chartered banks that are at the top of this page, which are the largest in terms of their numbers of branches, there are lots of other banks, the provincial banks, that are joint stock banks coming into existence without any royal charter that are quite large and that have, in the case, for example, of the Commercial Bank Company of Scotland, 31 branches, which is more branches than even any of the chartered banks.
- 2.114 To give you a sense, and this table makes the same point in a different way, and these, of course, as that final column on the right shows, these banks are also fully empowered to issue notes. If we looked at number of branches per bank, what we would see is, of course, it's large banks that are able to branch in England. If you're looking at the two different kinds of banks in

Scotland, so this is public banks, that is, the three specially chartered royal banks and the others, you'd see that it's true that there's more of a mix of size in the non-three specially chartered banks. They tend to be smaller.

- 2.115 But as I pointed out, there's nothing that's restricting them from being larger. In fact, a few of them are just as large. I think I'll skip over this discussion of all of the differences of the clearing operations and the way that the banks innovated, except just to point to this last point that I've made before, which is, all of the innovations that are going on, interest-bearing deposits, bank note issues, and clearings through central clearinghouse, registrations of collateral, I could go on and on, lines of credit being given, are all basically Scottish innovations.
- 2.116 Because the Scottish banking system is so much deeper and deeper with larger banks, they are more innovative, more competitive, they're also much more stable. We can look at some statistics to see the bank density difference. This is showing you that, throughout the period, Scottish banks, there are more bank offices per inhabitant in Scotland. But then when we look at crises, we also see that banking crises, the number of English bank failures per 1,000 banks over the whole period from 1809 to 1830, is 18 in contrast to Scotland, which is four. In other words, English banks were four and a half times as unstable as their Scottish counterparts during this period.
- 2.117 My point here is that when we're thinking about all of this evolution that's happening in the 17th and 18th centuries in Britain and France, if we were trying to approach the question of why some places in some times have more scarcity of credit, more abundance of credit, more instability financially or less instability, more instability of banks, greater failure likelihood of banks, do you think that we could possibly discuss this as anything other than a reflection of evolving sovereign risks between Britain and France, which were themselves the result of their war?
- 2.118 And so, what lesson do we think we learn from this? What we learn is, if you're fighting for your life as a sovereign, you do actually pretty innovative and interesting public finances, innovations, and you take a lot of risks. And the fact that we're doing that knowingly and willingly is especially illustrated by the contrast between England and Wales, on the one hand, and Scotland, where it's the same sovereign who's in charge of the regulation of both regions and is regulating them very differently as a direct consequence of what's the right strategy to wage the war, not what's the right strategy to develop the banking systems.
- 2.119 This illustrates what I call Calomiris' law of banking. By the way, I don't have more than one. I only have one law of banking, and that is, there's a chance that a country can establish banks that serve the private interests of credit once the country has solved the public risk of default. In other words, you cannot, you won't find in any country, an example of a sovereign who's in a very risky situation, deciding to charter banks and provide them with sovereign protection for the purpose of providing private credit, even though the sovereign is on the brink of disaster.

- 2.120 No, the sovereign is going to use its power, first and foremost, for itself. It has to. That's a survival thing. Once that has been resolved, just as we saw in the English and Welsh case in the 1820s, then it's not that the sovereign has anything against private credit markets developing. It's just, that comes after the sovereign has won the war, not before. When we look at a lot of countries today, that's a really important law to understand. It's not the only reason that private credit is lacking. Sometimes it's lacking for other reasons.
- 2.121 But it's certainly a very important reason, that if the government is impunious, it's not going to allow savings to be, resources to be assembled and deployed that aren't helping it. It's going to find a mechanism that ensures its survival first and foremost. That's what banking policy and financial regulation in the 18th century was almost entirely about. OK, we have time for some questions.
- 2.122 I guess with this last point that you mentioned, it's an open secret that the sovereign will, at the end of the day, look out for himself. If that's the case, then the privates and private investors, they act as if that were not the case, as if the sovereign were actually looking out for their best interest. If they were to act as if the sovereign were looking out for itself, the economy would probably reach a standstill. I guess my question is, with the evolution that we've seen at least so far, how has the investors' perspectives and actions evolved? Or why haven't they understood that the sovereign will, at the end of the day, look out for itself? I don't know if that makes sense.
- 2.123 I'm not sure that, I certainly didn't say that investors mistakenly believe that sovereigns will look out for private interest before sovereign interest. I didn't say that, and I'm not sure I think that's right. I think people understand the equilibrium that they're in. I don't think it's the case that people are misled. I think it was clear that, for example, in England, it was clear that the Bank of England had a monopoly, not because the sovereign was looking out for private interests, but because it was looking out for public interests. Of course, it was interested in certain private interests who were allied with the sovereign and who were given specific marching orders to have their mission serving the sovereign interest.
- 2.124 But I don't think there are a lot of examples where people are fooled. We're going to talk about Mexico, for example, in the 1980s and 1990s, a little bit later in this class. In the case of the chartering of the South Sea Company and the Bank of England, we talk about Tories and Whigs, respectively. When we talk about the banks in Mexico, often the phrase that's used is the six families. The political role of the six families, and each having control of a bank, and an industrial conglomerate related to the bank, and then certain political expectations about the two-way street of the access to the charter.
- 2.125 So insider games in developing countries where they're also subject to a lot of default risk in today's world are quite similar. I think it would be a mistake to say that people don't understand, let's say, in the 1980s in Mexico, that the six families have a privileged position and that there isn't such a thing as arm's length lending in the private sector. There aren't any banks that are created with the mission of arm's length lending. Not because the government was

against arm's length lending, but because the government was for its own survival. And so all of its banking policies with the **six families** were geared toward the government's needs.

- 2.126** What you'll see, for example, that we'll be pointing to is, when the government gets into trouble after the oil price collapse, so there's a huge expansion of investment in Mexico after the discovery of oil. But then in 1982, oil prices collapse, and you see lots of commitments to expenditure, followed by a big collapse in revenues. So what happens? Predictably, the government starts taxing the banks, starts taxing its friends who are running the banks, and does that with reserve requirements. Basically, forcing the banks to lend at zero interest to the government.
- 2.127** Then ultimately, it nationalizes the banks because they couldn't get enough from just forcing the banks to lend them more in this extreme period. And so they just nationalized the banks. And so that leads to the **Calomiris definition** in an autocratic society of what the word friend means. And friend is the person, is the last person you stab in the back. Your friend is not someone you don't stab in the back. It's just the last person you do. So you first try to get your money from everyone else. Then when you can't, you try to tax your friend, and then ultimately, you stab.
- 2.128** So I think generally, whether it's the ancient examples we're talking about or the current ones, I think the world understands what the game is. The important thing to come to grips with is the connection between **ex ante** thinking and **ex post** outcomes. So for example, if you're part of this network, whether it's the **PRI**, **six families** network in Mexico of 1980, or the **Porfiriato** in Mexico in 1900, whichever or the Korean **chaebol** network we're going to talk about, in 1990, or we could go on and on. The Indonesian network of 1990, or—
- 2.129** These centralized networks of private-public partnerships create rent-sharing arrangements where the private participants understand that they must bear certain risks, including the risk of being the last one who gets stabbed in the back, in exchange for having access to these **rents**, these privileges. That same arrangement that we've been talking about in the **Mississippi Company** or in the **Bank of England** or the **South Sea Company** that's part of this early modern world, that same set of arrangements lives on in all of these network autocratic arrangements in **emerging markets**.
- 2.130** I know from having taught **emerging markets** now for 30-some years, the markets are very, very aware of this. For example, there's a hedge fund. I won't name it. This entire business strategy is to get members of the autocratic crony networks from each country as investors into their international hedge fund. It's like a **diversification** strategy. You'll be one of the **six families** of Mexico, one of the 300 families of China, et cetera. If we have them all in our fund, they will help us manage our risks in all these countries.
- 2.131** They'll know when something bad is happening for us to avoid. They'll help us manage bad things if they do happen. They'll create good opportunities for us where we're protected because we're connected to them. It's that exact same thinking that's visible in these relationships between the elite and the government. And by the way, that was our story for

Rome, too, right? We have to, when I say the whole point of **financial crisis**, what do I say the motto of this class is? **Know yourself**. So Scotland isn't England, France isn't either of them, Mexico isn't Chile.

- 2.132** When you think about what **financial crisis** risk means, it's part of what I'm calling here an **adaptive** outcome strategy. It's an outcome that's part of an **adaptive** strategy of whatever the **political equilibrium** is in each of these countries. When I say **adaptive**, I don't mean it's ideal in some economic sense. I mean, maybe it's the best you can do, and it's what the society rewards, partly because it's the best they're able to do at that time.
- 2.133** So I actually think people are—I'm sorry it was such a long answer to your question, but I think people are quite aware of this. As I said, there's a whole hedge fund organized around the awareness that the sovereign discriminates in its regulatory policies and other policies to create **rents** for its allies. That's what autocratic **emerging markets** are all about.

3. The US and Canada

AT A GLANCE

Compares the crisis-prone United States with crisis-free Canada. Reviews the standard theory of bank fragility (demandable debt, first-come first-served, maturity mismatch) and argues it cannot explain why some countries never have crises, since banks can manage risk with cash and equity and governments choose banking structures. Identifies unit banking as the key US choice, which prevents diversification ex ante and coordination ex post, alongside greater tolerance for real estate risk. Shows that everyone knew this, from the National Monetary Commission to bank balance sheets, and traces the persistence of unit banking to the agrarian populist coalition in the game of bank bargains.

- 3.1 Welcome back. We're going to continue now our discussion by looking at another case of two countries that are right next to each other. Before, we talked about Scotland and its differences from England and Wales in the 18th and early 19th centuries. Now we're going to talk about the US and Canada, which are also very different. Let me remind you, in the case of Scotland and England, Scotland was allowed throughout the 18th century to develop, based on a free entry model, a very financially deep and stable and innovative banking system.
- 3.2 It had all three of those attributes. It was able to supply more credit than the English system could. It supplied a much more innovative set of institutions. They invented most of the important things that are necessary in commercial banking. And it also was more stable in two different respects, actually, I should have emphasized. One is the one I did emphasize, which is the failure rate of banks was much lower in Scotland than it was in England and Wales.
- 3.3 But the other thing to mention is all of the crises that we did see had their origin in England, not in Scotland. Scotland was a victim of shocks that were coming at it from England, and it managed them much better than the English banks managed those shocks. There's a sense in which this is often surprising to people to hear two countries, especially under the same sovereign, on the same island, with such different experiences in terms of the ability to have a supply of credit and a stability of the supply.
- 3.4 That also illustrates something that people often find surprising, which is it's possible to have more credit and more stable credit. I think sometimes in our heads, we may think more credit means unstable. That's not true. It can be true, but often the countries that have greater supply of credit often have more stable credit. If you design a system well, you have more innovation, more stability, and more credit.
- 3.5 Now, today, we're going to now turn from the English-Scottish comparison to the US-Canadian comparison. We're going to keep our focus on banking system stability. One thing that's worth saying, as a finance professor, I have constructed and published, and I teach courses just about the theory of bank instability. One of the things that I like to make fun of when I teach those

finance courses is that often the models that the finance professors are teaching make it seem like things are the same across countries. So the models are very general.

- 3.6 What's the standard view of where a **banking crisis** comes from? Well, there's a shock that happens, which tends to harm the value of bank assets. If there's, I can tell you in all of world history, if there isn't a shock that's a negative information about the value of bank assets, there's never a **banking crisis**. So of course, there's a shock somewhere. It doesn't come from our imagination, although some models of **banking crises** have imagined that **banking crises** can come from something, even when there's no shock. But that has no validity in history.
- 3.7 But then when they talk about, well, what else do we need? They often focus on the inherent fragility of banks. Well, if a railroad company got hit with a negative shock to the value of its assets, it's not going to have a crisis. It's just going to have, its stock price goes down a little bit, maybe even its bonds that it's issued fall in value a little bit, and that's it. There's no crisis. It's just bad shock.
- 3.8 But with banks, it's a little different because their debts are short-term, in fact, typically demandable. So what that means is that the depositors of a bank have the option to come get their money. And in fact, we have something in banking, it's a little different from all the other sectors, which is called the **first-come, first-served** rule. Let me explain why that is different. If you had a debt from a normal company, not a bank, and you came and got your debt back from the company that owed you the money, and then, let's say, two months later, the company declared bankruptcy, that would create the need for you to put the money back in the company.
- 3.9 That's called the **preference period**. There can't be preferential treatment of creditors just prior to a bankruptcy. If you got paid off, if they just decided to pay your debt off as a decision and then immediately declare bankruptcy, you might have to put your money back into the company. But in banking, it's never been that way. And in fact, the bankruptcy laws for banks have always been different for other companies. So when you think there's a problem at your bank, I'm going to show you tomorrow a painting about this. If you think there's a problem and you run to your bank and you get your deposits out, and then an hour later, the bank fails, you get to keep the deposits.
- 3.10 That is a choice. That's a legal institutional choice in banking that makes it different from others. You can understand that legal choice, you can justify it, you can rationalize it as being something that gives power to people to pay attention to what's going on in a bank so they get rewarded if they see a problem developing. That's the way I have argued theoretically to make sense of this legal difference.
- 3.11 But whatever the cause of the legal difference, the point I'm making right now is that there's an inherent fragility of banks that has to do with what's called **asset liability mismatch**, sometimes **liquidity transformation**, sometimes it's called **maturity transformation**. But I want to emphasize this mismatch. The assets are loans. They tend to have some kind of **duration**. Maybe some of

them are three months, maybe some of them are a year, maybe two years, whatever. They're not maturing every moment. They have some maturity.

- 3.12** They're also very illiquid assets in the sense that they're customized. They're not homogeneous assets that are traded in markets, typically. It would be very hard to sell them if you needed to. It's a little bit puzzling, many people have pointed out. Let's see, banks are willing to give everyone the option to come get their money back with no notice, and the assets they have are hard to sell, not so easy, and don't mature at the same rate, not every day, right? So the problem is, what do you do if a lot of people lose faith in you because of this big shock that's happened, and come in and ask for their money back?
- 3.13** And then you have to try to sell your loans. You can't do it. So you have some cash, but you don't have enough cash to pay off all your deposits. That's what's called **fractional reserve banking**. You're not just holding cash on your asset side. You're holding cash and loans. You lent out some of the money. And so the problem many theorists have noted is that, because banks structure themselves this way, they're inherently fragile relative to, let's say, railroads.
- 3.14** Now, I don't dispute that. In fact, I've spent a lot of time writing down articles to try to explain this. But the thing that I am pointing to is, first, that's not a very good model of bank failure. And the reason is because all the countries in the world have that same structure of banks. But not all the countries in the world have **banking crises**. The point of today, the remainder of today is, we're going to look at a country that never had a **banking crisis**, called Canada. But it had banks that were structured in exactly that same way.
- 3.15** The first thing to note is, don't always believe what financial economic models tell you, that banks are inherently fragile because of their structure. So that when they get a shock, that structure leads to some sort of magnification problem that leads the bank to fail. I'm not saying that never happens. I agree with the basic point. But what I'm saying is just as interesting as to ask the question, why does it not happen in some places and some times? Pretty interesting, I think. What it points us to is a few things.
- 3.16** One lesson it points us to is that it's possible for banks to structure their own portfolios and their own management so that this inherent fragility is not such a big problem for them. That's interesting, right? So it's not, as the model tells us, it's all very mechanical. Shock hits, we have the maturity mismatch, **banking crisis** happens. Nonsense. Banks can manage their risks so that they're fairly robust. How do they manage their risks? Well, they maintain an adequacy of equity capital in their funding side and an adequacy of cash holdings on their asset side.
- 3.17** On their funding side, what that means is, they limit their leverage so that instead of having 99% debt, they might have 90% debt or 80% debt. That's called risk management because the debt holders know that if you have a lot of equity, the more equity you have, the less likely any loss on the asset side is to translate into a **default risk** on the liability side. Banks can manage the problem. They manage it with holding more equity and also holding more cash.

- 3.18** If I had, let's say, 99% cash on my asset side, then even when bad news happened in the economy that made my loans less valuable, the depositors wouldn't care because I've got 99% of my assets are in cash. So why worry? What this is telling us is, these two very basic risk management tools of cash and capital are something that banks can use. The first answer to the question, "Why is it that a shock doesn't always make a bank fail?", is that banks can manage their risk voluntarily. They can choose to hold more capital and more cash.
- 3.19** The second answer is that banking systems can be regulated in different ways so that some of them are more stable inherently than others. One of the answers is private decisions banks make, and the other is public decisions that governments make. Those two things, put together, help us understand why banking systems sometimes are more fragile. That is, for a given shock, more likely to have failures than others. So for example, when we talked about Scotland and England, well, in Scotland, we forced banks to be small. We also didn't allow them, we limited the kinds of liabilities they could issue.
- 3.20** By forcing them to be small, we meant that they weren't very diversified. Small banks can't inherently diversify as well as large banks. They can lend across a wider range of loans, various sectors, various locations. So that turns out to be very important. And what we're going to see is that same fact comes out in the US, that in the US, we're choosing to have banks be smaller and less diversified. That's a very key part of why the banking system in the US is less stable.
- 3.21** Now, of course, that doesn't answer the deeper question—but then why would governments ever choose to structure banking systems to make them more fragile? Now, we saw the answer in England versus Scotland. It's because the government partnership with the **Bank of England** that served its financial interests to fund itself inherently led it to limit entry into banking of other chartered banks in England and Wales. And their way of limiting entry was to say, you could only have very small banks if it wasn't the **Bank of England**. It was a political decision. It makes sense, right?
- 3.22** And so we're going to find that same puzzle arises in our discussion of the US and Canada, that we're going to see that there's an absolutely clear difference between the two countries' experiences with respect to crises. The US is crisis-prone, accident-prone. If you've ever known someone who always seems to get into an accident, it's like the US has this problem. And in Canada, accident-proof. The two countries are right next to each other. They have very similar histories. Their geographies are a little different, one's north and the other south. We'll talk about that. But very similar.
- 3.23** So it's very puzzling, in a sense, to see two such similar countries, and their GDP paths are quite similar because they're so linked economically. But their banking systems have had completely different experience, not just for a decade, not just for two decades, but for 200 years. Well, that's pretty strange, right? It gets us, again, when we are thinking about learning, we start from this puzzle. Well, if I'm right next to someone, suppose my next door neighbor puts a really good fertilizer on his lawn, I notice after a couple of years, my lawn stinks, his lawn looks green. I'm going to go find out from him, what fertilizer did you use?

- 3.24 So there should be learning. So you'd think, what about US and Canada? They're right next to each other. OK, we understand that learning was prevented by the Calomiris' law of banking in Scotland and England? But why was learning prevented in the case of banks in the US and Canada? It's quite strange, and it's not so obvious. We will talk about it. I will tell you that people were very aware of the differences. There were people who were trying to learn from them, but they weren't able to. They weren't able, practically, to execute the changes that were necessary.
- 3.25 It was clear what the changes were. I knew which fertilizer I wanted to put on my lawn exactly, and I wasn't allowed to do it. This is a great example for us of how we think about this puzzle of persistence of crises. I want to emphasize that this comparison between the US and Canada, and between Scotland and England, actually, there's a much broader set of comparisons we could make. These are not the only comparisons. So as I mentioned earlier today, the period we're living in right now, since 1980, is the most unstable period for financial systems and for banking systems in particular in all of world history.
- 3.26 Another period that's quite interesting, 1874 to 1913, what's interesting is, there were so few crises. In the US, we had a lot of them, but even the ones that we had, which we're going to talk about starting tomorrow morning, this period, 1873 to 1907, the crises that happened then in the US, they were all pretty mild. So yes, we had a lot of crises. It's puzzling. Why did we have, of the 10 that happened throughout the world, we had half of them in the US. Half of the worldwide crises from 1873 to 1913 happened in the US. But interestingly, they're all very mild.
- 3.27 So that period was one in which crises tended to be, on average, much milder and also much less frequent compared to today. So, again, we're struck by differences. It's not that the period 1873 to 1913 was one of unusual economic stability. It wasn't. It wasn't that there was just a lack of shocks. There were plenty of shocks. It was just that banking crises, severe ones, didn't happen very much. The two countries that had what I would call the only severe banking crises of that period, each one, Australia, 1893, Argentina, 1890.
- 3.28 So from all of 1874 to 1913, two crises that are of similar severity to the typical one that we've experienced from 1980 to the present, there were only two countries that had crises that were similarly severe to the ones that were seen in the current era. How many have we seen in the current era? A hundred. That's pretty interesting, right? These comparisons, what are they telling us? Banking crises cannot be understood simply by looking at the structure and mechanism of what a bank is.
- 3.29 Because since Scotland, by 1750, had pretty much invented all of the elements of what a modern commercial bank should look like, all modern commercial banks, pretty much, are doing what Scotland's banks invented. They're paying interest on deposits. They're trying to register collateral interest if they can. They're using clearing houses to clear their transactions. They're issuing lines of credit. They're all doing and they're all structured with this maturity transformation, liquidity transformation mismatch.

- 3.30 We've got some explaining to do. Why is it? We do some cross-country comparisons, England versus Scotland. Completely different histories for over a century. Canada and the US, two centuries, completely different histories. Some periods worldwide, very low failure, low crisis experience. Some periods worldwide, very high. The reason I'm emphasizing this theme is that it's going to help us understand what the causes of this must be. It has to be something particular to the country time period. It has to be something about the structure of the country.
- 3.31 But the fact that we're experiencing more of this over time is also puzzling because it's telling us that this can't just be something we can mechanically unlearn. Because apparently, our learning is going in the other direction. Notice that this is inherently also inconsistent with that Minsky-Kindleberger thinking. It isn't that crises are happening constantly as part of business cycles. It's quite the opposite. Some countries never have them. The US GDP cycle looks a lot like the Canadian GDP cycle throughout this period. But one has crises, the other doesn't. Go and explain that, please, Hyman Minsky.
- 3.32 Or how come one era that has lots of economic instability, 1873 to 1913, has few and tend to be fairly minor crises, and another period has much more? This is pointing us in directions that we've been talking about from the beginning, about what crisis risk is stapled to other things. In some places, maybe it's adaptive, not that it's economically desirable, but somehow we're actually consciously choosing to live in a situation where there's more risk of this.
- 3.33 That theme is going to be borne out quite a bit in the next hour, what we're going to talk about, about the choices that were made in the US and Canada. But consistent with our strategy of doing deep dives, we're going to go into those details. We're going to watch the two countries evolving together over time and look at the choices that they're making, so that we're not speaking in generalities, but we're going to look very closely at why things turned out so different.
- 3.34 I want to emphasize, I mentioned that the GDP paths of the two countries are quite similar. If anything, the Canadian GDP path is more volatile because Canada, historically, has been a more primary commodity-exporting country than the US. I'm not saying the US wasn't as well. I'm just saying that if anything, Canada has been more that way than the US. And yet, our basic facts are, the US has had 17 major banking crises over these 200 years, and Canada has had zero. So that's a pretty striking comparison.
- 3.35 And of course, you might say, well, could this be the result of different regulatory choices? The answer is, obviously, it's a choice of different regulations between the two countries. And that's the same thing that we said when we looked at Scotland versus England. These were regulatory choices that made the difference happen. But that's not an explanation. The explanation has to be why those regulatory choices were made and why they persisted for 200 years, or not quite for 200 years, but for more than 150 years.
- 3.36 I also want to emphasize that the US's experience of being crisis-prone isn't just in the distant history. It's also in recent history. That fact I mentioned, that we're living through this period

since 1970, in which the frequency and severity of **banking crises** is the greatest in all of world history, that's true. And the US didn't get out of the crisis business during that period. In fact, we had two. According to all the definitions of these events, the US had two.

- 3.37** So let's notice, if we put the US in perspective, just in the recent era, we would see that 18% of countries in the world from 1970 to 2013 had two or more major **banking crises**, 18%, that a little more than half of the countries had only one, and almost a third of the countries had zero. In the zero category here, we would put Australia, New Zealand, Canada, of course, we know that, Hong Kong, Malta, I could go on. A few, not that many, but some. You would look at those and say, OK, well, those sound like, politically—this was Hong Kong before mainland China took it over, right? That's going to be a new political situation.
- 3.38** But during this period, Hong Kong, Singapore, Malta, and I could go on, are all part of that zero crisis pool. Developed, fairly mature economies with sophisticated legal and financial systems that are operating in what you might call a very predictable and a financial system geared toward success for its own sake. Then in the one-crisis camp, we have a bunch of countries. Then in the two-crisis camp, who are the countries? Well, Democratic Republic of Congo, Argentina, Chad, and the United States of America. There are a few more, but they all sound more like Chad, Democratic Republic of Congo, and Argentina than the United States.
- 3.39** The US is the only country, sort of mature, developed economy, even in the recent period, is in the red zone. No matter how you think about the US, historically, more recent era, cross-country comparison, a match comparison with its next-door neighbor, or comparison with all countries in the world, including less developed countries, the US stands out as a fairly unusual case. So have I convinced you yet that there's something to be explained here that's not just mechanical economic modeling, that has to explain this peculiarity of the United States? I hope so.
- 3.40** And there are different pieces to this. The most important piece to it, I'm going to argue, is structure. It's going to be what we're going to describe as the choice of **unit banking** in the US. Through most of US history, and especially until the 1980s, most banks in the US were constrained to operate one office. That's what **unit banking** means. So if you're in, let's say, Champaign-Urbana, Illinois, in 1970, you, as a bank, are allowed to have one office, your headquarters. All your loans, you have to make out of that office. All your depositors come there. You only have one location.
- 3.41** By the way, if you've ever been to Champaign-Urbana, you know that there are two things going on there, other than the university, and they are soybeans and corn. So as a bank, the entire livelihood of everyone that you're lending to, because banks are inherently not able to lend to people they don't know from across the other side of the world or even the other side of Illinois. So lending tends to be very local because information is a crucial part of lending. You have to be able to gather information, control the behavior of your lender, of your borrower, meet with your borrower.

- 3.42 And so, yes, it's true that the environment of Champaign, Illinois, even if you weren't a corn or soybean farmer, if you're operating a market, who are your customers? The people who work in soybean or corn. So every loan you make is heavily dependent on the price of soybean and corn, right? If the prices of soybean and corn fall, land values are going to fall. And when land values fall and when the price and the incomes fall, some of your loans aren't going to be good.
- 3.43 You might say, "Well, this is a very primitive idea, professor. Haven't you been to school? Don't you know better? That's not a good way to structure a bank. You should allow the bank to be diversified so that it doesn't just lend to people who are dependent on the price of soybeans and corn for their risk. But how about also different crops and different kinds of industries and different locations?" The answer is, well, that's what **unit banking** doesn't permit.
- 3.44 There's another consequence. If you have, let's say, 10,000 or 20,000 banks in your country, and they're all lending in different markets, that means that it's also very hard for them to coordinate their behavior. Suppose that a shock happens, and we have a need to coordinate our response to it. I have some extra cash. You need, you have something I need. We work cooperatively. Banking systems did that quite a bit in response to big shocks.
- 3.45 Now, if you're in Canada, throughout this period we're talking about, the number of banks that have to get in the room to make some cooperative arrangements to help each other through a crisis is, in today's world, it's five. Historically, maybe it was a dozen. Could have been as many as 20. But that's not a very large number. They're all operating through the whole country. They know each other very well. Their operations they know are all diversified, and they're able to sit down in one room and reach agreement about how to manage things to mutual benefit.
- 3.46 So they're **ex ante**, they're more diversified, and **ex post**, they're able to coordinate. And for example, in the case of when shocks would hit in Canada, often the **Bank of Montreal**, just a private bank, not the central bank, would coordinate the other banks so they would decide, oh, you know, Bank X looks like it's going to fail. We have to decide, when Bank X fails, will the depositors in all of our banks, how will they react to that? Will they react to that and say, "Oh, I'm worried that there could be connections between Bank X and some other bank that might bring those other banks down"?
- 3.47 Or maybe not. Maybe we believe that they won't react that way. What makes us think one way or the other? Because we know that it's generally known what the connections are among the banks. And so the market will sort that information. And sometimes people will think, oh, because Bank X failed, Bank Y is more at risk. But sometimes they won't think that because there's no reason to. So in the history of the Canadian private bank operations that I was just describing, in two cases, they decided, because they were worried about the knock-on effects, they decided, before the bank failed, to just acquire it as a group, which basically shared the risk so that there wouldn't be any knock-on effect.
- 3.48 That's called managing crisis risk, in two cases. But in other cases, they just let the bank fail. No problem. Isn't that interesting? I find that extremely interesting. But that thinking isn't possible if

you have 20,000 banks each operating in a geographically isolated way from each other. How could you possibly coordinate your behavior? How could you even gather information about the other banks? You probably don't even know the name of all 20,000 banks, much less what their operations look like.

- 3.49 But if there are only 10 or six or eight or even 20, and they're all operating on top of each other throughout the whole country, you have information, you have relationships, you have ability to meet and coordinate. So I want to emphasize, these are two different aspects of the unit banking choice. It makes diversification *ex ante* much harder, and it makes coordination *ex post* almost impossible. That is a really important difference—I would say the key thing between the US and Canada. I've mentioned the banking structure choice.
- 3.50 The second point, of course, is different real estate risk. We're going to come back to this and talk about, especially in the current period, when we look at the US crisis of 2008, we're going to see that the US has traditionally, but especially in recent years, been much more willing to tolerate major risks in real estate, which, in fact, we are subsidizing. We've already talked about why that might be. Obviously, it serves a political purpose. But that's also been a historical difference between the US and Canada, banking systems.
- 3.51 I also want to mention that you might think, well, wait a minute. Maybe it's that the Federal Reserve, as a central bank, operates differently from the Bank of Canada as a lender of last resort. Here's the interesting thing. The Federal Reserve was created in 1913. The Bank of Canada was created in 1935. Almost all of the period of Canadian banking stability is a period in which there was no central bank in Canada. It's not about central banks operating wisely to manage risk in the banking system. Through most of Canadian history, there was no central bank. If anything, the US should have shown more stability if it were all about central banking.
- 3.52 Here's one of my favorite graphs. The score is, since 1840, the score is still Canada, zero, US, in this case, 12. If it was since 1790, it would be Canada, zero, the US, 17. But the point is, this is a big difference. If you look at the total amount of bank credit, because the GDPs are tracking each other so closely because they're adjacent economies, it's remarkably similar. Those blue and red lines look like they're right on top of each other for most of the history. So the total amount of credit that's being provided relative to GDP up until the 1990s looks almost identical.
- 3.53 And then after the 1990s, what happens? The US actually seems to fall below Canada. So now we have the same example that we talked about in the case of Scotland versus England. Canada is able to provide more credit through its banking system with greater stability. I could also talk about innovation. There was a World Bank study that showed in the 1990s, it came out and showed that Canadian banks were much more innovative, even. Now, a lot of people have argued against that, but I'm just saying that at least there's an argument to be made that the Canadian banks are even more innovative. So this Scottish-English comparison seems to work very well for the US and Canada.

- 3.54 The other thing that's interesting to notice is, not only are the GDPs tracking each other, but a lot of other economic outcomes are quite similar. So for example, here we're looking at inflation choices. So this is looking at price index from 1952 to 1980. There was a big inflationary experience in the '60s and '70s. I'm just showing you here the rate of growth of prices from 1952 to 1980 in Canada and the US were quite similar. So **monetary policy**, which was the problem, the cause of this inflation, was a similar experience in both countries.
- 3.55 This reflects the fact that Canada is our neighboring economy. They're tracking our exchange rate policies quite similarly, and so their inflation policies are also quite similar. Canada and the US aren't different in every respect, and we'll talk more about their histories because you'll see they're quite similar in some respects. I mentioned before that that tolerance for real estate risk is much higher in the US, and you can see it here. This is the percentage of mortgages in arrears. You can just see in Canada, mortgages just don't go into arrears very much.
- 3.56 And during the recent 2008 crisis, Canada still looks like the cardiogram of Iraq, and the US looks like something else, right? So you might say, oh, I know. It's just those Canadians are so boring. They just don't like risk. Trust me, that is not true. I'm sure that it's not just because Canadians are boring, and I don't think they are boring. In fact, did you know that the French Canadians tried to start a revolution in the 1830s? Yeah. It's not as boring as you think. And by the way, there was almost an attempt at an Irish invasion of Canada from the US in, I think, 1865. There's a lot going on that you might not know. It's not all so boring, right?
- 3.57 But it's boring from this calculation. Here's another interesting thing. Despite all this subsidizing of real estate risk in the US and all these real estate crises, as of 2010, the percentage of people who owned their own home were identical in the two countries. This is generally true. As I explained earlier today, it's because **mortgage risk** subsidies tend to be impounded into the price of houses. So that makes it, when you subsidize mortgages, yes, people become more mortgage-indebted, but that doesn't mean that more people can get homeownership. It just means that house prices become more expensive.
- 3.58 It's very hard to explain with economics why the US choice across all the things we've talked about—**unit banking**, historically, **mortgage risk** subsidization in the 20th and 21st centuries—why does any of this make sense? The answer is, it doesn't have to make sense economically, but it does have to make sense politically. It can persist if it makes sense politically, because that's how societies are responding to the events around them. They're making political choices.
- 3.59 One of the things I want to point to here is that much of Canada is up there where the polar bears live. Most of the economic value created in Canada is within 100 miles of the US border. In fact, if you're thinking about Canada initially, around the Great Lakes region in particular, most of what's going on in the 18th and 19th centuries in the development of Canada have to do with the sort of horizontal line that's along the border with the US, and especially relating to the Great Lakes.

- 3.60 What I want you to think about there is that, even though there's commonality in our location, the orientation of the two countries is orthogonal. That is, the US is starting off with a vertical orientation that has to do with the access to the Atlantic, and Canada is starting off with a horizontal orientation that has to do with the Great Lakes and not being in the North Pole. That is not a trivial fact. There's more to it than that that explains fundamentally where these countries are different. But I'll hold you in suspense a little longer.
- 3.61 Another thing that's puzzling is, in the US, we have always known that there are differences between the US and Canada, and we've always regretted that we're such a bunch of losers when it comes to **banking crises**. For example, we established something in 1910 that gave its report. I'm sorry, it was established earlier, called the **National Monetary Commission**. Now, when I was a graduate student, part of the process of becoming a scholar was to read all of the books that were commissioned by the US **National Monetary Commission**, which were all published in this enormous series in 1910.
- 3.62 There were three books about Canada. There were, I think, three or two books about Germany. There were three books about the UK. There was a book about Japan. There was a book about Mexico. There was a book about Italy. My point is, it's a very large collection. I own the collection. They were written by the leading lights on banking structure and regulation and instability in the world at the time. They were written to try to explain why some banking systems function better than others. They are still used as authoritative sources in the history of banking. They commissioned an amazing array of books that today's economists and historians are still learning from.
- 3.63 OK. And so the three books about Canada, do you think they mentioned the favorable features that I've already described? Because what was the point of the **National Monetary Commission**? It was commissioned after the **panic** of 1907 in the US to try to understand why was the US banking system so crisis-prone. So let's just look at those three books for a minute. OK, so those three books said all the things which I've been saying. Did the **National Monetary Commission** understand the importance of **unit banking** regulation in the US as a cause of bank instability? Of course they did. Everyone understood it.
- 3.64 When they made their recommendations to Congress, did they mention **unit banking**? No. They mentioned wanting to create a **Federal Reserve** system to create a **lender of last resort**. I'm not saying that was a bad idea to create that, but isn't it interesting that they didn't mention the key fact that was different about the US and that was causing all the trouble? Why do you think that is? Think about that just for a minute. Let me suggest an answer, because they knew that that was a political determinant that had been a precondition, which they could have no say in. It was decided by individual states and separately by Congress.
- 3.65 It was a very complicated decision process. Individual states had jurisdiction over state-chartered banks in their state, and then the national banking system had jurisdiction over national banks throughout the country. There was divided responsibility. It was all heavily politicized. We're going to talk about that heavy politicization. It would have been not only dead

on arrival, but it would have harmed their other recommendations. It would have harmed their other recommendations. If they had said, "We need to get rid of unit banking," it would definitely have been politically infeasible, but it also would have tainted them.

- 3.66 I'll show you some cartoons in a minute so you can see what I'm talking about. It would have made them into, from a populist political standpoint, it would have made them look like the lackeys of Wall Street. And of course, many of the people who were involved were sophisticated financial people who did have connections to Wall Street. So the National Monetary Commission doesn't mention the problem. It would be like you go to the doctor, you have cancer, and the doctor says, "I see you've got the beginning of a bunion on your foot." It's like, "Yes, doctor, but —and I'm glad you want to fix the bunion, but what about the cancer?" "We can't talk about that."
- 3.67 How crazy, right? But that's what we did. That's what we did. Don't think that people weren't aware of it. Not only was the National Monetary Commission aware of it, everyone in Congress was aware of it. That's just the way it works. Here's another indication that everyone was aware of the difference. Remember, I mentioned that banks individually manage the risks that they're aware of by holding either, if they're worried about a risk, they hold more cash or they issue less debt and therefore rely more on equity as a source of funding.
- 3.68 If we look on average at the banks in the US and Canada, in the US, their cash over assets ratio, 0.45, in Canada, 0.27. Lower. And what about their equity to assets ratio? In the US, 0.2, in Canada, 0.19. So when you look at those balance sheet ratios, what are they telling you? They're telling you that banks in Canada perceive that they need less cash and less capital, less equity capital, to deal with their risks. What does that tell you? It tells you that the banks in the US knew that they were facing more instability than the Canadian banks.
- 3.69 In fact, to really deal fully with that greater instability, they would have had to hold a lot more equity than the Canadian counterparts and a lot more cash. So the point is, this is true. The US bank crisis proclivity is true even though the US is holding, banks are holding more cash and more capital. And what does that tell you? They knew, because of lack of diversification ex ante and lack of coordination ex post, that they were much more crisis-prone, and they partly at least mitigated the problem by holding more cash and capital.
- 3.70 So not only did all the politicians in the National Monetary Commission know about this difference, but in the markets, the banks themselves knew about the difference, and of course, everyone knew. It's really useful in a system like this at the time to make these comparisons because they're telling us about the risk management choices of the different countries. And part of the advantage the Canadian banks had from the greater diversification was that they didn't have to do costly things. They could be levered up more, which made their funding cost cheaper. They could hold less cash, devote more to interest-earning assets.
- 3.71 Now, I mentioned the cartoons. So if you had been in the National Monetary Commission and you'd thought of raising the point that you wanted to get rid of unit banking, you had to keep in

mind, what was the political environment that had given rise to **unit banking** in the US? It says here, "Christ, who overturned the money changers' tables, evicts financial interest from Congress." So these are cartoons that are all about how the real problem is that Wall Street, big banks in Wall Street, by the way, big banks at this time? What were they? They had a few offices.

- 3.72** So there was a political opposition to creating a Canadian-style system in the US that went back a long way. We have to try to understand that. Here's another one. You can see this is Gulliver's travels. It's all about how Wall Street is holding the people down by manipulating the political system. I just want to emphasize the political opposition to **branch banking**, to preserving **unit banking**, doesn't necessarily line up with the economic logic of what's causing **banking crises**. There's a very popular view out there, especially in agricultural areas, about preserving **unit banking**.
- 3.73** In William Jennings Bryan's populist and Democratic Party, he was a nominee for both in 1896, running for President, preserving **unit banking** was one of the three key platforms of his campaign. When you had referendum in the state of Illinois in the 1920s to ask the question, "Should we relax **unit banking**?", the referendum lost. It was not popular in agricultural areas to have **branch banking**. That's very puzzling. How can it not be popular? Because in some sense, it should be extremely popular in rural areas, you might think, because they're the ones, because of undiversified banks, whose banks are failing. When the banks fail, they don't have a bank anymore.
- 3.74** They're the ones who have big contractions in credit when banks get into trouble. The answer, in theory, and I think there's evidence for it in research that I've done with **Carlos Ramirez**, is to think about this as a **credit insurance**, that, yes, it's true that in really bad states of the world, your bank disappears and you're without credit. But in moderate states of the world, where, let's say, you're a soybean farmer and the price of soy falls, you're really in trouble.
- 3.75** But maybe in a **branching** system, people might decide to give credit. If it were a nationwide system, they might decide not to give credit to you for, let's say, a moderate decline. Whereas if the bank is still alive and doing OK, it has no choice but to lend to you because you're the local bank. So location ties the bank to make a commitment, a **credit insurance** commitment, to have to lend. Not in extreme circumstances, but in most circumstances. And that, I think, lines up very well for understanding this.
- 3.76** That is, the US banking system, from about 1815 until about 1980, the political bargain that's winning, that's running the show, is dominated by agrarian interests that see **unit banking** favorably and that don't like the idea of **nationwide branching** or even statewide **branching**. It's not just **unit banking** that this group favors. Also, when things go south, they tend to advocate at the level of their states, debt moratoria, particularly for agricultural borrowers.
- 3.77** They tend to advocate **deposit insurance**, something we'll be talking about tomorrow, because those eight states that all had failed **deposit insurance** systems that we're going to talk about,

they were all driven by the same coalition that was pushing for them. What we're going to see is that, after 1980, politically, that coalition lost power, and we had a new rise of what we're going to call an urban populist group that became the winners in what Steve Haber and I have called the game of bank bargains.

- 3.78 So unit banking is there for a very long time, over 150 years, and then it gets thrown out because the political coalition that's able to dominate the choice shifts. We'll see a lot of the outcomes related to mortgage risk subsidization have to do with the power of that urban coalition. So there's a political shift in power. But if you're looking at the period prior to 1980, it's all about the dominance of agrarian populism and its views on unit banking.
- 3.79 Now, Canada wasn't completely devoid of rural populists. In fact, the populist movement in Canada in rural areas mirrored that of the US. But there was a key difference. We're going to talk about that starting tomorrow. The key difference was that the structure of the Canadian Constitution did not allow rural populists to win the political controversy over the structure of banking, whereas the nature of the US Constitution structure did. When we come back tomorrow, we'll pick up here with a discussion of, why is it that populist interests, which were similar in the two countries, end up with two very different outcomes in the banking system?
- 3.80 We'll begin here to talk a little bit about how the Canadian Constitution, as opposed to the US Constitution, really shaped that difference. Then, of course, I won't stop there because we have to try to understand, why are the two constitutions different? And we're going to. Have a pleasant afternoon and evening, and I'll see you tomorrow.

4. US Banking Evolution

AT A GLANCE

Explains why Canada resisted unit banking through its anti-revolutionary constitution, which centralized banking policy nationally, appointed a conservative Senate, and excluded American banks, whereas the revolutionary US left chartering to the states. Traces the slow end of unit banking in the 1980s through demographic shifts, ATMs, shadow banks, and global competition. Uses a Russian bank-run painting to introduce market discipline, then analyzes the six national banking era panics (1873-1907) as rational, seasonal responses to recession news in a unit banking system, managed by the New York Clearing House, with tiny losses compared with the era since 1970 or Argentina's twin crisis of 1890.

- 4.1 Welcome back to our discussion of the history of **financial crises**. Where we left off last time, we were noting how much of the history of US bank instability in the period from about the 1820s until the 1980s reflected the very peculiar, in fact, unique choice in the US to build a banking system based on locally isolated **unit banks**, called **unit banks** because they only had one office. They had to do all of their business out of that one location.
- 4.2 In the contrast between the US and Canada, where the US has many **banking crises** during that period and Canada has none, as we also noted, people in Canada and people in the US were both aware that this key structural difference was the explanation for why the two countries had such dramatically different banking experience. The question then is, how come the US picked its path and how come Canada didn't?
- 4.3 Where we left off, I was reviewing the fact that actually there were some attempts in Canada that tracked the history of banking legislation and the powerful role of those **rural populists** who were the ones responsible for pushing for **unit banking**. That there was a parallel track of similar interests in Canada that were pushing for similar other things. In fact, Canada is an agricultural economy, and the US during this period was largely an agricultural economy. It's not surprising that people in Canada, at least some people, were pushing for some of the same kinds of rules as we had in the US.
- 4.4 For example, there's a populist push to create, even though they already had, the existing **nationwide branching** charters. Those charters were for only limited time; they had to be renewed constantly. In the 1850s, there was a push to create **unit banking** charters, which posed a threat, an alternative, to the structure of the existing nationwide **branch banking** charters in Canada. But that was repealed in the 1860s. The interesting point is, there were some attempts to at least create the beginning of the camel's nose under the tent of **unit banking**.
- 4.5 But whenever it got to a sufficient point, it got pushed back in favor of preserving the existing **nationwide branching** system. There was another push in 1868 to try to imitate the structure of the US system. But again, it gets pushed back, in fact, to the point where the finance minister who had suggested this ends up having to resign as a result of the political decision that, "No,

we're not going in that direction." In a minute or a day, we'll see that, in the early 20th century, there's also a movement to create agricultural real estate lending subsidies in the US.

- 4.6 In 1911, there's a similar kind of attempt to expand real estate lending in Canada, but it gets pushed back. Then finally, we're going to talk about, in the 19-teens, the creation of state-level **deposit insurance** systems in eight states. There again were pushed by the same coalition of agricultural populists who supported **unit banking**, and because of the fragility of **unit banking**, tried to create a solution to that fragility without undermining the structure of the unit system, by creating state-level bailout funds through **deposit insurance** for **unit banks**.
- 4.7 In Canada, too, in 1923, there was a similar attempt, but it failed. What's interesting through this whole process is to notice that Canada is, in some sort of, the sense of its geography and its population, it has this **rural populism** definitely as part of it, especially in the West in Canada. They're pushing, they're watching things in the US. They have similar preferences to the agrarian interests in the US. They're pushing for similar things, and they're failing.
- 4.8 The question, of course, is, what's the crucial thing that makes them fail? To get at that, I want to talk a little bit about the history and the constitutional history of Canada and the US, so that you can see that really it's the constitutions in the two countries that are so different. But that constitutional difference, which really explains why we end up with **unit banking** in the US and nationwide **branch banking** in Canada, that constitutional difference itself reflects a historical difference between the evolution of these two countries that's very understandable.
- 4.9 I think this example is really helpful for us. The broader theme of our class is to try to understand historical learning and where historical choices come from and how they can persist. Constitutions have an important role in that, but they are themselves the outcomes of political and historical processes. We have to respect the importance of the Constitution, but then we also have to try to explain it, not just treat it as if it fell from heaven.
- 4.10 Let's start with Canada. I think the most important thing to remember is that the US is a revolutionary country that fights a revolution to leave the British Empire. Canada, by the way, still, Canada Day is celebrated every year. Canada is the country that's the closest to the United Kingdom of any other country. The currency has the sovereign's picture on it. Canada, you might even call the anti-revolutionary country. Let's talk a little bit about how Canada becomes the anti-revolutionary country, the country in North America, it's the opposite with respect to its relationship to Britain of the US.
- 4.11 First, let's recall that Canada didn't start off being British at all. Canada started off being French. It was colonized by the French. Through an amazing reversal of fortune in 1763, Canada becomes a British colony rather than a French one. That's pretty awkward in a sense, because imagine you have a country at that point, Quebec, that's full of people, all of whom are French, and now they have a new sovereign who's not French. From that moment, pretty much, you might even say till the present, there has been repeated attempts grown out of that sense of

conflict or resentment to create **Quebecois** independence. That is a theme of Canadian history that's going to be very important.

- 4.12 Now, there is another important change that happens in the 1780s. Once the US Revolution is successful, there are a lot of people, especially in New England, who aren't very happy about the US Revolution. They're called **loyalists**. They decide that they would prefer to live in a place that's governed by Britain, not by these new, perhaps rather extreme American republicans. So they move to, many of them move to one of the locations in Canada that's become very important since then, called Toronto. You have the French bastion, of course, centered in the great city of Montreal. Then after the American Revolution, **Loyalists** populating Canada, especially in Toronto.
- 4.13 Notice this is a very different history from the US. You have some people who are very opposed to British rule because they're French Canadians, others who are leaving the US in order to be part of a continuation of a royal Canada, as opposed to a republican US. As we see the story evolve, those resentments by the French Canadians play out in a quasi-attempt at revolution in the 1830s. Now, it's not that this was ever, I think, a really serious threat to liberating Quebec from the British. But what it did was, it made the British realize that they had to reorganize the authorities there to prevent French opposition to British rule from paralyzing the economic development of Canada.
- 4.14 And so the key thing that's going to happen there is to vest, especially economic decision-making power, in a national government. One of the strategies of the separatist **Quebecois** movement was recognizing the horizontal orientation of Canada to try to block economic development through their own ability to decide what happened in their province, which had direct implications for what was possible in the western part, Toronto and farther west. And so part of the calculation was to be obstructionist about allowing things that would happen that would expand Canadian development, especially in the west.
- 4.15 So the government in the UK starts realizing this. And ultimately, when you have the Canadian Constitution in its final form in 1867, what it does is something opposite of the US. It creates a, just as we have a **Tenth Amendment** that reserves all these rights of the state. And I'll talk about where the US's take on populist democracy comes from, this post-revolutionary constitution in the US. But in Canada, they have the opposite of the **Tenth Amendment**.
- 4.16 So first of all, they put all their economic decision-making power into the national government, including the decision-making power about the structure of the banking system. This is very important. We'll see the US does not do that. And so any decision about what's going to happen in the banking system isn't made in the local provinces or in Canada, it's made by the national government. And then they have this anti-Tenth Amendment that says, and by the way, if we forgot to give any power to the national government, if they want any, they can have it.
- 4.17 So it's really the opposite of, that the default strategy on things is to preserve a decentralized control of economic decision-making. In Canada, it's going to be very centralized. That is very

much, I think should be seen, as a way to check the influence of regional separatism, especially related to the French. I think that this is a really important and understandable part of being the anti-revolutionary country, wanting to promote Canadian development on a nationwide basis, check the **Quebecois** influences, and especially, given that horizontal orientation, there is a natural regional interdependency to building railroads, to promoting infrastructure in Canada.

- 4.18** Part of the Constitution that's also important, and in many of those examples that I talked about, where there were attempts in the rural areas to try to promote legislation, national legislation, that would favor **unit banking** or related policies, one of the things that the Canadian Constitution does is, it makes the appointment of senators a royal prerogative. That is, senators are appointed by the Queen or King of England. And that's still true today. Many people are surprised by that. Now, of course, they do it with consultation of the local authorities in Canada, just as a matter of practice, but they don't have to.
- 4.19** By the way, as recently as the 1990s, when **Mulroney** in Canada needed to get support for a particular piece of legislation in Canadian Senate, and he lacked that support, he asked the Queen to appoint several special additional senators. It's up to the Queen. And she did. And that helped him to accomplish his political objectives. But during all these episodes where the populist interests were trying to get changes to happen in Canada in the banking system that tracked their desires, it was the Senate that was typically the place that stopped them.
- 4.20** So the Senate also has a property ownership requirement as well as an appointment. So it's just naturally a more conservative branch of government and is designed, of course, to protect national interests as seen by the sovereign, initially, of course, and that has declined a little. Canada is much more of an autonomous country, of course, than it was in 1763 or even 1840s or even 1860s. But I think you get my point. It was designed, it's not a coincidence, Canada was designed to be more stable and more national in its decision-making and more loyal to the interests of the Crown in England.
- 4.21** Another thing that's relevant is this proximity to the US. The US is a huge country. Canada, of course, by population, is a much smaller country. There's always a concern in Canada that somehow they're going to get gobbled up economically by the US. Their economic growth or business cycles are very much connected to the US because the US is much bigger than they are. But there also was a concern that somehow the US financial interests would move into Canada and dominate the Canadian financial system. That was a helpful influence, too, because it led to a implicit agreement, politically in Canada, to keep American banks out of Canada.
- 4.22** Now, they're allowed to be in Canada in principle, but in practice, they're not allowed to really build major businesses in Canada. So that the big nationwide banks in Canada always knew that there was a threat of US entry. And so part of the discipline that's implicit there is that there's going to be an agreement between the government as the chartering authority and the banks to allow them to operate these very large nationwide franchises. And there's also going to be an additional agreement not to allow large US companies to come in and dominate Canada.

- 4.23 I think that's very relevant because what it means is that the Canadian banks, in some sense, are very mindful of the fact that they have to be part of a cooperative process with the government, and that that's really another aspect of this Canadian system that's there from the beginning. For the government's part, it also ensures discipline on the banks by having short-term bank charters. Charters have to be renewed. Currently, charters are renewed by Canadian banks every five years. So your charter is gone in five years, and you have to negotiate the new charter.
- 4.24 And that new charter, the good news there is, it's going to reflect all the things that have changed that you might want to change about the rules governing banks in the interim period. It's a very interesting political arrangement where the government and the banks are involved in a very different kind of political process of nationwide decision-making with this exclusion of the American banks and the very short-term Canadian bank charters. It's a very different situation from the US.
- 4.25 I just want to remind you, the slightly different map from the one that we looked at before, that this other difference, of course, is that the US, initially, is established as a set of colonies all on the Atlantic seaboard, all with immediate access to the global economy through the Atlantic Ocean. But in Canada, that's not true, that western development can only happen if you can access the Atlantic seaboard from going through the eastern parts of Canada. And so that's where that crucial dependency of western development on a nationwide consensus is important and is a driver of the Canadian preference for nationwide institutions governing decisions like the chartering of banks.
- 4.26 OK, so just to summarize a little bit, banking policy in Canada is determined by the national government. In the US, as we'll see, it's left to the states. Very different. And so in Canada, it's not possible to build local **rent-seeking** coalitions that are controlling bank regulatory policy. Whereas in Canada, it's not possible. In the US, it is very possible. And in fact, it's the quintessential aspect of the US, the fact that bank chartering decisions are determined locally within states, not nationally.
- 4.27 Although in the US, bank charters initially, like the Bank of the United States, which is chartered under Alexander Hamilton's leadership in 1791, it's chartered with a 20-year charter. But in the US, more generally, bank charters become permanent. And in fact, in the US today, that's the case. So a very different kind of situation.
- 4.28 When we think about the instability consequences of **unit banking**, there are really two different parts. One has to do with lack of **diversification** of risks in advance, so that when there's a precipitous decline, let's say, in a commodity that's relevant for your local economy, like soybean prices fall, that means that the banks in the local area are likely to suffer extreme stress. **Diversification** is crucial for reducing bank failure risk.
- 4.29 But it's not just that. It's also that, if you have a structured banking system in the US with maybe 20,000 individual banks that are regionally isolated from each other, they can't really coordinate

their behavior. Whereas, as I mentioned yesterday, in the case of Canada, there were two episodes in 1906 and 1908, where the Canadian banks, under the leadership of the **Bank of Montreal**, which is the oldest of their chartered institutions, decided that it looked like the shocks that were disrupting the banks that were failing in 1906 and 1908 were significant enough and worrisome enough that it made sense for the banks to simply acquire those failing institutions. What you would see, if you were just a member of the public, is one day it would be Bank X, and the next day it would say **Bank of Montreal**.

- 4.30 Then the costs of that cleanup would be borne on some kind of a prorated basis with the other banks that were involved in the cleanup, and then they would distribute also the branches of the rescued bank accordingly. It was a very orderly process that could be coordinated, and I want to emphasize this because there was a small number of them who could sit together in a room, who were familiar with each other, who knew each other, operating all over the entire country. Very different from 20,000 banks operating in local environment, not really even knowing each other across the country.
- 4.31 Now, let's talk about the US. Of course, the key point here in the beginning in the US is, the US is a revolutionary country. The US is getting rid of a sovereign that is controlling it from some distant centralized place. You may not be aware of this, but one of the, let's say, four or five most important issues that were points of contention that helped to influence the desire for the revolution was imperial policies limiting the chartering of banks.
- 4.32 In America, in North America, in the US colonies, what would become the United States, the idea that you wanted to have centralized power over economic decision-making was just inimical to the country that had just fought a revolution to get rid of that sort of thing. Even though initially you had **Alexander Hamilton** playing a leadership role as the finance minister, secretary of the Treasury, who tried to lead bank chartering policies; and initially, just by sheer force of his leadership and his role in the revolution, **Hamilton** managed to get through, over some pretty serious objections, the chartering of a bank called the Bank of the United States in 1791.
- 4.33 And also managed to, with his federalist party allies, to try to seek a vision of banks—the hope was that the Bank of the United States would become a nationwide, and it was, a **nationwide branching** bank. But then also within each state, the federalist party officials that were guiding things were also thinking about establishing banks in each of the major cities, and that was with very few charters. Even though it wasn't quite the same as Canada, because after all, the US Constitution did not create national authority, it created state authority. So it made possible that the outcomes of bank chartering would be determined at the state level.
- 4.34 But **Hamilton**, of course, created a **nationwide branching** bank, but it had a 20-year charter. And as soon as that 20-year charter expired, it was allowed to lapse. A second **nationwide branching** bank was established in 1816 with a 20-year charter, and that reflected the fact that during the War of 1812, the fiscal role in assisting the national government of the first bank in the United States was sorely missed. So they established the second bank in the United States. But it was

just as contentious from a populist standpoint as the first bank of the US, and its charter was allowed to lapse.

- 4.35 In fact, Jackson vetoed the rechartering of it in 1832, and that was the end of this idea of trying to establish any kind of nationwide system of banks. So the Constitution in the US did not have any of that nationwide power allocation. It left them to local governments. The local governments, even in the states that were initially dominated by federalists that saw some advantages to trying to build a nationally chartered banking operation with branches, they were ultimately defeated.
- 4.36 And so it was baked in the cake of the structure of the American Constitution, reflecting Americans' revolutionary past, to move in a direction completely different from Hamilton. And what was that? That was unit banking. The final phase, which we'll talk about tomorrow, is after the 1980s, when we get rid of unit banking. I think it's worthwhile just talking briefly about that. So how does this system come to an end? You can see that part of the story is that the Great Depression, which leads to the failure of a lot of small banks, is a topic we're going to turn to in the next hour.
- 4.37 Well, that really reduces the number of banks, and it has a slight effect on the number of branches per bank, because in the extreme circumstances of the Depression, some of the states allowed consolidation to happen with some limited branching. And so that means you might allow branching within a city, but not anything like statewide branching, not anything like nationwide branching. And so you can see there's an initial Great Depression movement to reduce the number of banks, but that's mainly because banks are failing under this extreme shock.
- 4.38 Some of the banks that had been allowed, especially during the Great Depression, or in the case of some states like California in the 19-teens, for other interesting economic reasons, to establish branching systems, they did grow with some branching. You can see that lower blue line is showing some growth in the number of branches per bank. But there's not a lot of pressure to get rid of the unit banking system, and the number of banks stays pretty much constant.
- 4.39 It's really not until the 1980s that you start to see an inflection point there in terms of the reduction in number of unit banks and now a big acceleration in the number of branches per bank. What this is reflecting is the political decision to finally give up the commitment to the unit banking system in most of the states. That happens initially with state-level decisions to allow neighboring states to enter your state and to expand branching networks across states. This whole thing evolved in a slow process.
- 4.40 In the 1930s, you had many states allowing to some minor relaxation of branching within their states. Then the beginning in 1970s and the early 1980s, you started having states allowing more branching within their states and even neighboring states to operate banks across state lines. And then eventually, it's as, it's as recent as 1997 that the US, as a country, finally has

unlimited nationwide **branch banking** allowed by law. That's how recent it is. But it begins that process really seriously in the 1980s.

- 4.41 Some of it reflects distress. So there are some real estate problems. There's a big run-up in interest rates that we'll talk about the **thrift** crisis tomorrow. That means the **savings and loan** collapse of the 1980s. But there were similar stresses in oil and gas lending in banking, in real estate lending and banking, at commercial real estate, as well as residential real estate lending. And so those stresses led to similar consolidation, just as you see the number of banks falling in the '30s as a result of stress. Again, you'll see that in the '80s.
- 4.42 The interesting thing was, this time, the politics the first time did not decide to create nationwide **branch banking**, but this time in the 1980s, it did. It took a while. It happened in stages. First, relaxation that allowed banks in neighboring states to establish, and then eventually, after you move beyond these regional relaxations, to this full nationwide relaxation of **branching** limits, which was passed in 1994 and fully enacted by 1997.
- 4.43 But the question, of course, is, what changed? Why is it that the same kind of influence of bank stress that you might think would encourage people to give up on **unit banking** to finally learn the lesson that **unit banking** is a bad structural feature? But I want to be careful there. If you're a rural populist, you don't want to learn that lesson. You like the existing system. So I don't want to make it sound like they're stupid. They're not. So, when we think about what's changing, we have to think about a change in power somehow, the allocation of power that's leading to a diminution in the influence in that **game of bank bargains** that decides how the power over banking is going to be allocated.
- 4.44 There's got to be some change, not just the distress of banks, because we saw that in the '30s as well as the '80s. So the question is, what changed? We could spend a whole course talking about that topic because it's pretty interesting to look at how things are changing in the technology of banking and other aspects in the 1970s and 1980s. And I just want to mention a few different things.
- 4.45 So first, of course, there's a demographic change. There's continuing technological improvement in agriculture that's leading to fewer and fewer people living in rural areas. People are moving to the cities. That's a process that's still ongoing in the US. What that means is, allocation of power is shifting from rural areas because population is shifting toward urban areas. So the people who are in charge, well, that reflects where they live.
- 4.46 People in urban areas always favored **branching**. They didn't have any reason to favor **unit banking**. And both manufacturing interests and even people who weren't themselves running manufacturing firms in cities, from a trade standpoint or even just wage employees, were not advocates of **unit banking**. So the population shift is a big part of it. Second, there were some important Supreme Court decisions related to a new technology called ATM machines. And so the question is, are ATM machines going to be considered a violation of anti-branching laws?

- 4.47 And if the Supreme Court decides not to see them as a violation of anti-branching laws, well, in some sense, the genie's out of the bottle, because now you're starting to erode the franchise of **unit banks** by creating new sources of competition through this new technology for ATM machines. So that was part of the contribution. Another thing was that shadow banks were creating a lot of domestic bank disintermediation, so that the chartered banks were not just losing out because ATM machines were arising and allowing people to work with larger banks in other locations and do their transactions locally through ATM machines.
- 4.48 That was part of it. But another part of it is, if you wanted to borrow, you could borrow from finance companies. If you were a relatively wealthy person, instead of holding a bank CD, you could hold a commercial paper. So the world was changing with new non-bank intermediaries. Also credit unions are becoming important. Banks are facing a lot of competition. And then fourth, they're losing a lot of global market share. So if you looked at the US, this is not something that we currently really think a lot about because it wasn't a very important long-term shift, but I was working on banking in the 1980s.
- 4.49 It looked like British, Japanese, and German banks were coming into the US in great force and entering precisely because of the weakness of our system. From the standpoint of doing global banking transactions, you can't do those if you're a tiny little **unit bank** because you need to be able to manage relationships with very large corporate customers operating through the whole world.
- 4.50 There was a sense—and I know **Alan Greenspan** in particular was outspoken about this— that if the US was going to have a banking system that was participating in global transactions, which were becoming increasingly important because of the increased importance of global trade and global capital movements in the 1980s and beyond, that if we were going to have a global banking system, we had to allow banks to become big. In addition to the effect of the crises in the 1980s, these other factors were absolutely crucial to leading to that political shift toward now what I'm going to call the **urban populist** banking coalition.
- 4.51 I want to emphasize, as we'll see when we talk about it tomorrow, the point is not that we get rid of local coalitions controlling banking system outcomes in the US. We don't. What we change is, instead of it being dominated by rural interests who favor **unit banking**, it becomes dominated by urban interests in the context of a nationwide **branch banking** system. That's going to be fun to think about. It's going to be a completely different **political equilibrium**, but nevertheless, it's going to be a very **political equilibrium**.
- 4.52 I think we should just continue our discussion because this discussion of American bank fragility is very much what our next session is about. I think we'll just continue seamlessly into the next one. Our topic here is to get into a little bit more in detail about these crises that are happening in the US from 1873 to 1907 and from early 1930s as a group. I'll talk about some of the intervening period a little bit, too. I want to start with a little art, just to create a little change of pace here. This is a painting that a friend gave me a photograph of this painting which I then framed. This is a painting.

- 4.53 It's actually the first version of the painting. The second version is very different from this, but you can tell it's the same scene of a bank run in Russia in the 1880s. If you look carefully at this painting, there are really three kinds of characters in this painting. First of all, front and center, is this fellow. He's the one who's looking right at us. What he's doing is, if you can tell, is, he's putting something into his breast pocket. What is he putting into his breast pocket? Cash. He was the one who was at the front of the line, and he got all of his money back from the bank, put it in his pocket, and is feeling pretty good about the day, all told.
- 4.54 Second character is this fellow. He's the banker. And what is he doing? He's glaring at our hero, the guy who got all of his money out of the bank. He's glaring at him because he sees his behavior as having been contributory to all these other people who are standing in line trying to get their money. And as you can tell from the facial expressions on some of their faces, they're not getting their money. They're quite upset. The bank has run out of cash. The bank is now suspending conversion of the deposits into cash. And of course, their faces indicate that they're not at all happy, that this is a very disruptive event.
- 4.55 So I think this is quite a wonderful, almost operatic drama about bank runs that we want to put front and center into this discussion that we're going to have about the problems of run risk in the US banking system. You could look at this painting in one of two ways. You could look at this as, this was a tragedy that should never have happened. Maybe it was all unnecessary. Maybe this fellow created a run out of no good reason. And that we want to try to design our system to prevent people from doing that. We don't know if that's true.
- 4.56 The alternative is, maybe this fellow over here was not running his bank very well. And that what our hero did was, because he could smell a rat in the bank, that is, that the bank loans didn't look like they were very carefully managed, or maybe the banker was employing his incompetent son-in-law as the chief operating officer of the bank or who knows what, he saw problems coming. He smelled a rat, acted. Others saw what he was doing, followed him, and he accomplished his objective, which was to make this fellow cease and desist.
- 4.57 We don't know if the bank might be able to reopen. We don't know a lot from this painting, but what we do know is, it's not going to be a very pleasant day in this person's life. That might be a positive thing. We might call that **market discipline**. We might call that the whole point of allowing someone like this, properly incentivized by the ability to be rewarded for being the first in line by getting his money out in full, we might be thinking that whole incentive of allowing this to happen helps to preserve discipline by a reasonably informed person over a possibly incompetent or crooked one.
- 4.58 Those are two very different interpretations. What I want to do is help you think through, what does the evidence from **banking crises** tell us about how we should be thinking about this? Maybe surprisingly, I'm going to end up arguing that, at least during the historical period that we're going to talk about in the US, despite the instability of these crises, that there's a lot of evidence that this process was actually a favorable one of **market discipline**.

- 4.59 The other thing I should mention is that these depositors, now they're Russian depositors, but if they were American depositors, let's say, in national banks that had failed as a result of a run, what was their expectation of loss? Well, I'll give you a rough number— about 10 to 15% on average, prior to the 1930s. So it's not like they lost everything, but they lost about 10 to 15% of their deposits. If this bank did not reopen and had to be liquidated, typically they were liquidated through something called **voluntary liquidation**.
- 4.60 That is, the banker, that regulator comes in, has a conversation with the banker. The banker doesn't have enough to recapitalize the bank. The banker has also some strong incentives through something called **double liability** to shut things down before things get worse, because if they don't, he might have to contribute money after the fact. He and his stockholders might have to contribute money after the fact. And so most of these cases where the bank is forced to shut down are **voluntary liquidations**. If they're not **voluntary liquidations**, then they are another thing called **receiverships**, where the regulator appoints someone to liquidate the bank. Those tend to be more severe cases in terms of their losses.
- 4.61 But depositors are typically expecting to lose, even if the bank can't reopen, they're going to lose maybe 10 to 15%. This is for the national bank failures prior to the 1930s, and sometimes less, sometimes zero, often zero, and sometimes 5%. I want to emphasize, you didn't lose all your money, but you did lose some. The reason that that's very relevant is, we're going to go through in our own minds a calculation about what were depositors thinking when they were deciding whether to, once they saw that there was some money leaving the bank, and it was something that people were monitoring, it was something newspapers talked about, and it was something that you could observe, would you get a run on the bank?
- 4.62 I also, in discussing this, want to emphasize, reductions in deposits aren't always runs. Sometimes the bank might be just doing a little bit too much risk-taking, and that by having some deposits withdrawn from the bank, that bank deleverages itself, which reduces the risk of default. If it's done in a gradual way, so that if, let's say, depositors come in and ask for 2% of deposits back this week, well, then the banker, how do I handle that? Oh, I have, some of my loans are coming due every week, right? So the loans that are coming due this week, I use the proceeds from those loans to pay off the depositors, and then I just don't renew any loans this week. The effect of that is actually to reduce my risk.
- 4.63 If we're seeing reductions in deposits that are happening gradually, which is generally what's happening, when depositors are withdrawing funds, noticing maybe that a bank is taking a little bit too much risk, it's often happening very gradually, and it's a very stabilizing event because that **market discipline** leads the bank to reduce its loan risk and to reduce its leverage. But if there's a sudden reappraisal of the circumstances of the bank, for example, suppose that some other banks, this fellow here might himself be involved in a lot of banking operations and might be aware that there's a concern that the banker in charge of this bank has been doing some rather shady things.

- 4.64 Under that circumstance, there might be a sudden **reappraisal of risk**, and that sudden **reappraisal of risk** might lead to very large withdrawals and a general view that this bank is losing the confidence, particularly of other bankers. In that case, then people start lining up and you might get a massive withdrawal, and it might be too late. They may not get any money. There were also some instances where there were rumors that were spread for malicious reasons about banks. Here in Tarpon Springs, Florida, I think it was around 1905.
- 4.65 I say this because I'm a Greek American, so I can tell you about this. There was a Greek American —Tarpon Springs, as some of you are aware, is a big Greek community in Florida— who, I guess, maliciously spread the rumor that a bank was in trouble, and it led to a massive withdrawal of deposits in the local Greek community. You could see how people could spread rumors maliciously like that. But what happened was, the banker at that time, who was completely innocent of any wrongdoing and whose portfolio was very solid, well, the other bankers that he had relationships with, not just in Florida, but in other places like New York or Chicago or New Orleans, or I'm not sure where, those other bankers sent him cash because they knew he was fine.
- 4.66 So what's interesting is, the best way to think about this is, there's a group of people who we might call informed depositors. They're often other bankers that the bank has relationships with. And then there's some uninformed depositors, and then there's the bankers. It's really a three-sided relationship. If runs are completely the result of malicious gossip, they don't tend to produce a lot of bad outcome. I mean, they could sometimes, but we don't want to think that that's the case.
- 4.67 In fact, in the case of this Tarpon Springs run in 1905, the person who mainly suffered from the run was the perpetrator, because people were able to reconstruct how each of them had heard the false rumor about the bank, and they were able to trace it all back to this one originating malicious gossip. I don't think they strung them up, but it was a close call. OK, so I raise this so that you can see that there's a lot of discussion, particularly in the US, because we're the crisis center, right? We're the place where crises are happening, and we're going to see we're also a place where the word "**panic**" becomes particularly important.
- 4.68 Because during this period, 1873 to 1907, there are six **banking crises** in the US that are of some kind of major importance. 1873, 1884, 1890, 1893, 1896, and 1907. Those six different events are happening over a fairly short period of time. What's interesting, as we'll see, is that none of them ends up being a severe **banking crisis** in terms of many, many banks failing, with possibly one exception. We'll talk about 1893. But for the most part, these **panics** are times when there's a lot of disruptive withdrawals, like that painting we just we saw, but not leading to lots of bank failures.
- 4.69 We want to try to understand that peculiar US phenomenon of this high frequency of **panic** events that don't result in lots of failures. We want to see, is that the result of some random, irrational, malicious actions, or is it more systematic, and can we make sense of it? If we think of it as being more systematic, we might think, here's a model of how these happen. Suppose that

you have a system structured of **unit banks**, and you know that some sort of adverse aggregate shock has happened. Let's say that they're starting to see some indicators in the news that a recession may be brewing.

- 4.70 You also know that this is a particularly vulnerable time of year for the banks. What makes a time of year vulnerable? Well, because the US is still a primarily agricultural economy, spring planting and fall harvest seasons are times when banks tend to be doing the highest levels of their lending, and therefore, they're at the highest risk points. Bank risk has a seasonal feature to it because bank lending has a seasonal feature to it. Banks' risk exposures are at their maximum in the spring and the fall.
- 4.71 Now, imagine that you hear a rumor that the recession may be starting, and it's a time of seasonal high bank risk exposure. Well, it turns out that all six of these events happened at exactly those times. There are no exceptions. All six of them were happening at cyclical peaks when bad news was circulating about liabilities of failing businesses, not banks, but businesses that were in trouble, which is a great indicator of recession. They all happened either in the spring or the fall at a time of high risk exposure of the banks.
- 4.72 Now imagine, why would that be? Because as a depositor, you might say, OK, well, **unit banks** are not very well diversified. I know that there's a lot of risk that a recession is happening, and I know that this is a particularly vulnerable time because of the seasonal exposure. So maybe just to be safe, I'm going to go withdraw money from the banks. So notice, that is not random. That is very smart. It's an outcome that's the result of **unit banking**, the combination of **unit banking** and **market discipline**, with depositors who are actually fairly savvy.
- 4.73 But how do we square that with the fact that it turns out that, after the fact, generally speaking, there are not lots of bank failures? Well, let me help you think that through. This is an exercise I always perform with my MBA students. I'll say to them, all right, suppose here we are, we're having class, and you know you've got your lunch break coming up. Suppose I told each of you, let's get rid of **deposit insurance** for a minute. I told each of you that you have deposits sitting in uninsured bank accounts.
- 4.74 Let's say that there's a 10% chance of a 5% loss to you as a depositor. That is, there's a one in 10 chance that your bank is going to shut down today. If that happens, your expected loss on your deposits is 5%. That means the total expected loss, which is the multiplication of the probability of the event, plus the loss given the event, is half a percent, or in finance terminology, 50 basis points. So your expected loss on your deposits is 50 basis points, which is 10% times 5%.
- 4.75 Now, then I would say to people, how many of you would choose, instead of having lunch, to just go withdraw your money from the banks? Typically, the outcome I got was almost everyone. Notice that it doesn't take much risk to make people want to take their money out of their deposit accounts, even though the probability of a bank failure might be very low. That's the key thing to notice. Bank depositors aren't just pricing risk. They're intolerant of risk.

- 4.76 Now, think about each of you. You have a bank deposit, and this is true in your portfolios. You may have some stock, you may own your condo, you may own some other assets. What's the function of your bank deposit? Your bank deposit is there so that you can manage whatever payments come up over the next month or two. Bitcoin is not a good strategy, right? Neither would it be just to hold it all in stocks. What is it that you want from that deposit?
- 4.77 You want predictability, you want stability, you want to know that if you have an expense that comes up, maybe a family occasion, maybe a health problem, maybe who knows what, you want to take a vacation suddenly. You want to know that you can access those funds predictably. And so when we think about deposits, what makes them different as financial assets is that people are **risk-intolerant** about them. And that **risk-intolerance** is reflected in the fact that I'm saying to you, OK, there's a 10% chance you're going to suffer 5% loss, and you go get that money out.
- 4.78 Whereas if I told you, look, the stock market has ups and downs, there's a 10% chance that you might suffer a 5% loss on your shares this week, you'd say, "Well, yeah, I know that's true just about every week, right? There's a 10% chance that I could suffer a 5% loss. That doesn't mean I'm going to change anything because things could go up, things could go down." So notice the behavioral response in deposits partly reflects their role in the payment system as a place that people turn to be able to reliably execute payments in the short term that might be coming up for them. That's why people often react.
- 4.79 But that doesn't mean, just because people will react, doesn't mean we should be surprised that after we get these events, so a recession happens at a high time of bank risk exposure in any one of these six events, 1873, 1884, 1890, 1893, 1896, 1907. We say, oh, well, but now we don't see too many banks failing. In fact, if you put 1893 aside, you can barely see any increase in bank failures in any of those other episodes.
- 4.80 But that shouldn't surprise you because all that's happening is, people are being extremely **risk-intolerant** about an observable risk and knowing that their banks are **unit banks**, and therefore more vulnerable to failure from this observable shock with unobservable consequences in the banking system. Why doesn't that happen in a **branch banking** system? Because just the fact that a recession has happened in a highly diversified nationwide portfolio isn't going to really lead to those same kinds of concerns. That's why these **panic** episodes are a US phenomenon during this period, 1873 to 1907.
- 4.81 Now, you I might say, how did I pick these? How did I pick these six? I think that you should always, as students, whenever you're hearing a professor talk about something, you should ask those kinds of questions. Let me tell you how I picked them. I wanted to try to see— so first of all, everybody picks them. So it's not like I picked them weirdly. So all the books about the **national banking era panics** always talk about these six episodes. But can we objectify that? And the answer is, we can.

- 4.82 Because if you're getting this observable shock that's leading to concerns by depositors about where's the problem that might be coming out of this high vulnerable moment where there's a recession shock, let's look at the banks' behavior themselves. What's a bank behavior that's indicative that we've entered one of these moments of systemic risk? Well, the **New York City Clearing House**, which is at the epi-top of the bank pyramid in the US—so the banks have relationships with each other, as I said.
- 4.83 Why? Because trade is flowing from the periphery to New York in particular. And so all banks have accounts with New York City banks. And New York City is operating at what's sometimes called the top of the **reserve pyramid**. Because all banks maintain balances with them, so that when they need to pay for the movement of goods to New York and out of New York, because the people who are moving the things often have relationships with the peripheral banks. So as the, the banks all have relationships with New York, New York is sitting on, at the top of the banking system.
- 4.84 Even though they're small **unit banks**, they still have a lot of control through their management of other banks' cash balances. The New York City banks are always the place that's kind of like, they can feel the wind coming because they're seeing movements in their own cash balances. They're seeing they have concerns about systemic things on a nationwide basis. They're running something called the **New York City Clearing House**. The **New York City Clearing House** is a group of, let's say, about 30 New York City banks who are operating as a group to manage things if **panics** start to hit.
- 4.85 How do they do that? They do it through collective action. The first thing that happens is, the New York City banks who are members of the **Clearing House**, get together and say, "We smell something in the wind here and we don't like it." They have a meeting. They say, "We'd better authorize the emergency production of **New York City Clearing House** certificates." It's like their own version of paper money to be used to pay off withdrawals if necessary. If there's a sudden increase in demand for cash withdrawals, we can help to mitigate that problem by creating a source of cash.
- 4.86 How do you make that cash credible in everyone's mind? By making it the collective obligation of all the New York City banks that are members of the **Clearing House**, not any individual bank, because everybody knows the **Clearing House** as a whole is going to be able to survive, no matter what. So the **Clearing House** can actually issue paper money to supplement the existing amount of cash and help weather the storm of a lot of cash withdrawals.
- 4.87 So there were only six times during this period that the New York City bank **Clearing House** met to discuss those kinds of problems and to undertake some observable specific action, either to authorize those certificates or to actually print them. What are those six times? 1873, 1884, 1890, 1893, 1896, 1907. When we think about these **panics**, all the logic works really well. They all are happening at seasonal loan demand peaks. They're all happening at recession shock moments. When they happen, they're all being responded to by some sort of collective action in the New York City bank **Clearing House**.

- 4.88 Now, as I already said, given this way of thinking, it shouldn't surprise you that they didn't tend to produce very large systemic bank failures. To give you a sense, the largest of the group in 1893 produced bank failures with losses that were about 0.1% of GDP. There are many different ways to scale the size of bank losses, but you'll see why I'm doing it this way. In other words, if you take the failed banks and you take their negative net worth, the losses that they imposed on depositors, and you divide that by GDP, that number is 0.1% of GDP. That is by far the largest of the six.
- 4.89 In what sense is that a small number? Suppose we looked at the 100 **banking crises** that have happened globally since the 1970s. There are actually a little more than 100 now. But until 2013, there were 100. What was the median comparable figure of negative net worth of failed banks divided by GDP? The median was 16%. What was the number during the **Great Depression**? About 2.5 to 3%. What was the number during the 1920s, which was a time when we didn't have nationwide bank failures, but we had agricultural shocks after World War I?
- 4.90 Agricultural shocks, especially to crops like cotton or to wheat, that resulted in such severe declines in certain regions that some banking systems in some states really got decimated. That was aggravated also by that creation of state **deposit insurance** systems that I talked about. In the 1920s, the bank failures throughout that decade are about leading to loss of about 1% of GDP. So the **Great Depression**, about 2.5% to 3%; the whole 1920s, about 1% of GDP; in 1893, 0.1% of GDP.
- 4.91 So now I want you to try to understand why I'm so excited, in a negative way, about the experience of the world since 1970. The median of those 100 has a loss of 16 country events has a loss of 16% of GDP. This is a different world that we're living in. When we think about what the losses were in the global **financial crisis** within the US, they're looking more like of the size, roughly, of the 1920s or some of the 1930s. These may seem smaller because for developed economies, when they have a **banking crisis**, it tends to be a smaller loss relative to GDP because the economy isn't so bank-dependent.
- 4.92 There are lots of other financial transactions going on, stock markets, bond markets. They're much financially deeper, they're much bigger, much more developed. Nonetheless, the point I'm making is, in **emerging markets**, those are huge numbers. In developed economies, in Europe and the United States, the crises that have happened since 1970 have been much, much larger than such a crisis in their own past and on the scale of things like the **Great Depression** in the US.
- 4.93 But my point here is to illustrate that these **panics** in 1873 to 1907 reflected a **unit banking** system suffering a kind of observable shock with unobservable consequences because of the structure of the **unit banking** system that leads to this kind of temporary disruptions. But once we sort out what the effects are of the recession, then we resume business as normal. Let's talk a little bit about how we sort that out. In three of these cases, we actually had **suspensions of convertibility**, 1873, 1893, and 1907. What that means is that, in New York City and in other locations, there was enough sudden pressure that actually the banks as a whole said, "Time out. We are not going to keep paying out deposits right now."

- 4.94 Then they said, "What we're going to do is, we're going to implement some collective action measures, print some certificates to improve the position of our banks through their ability to meet cash withdrawals going forward. We're going to publicize our bank conditions better. We're going to get our auditors to make public expressions about our condition. We're going to engage in a lot of activities. We're also going to not roll over loans right now so we can shore up our cash positions."
- 4.95 All these things are obvious. If you were suffering this unwarranted withdrawal of cash coming from this observable recessionary shock with unobservable consequences within a **unit banking** system, these are things you would do. How do they know? How do they know that it's time to reopen their doors, that it's time to not worry about deposit or withdrawals? The answer is, there's a market for bank **cashier checks**. If you've ever gotten a cashier check from a bank, it's a commitment by the bank to pay that's not dependent on whether money is in your checking account.
- 4.96 Those kinds of checks circulated as cash during these suspensions in a secondary market, and the discount on those was observable. How big do you think the discount got? Let's say, like, the day after the suspension in New York City? The discount on a particular New York City bank's commitment to pay, how large do you think it was? In my experience, students usually respond, "Oh, like 5%, 10%?" The answer is about a quarter of a percent. Why? Because just as I said, your expected loss was actually very small. It was a number like 5% or 10% or even less.
- 4.97 Even though people are lined up to take their money out in mass all around the block, if you ask them what their expectation of loss was for the liabilities of that bank, the answer was, well, very small. Once that number in the secondary market fell from 20, 30, 40 basis points, which is what it started at the day after the suspension, once it fell to zero basis points, then you knew, OK, panic's over. Our efforts to relieve people, to show people that we have shored up our position and that we're not subject to a lot of risk, have been successful. Now that the storm has passed, we can open up our doors.
- 4.98 That's how **panics** were managed during this period of the national banking system. This whole period worldwide, of course, the US was unique. We're the place where these kinds of **panics** are happening. In other countries, systemic nationwide **panics** like that aren't happening, typically. They're not happening, why? Because they don't have a **unit banking** system. Now, some countries are having some serious bank insolvency problems that are reflecting major shocks. Those are a very different story.
- 4.99 There are a few of them, and I'm giving you the list, as well as telling you what the actual outcome was, approximately, the data aren't so great for us to be too exact here, in terms of negative net worth of failed banks relative to GDP. You can see that the biggest ones were Argentina in 1890 with about a negative net worth of failed banks to GDP of about 10%, Australia in 1893, similar magnitude, Norway, 1900, about 3% of GDP, negative net worth of failed banks. In Italy, about 1%.

- 4.100 And I haven't been able to figure out what the Brazilian number is for the so-called encilhamento, but it's clearly less than the Argentina or Australian experience. What more can I tell you about these crises? Well, first, let's just notice that there are only, I think, six of them. Yeah, six. That compared to the 1970 to 2013 severity, both in terms of their frequency, there are far fewer of them, and in terms of their severity, on average, it's much less.
- 4.101 If you add the US cases in, where it was basically zero loss, with the exception of 1893, where it's 0.1%, you can see that this is going to average out to a number like 3%, not 16%. It's several times less frequent an occurrence and several times less costly than the current era. It's important to recognize this. Another fact that we could talk about is that the causes of these are pretty clear. One of the most important ones, Argentina, is the one that looks the most familiar to people who are looking at the recent experience.
- 4.102 The reason is that Argentina was establishing through something called **cedulas**, which just means certificate in Spanish. But these were particular certificates. They were certificates relating to guaranteed mortgages. So sound familiar? So Argentina, of course, the political group that was most important in Argentina were the Argentine major agricultural and ranch landowners. And the **cedulas** were government guarantees, effectively, of mortgages that were issued by either the main bank of Buenos Aires or the main bank of Argentina that were basically funding, they were making loans to those kinds of influential Argentines.
- 4.103 Now, they were limited in the number that they could do, but they all exceeded, the both of the banks that were issuing those **cedulas**, that were both government-created entities, both exceeded the legal amount that they were allowed to offer. Because these were guaranteed by the government, we know that they were viewed as effectively riskless and, therefore, very similar to Argentine sovereign debt. In fact, in the London debt market, the **cedulas** traded at the same price, reflecting the same interest rate discount as the Argentine sovereign bonds.
- 4.104 In other words, if your bank wanted to make a loan, you could then sell that loan. You make the loan, you have a mortgage, and you can now sell that on the London market as if it were a treasury bill. Now, how do you maximize your profit as a banker in those circumstances? By taking on the riskiest loan customer. Because what you do is, you go to the loan customer and you say, "Give me a high interest rate because you're very risky, and the market is making you have to pay a pretty high interest rate."
- 4.105 But then once I've pocketed that high interest rate, then I can sell it on the market as that same promise, but discounted now by a much lower interest rate. So the government **risk subsidy** accrues to the banker, maybe to some extent also to the borrower, because the borrowers may know that the bankers can be bargained with. So there's going to be some sharing of the rent coming from the subsidy between the bankers who issue the **cedulas** and the landowners who are borrowing the money.
- 4.106 That is, the borrowers will get a lower interest rate to some extent, and the lenders will be able to flip the **cedulas** in the London market and score a big gain. And who's paying the cost? The

taxpayers. Now, when we think, why am I telling you all this? Because the Argentine case is unique. It's the big **twin crisis** during this period. It's the one that has both a big loss in the banking system, because guess what? These are risky mortgages. So there's a big loss in the banking system.

- 4.107 But then, because those losses are shared by the government that's guaranteed those banks, that means that the government has a big expenditure problem, which means that the government has to print money, which means that the currency collapses in value. The Argentine case is the only one of these. It's certainly one of the largest in terms of the magnitude of the shock, but it's also the one that brings with it a big sovereign problem which is reflected in a depreciation of the currency.
- 4.108 In that sense, it's the most familiar because this is a pattern that you will see reflected in the last 30 or 40 years very frequently, that comes from the government protection of the banking transaction that both promotes more risk-taking in the banking system and with greater fiscal consequences for the government, which sometimes lead to the depreciation of the currency. When you have a simultaneous depreciation of the currency and a bank crisis, we call that a **twin crisis**.
- 4.109 **Twin crises** were almost never happening. Argentina is the only example of a severe **twin crisis** during this period. Italy is a much, much lesser example. Most of these, even Australia's crisis, even though it's a **banking crisis**, it doesn't lead to any consequences for the sovereign. The Australian government doesn't pay these losses. The depositors just suffer them. Australia is part of the **gold standard** system. And so Australia and Norway do not leave the **gold standard**. Argentina is the big outlier there. Italy has a much milder **twin crisis** experience, but that's a very particular event, which we won't go into.
- 4.110 So the point of all this is, we don't want to think about just **banking crises** within the topic of **financial crises** as all the same. The American twin, I'm sorry, the American **panics** during the **national banking era** were different from these other events. And even within these other events outside the US, they're different. Argentina was unique because that's the place where the bank crisis was a **twin crisis** because there was a clear link between government subsidization of risk in the banking system and the failures of the banks.
- 4.111 This next slide that I prepared was prepared mainly with the discussion that was happening after the Asian crisis of 1997. Because after that crisis, many economists, especially macroeconomists, who tend not to study banks or bank failure histories very much. Now they do more, but still that's relatively rare. Macroeconomists like to draw big graphs with things moving all over the place, but there's not a lot of discussion of the microeconomics of the financial system and what's going on in the banking system.
- 4.112 After the 1997 crisis, a lot of people were saying, oh, all of this instability that's happening is the result of the fact that we have this new globalization going on. These Asian economies, like the Mexican economy in 1994-95, and many of these economies that we're seeing, the reason for

all this instability is this fickle global investors and global big movements of capital. Now, so, as I say that, recognize that the period that's most like the period in the 1980s through 2013, in terms of a lot of global capital flows was 1870 to 1913.

- 4.113 So in fact, the period in the 1980s and 1990s was basically just restoring the level of global capital flows that had existed prior to World War I. So in case you're thinking that somehow there's an alternative explanation for all these things, these crises, that it's the result of fickle global capital flows, I would just say, no, it's not. And having lots of global capital flows doesn't seem to be a bad thing at all, per se, when we look at the period 1870 and 1913.
- 4.114 Now, part of the story for why we didn't observe more **twin crises** is because, and we want to talk about these as distinct but not really separate phenomena. That is, they're related. I'm going to talk, when we come back from our break, about how we think about the meaning of **market discipline** in a banking system. We've already talked about it a couple of times, but I want to be even clearer about it. So **market discipline** means that banks in their risk management are worried that depositors, especially informed depositors, if you recall the painting that we had up before, might become informed about problems at banks that might lead them to start withdrawing funds.
- 4.115 But that's often a self-correcting mechanism that leads to deleveraging of banks, restoring banks to lower risk. Because banks are subject to **market discipline**, they tend to be worried about managing their risks. That's very favorable. Argentina during this period was, of course, the exception because they weren't worried about managing their risk because the government, through the **cedulas** policies, was actually bearing the risk for them. We're going to see that that is exactly the same as the problem of **deposit insurance**.
- 4.116 But there was another commitment to discipline during this period, which was **fiscal discipline** under the **gold standard**, because the only way that you could actually maintain your role in the global economy at full tilt was to be part of the global **gold standard**. That meant that, how do you maintain a commitment to convert your currency into gold? Well, you can't depreciate your currency, can you? That means you can't be running deficits that require you to print money. It limits the total amount of government debt that you can issue, which limits your deficits, cumulative deficits, which makes you have to limit your spending.
- 4.117 If you are committed to **fiscal discipline**, of course, that's going to help make you committed to **market discipline** for banks. Why? Because you don't want to be like Argentina who has given commitments to banks that turn out to cost you a huge amount of money. That's the sense in which these two commitments are related. If you want to participate in the global standard fully, you have to be disciplined in your fiscal affairs, which means you're very likely to be better off if you allow **market discipline** in the banking system.
- 4.118 What's the key combination that makes this pre-World War I period different from the post-World War II period? Barry Eichengreen, my friend and a professor at University of California, Berkeley, has written about this. And of course, he's not the only one who's written about it. But

what's really different is, of course, the rise of democracy and the rise of what you might call consequent rise of governments being very concerned about cyclical variations.

- 4.119 If your government is concerned, maybe because of an election coming up, about unemployment rising a little bit right now, or maybe even more than a little bit, then that's going to undermine your ability to be committed to these disciplines because these disciplines are long-term commitments. But what if your politics changes you, from having long-term commitments to having more short-term commitments? Then all of a sudden, **fiscal discipline** and **market discipline**, both of which are long-term commitments, might become inconvenient.
- 4.120 You might say, well, yes, I understand that some of these banks are looking pretty dicey right now, but there's an election coming up. Maybe I should offer some protection to the banks. At least that will postpone the problem so it doesn't happen until after the election. Guess what? There's a huge amount of empirical evidence that governments do that in today's world. That banks are in trouble, governments go in and put a Band-Aid on the gunshot wound to get them through to after the election.
- 4.121 But similarly, unemployment is starting to rise. Maybe I should print some money. OK, it's going to be inconsistent with maintaining my exchange rate like a commitment to the **gold standard**. But that's OK because it's going to get me through the election. So that in a world of, where there's much stronger short-term electoral democratic pressures, commitments to long-term policies like **fiscal discipline** and **market discipline** in the banking system are going to be much weaker.
- 4.122 It's not surprising that the politics has changed globally, which has been what's underlying this huge shift from commitments to long-term discipline to a lack of those commitments. I don't want to go too crazy here. I mean, it sounds like I'm saying, "Bring back the Tsar, bring back Porfirio Díaz, too." I'm not saying that. However, the finance minister of the Tsar, **Sergei Witte**, and the finance minister of Porfirio Díaz, who was **José Limantour**, were both some of the wisest, smartest people about how to establish policies that created and maintained long-term healthy commitments to healthy banking systems and healthy fiscal affairs.
- 4.123 Why were they able to do that? Because they weren't subject to short-term electoral pressures. I think we just have to be mindful. Remember, what's the motto of this course? **Know yourself**. I'm not telling you that we're going back to this world—quite the opposite. I'm telling you, we're in this world. Let's own it. Let's understand what that is. Let's understand that these **banking crises** that we're talking about now, that are a symptom of a new problem that reflects a new kind of politics.
- 4.124 Let's take a break now. I'll take a question, if anybody has one, and then we'll take a break and come back and talk a little bit more in detail about that mechanism of **market discipline** in the banking system. That will get us into then talking about also what's new about the **banking crisis** in the US of the 1930s and how it's very different from the **banking crisis** that occurred in the US from 1873 to 1907, and also different from the **banking crises** in terms of its causes, very

different from the **banking crises** that we saw in other countries happening from 1873 to 1913. So first, any questions?

- 4.125 Do you think the forefathers were able to foresee the repercussions of this decentralized system? If they did, was it worth it? Was it growing pains of having state sovereignty?
- 4.126 I think I can say pretty confidently they were not able to foresee this. Why? It would have been just a Herculean thing for them to be able to foresee. First of all, **Hamilton** seemed to be winning. If you look at it, when people were chartering banks in other states, they would often send **Alexander Hamilton** a letter saying, "How should we do this?" He would give them very specific advice.
- 4.127 The way they were thinking about it, because it was mainly the federalist party members in the different states who were leading the way in banking, they were thinking of this coming out of urban areas like New York and Philadelphia, establishing banks that would eventually grow and become banks like the ones they're familiar with in Europe, which we're, with the exception of the **Bank of England**, because that's a different entity involved in fighting the French, they understand that they all were familiar with the Scottish experience in particular.
- 4.128 So they would have thought, no, what we're going to do is eventually, as we grow, we'll charter these. We'll have a nationally chartered bank, and then we'll have state chartered banks. And they're all growing out of the experience of these major cities, Boston, New York, Philadelphia. It wasn't really until the War of 1812, after that, that we started even having the chartering of these **unit banking** laws. It starts in Pennsylvania, for example. I think it was 1816. It was in agricultural areas, and there's a lot of protests where people are saying, "Hey, we need banks chartered out here, not just you guys in Philadelphia."
- 4.129 **Hamilton** couldn't, I think, have foreseen that political pressure because everything looked at that point like he was winning. There had been opposition to what was going on, but his strategy and people who were following his advice were pretty much winning. The other thing I would say is, even if you could have guessed that there was going to be this outcry in the rural areas, in those states, there was much more of a balance between rural and urban interests. Like Pennsylvania, you had Philadelphia. The western part of Pennsylvania was very underdeveloped. Pittsburgh hardly existed. That was new territory, still fighting the Indians.
- 4.130 And so, even if you could have imagined some agrarian uprising about chartering some banks, there would have been a balance. They wouldn't have been in control of the whole thing because those initial states were places that had those Eastern seaboard with major cities. But the way things evolved, the US went west. And as we go west, we've got places like Iowa, Nebraska, Kansas, lots of other places like that, Wyoming, Colorado. Those are states in which there really are no major cities to rival, and the domination is going to be by the agrarian interest. You would have had to foresee a lot to see that coming. I think the answer is no.
- 4.131 But I would also point out to you that, by the 1780s, with the Northwest Ordinances, there was already an understanding of creating a kind of, let's call it a Jeffersonian vision, which **Jefferson**,

Madison, and Monroe had all talked through, and there's some clear discussion about that as they're drafting the organizing principles for what would be the Northwest Territories, which is basically the American Midwest.

- 4.132 They very much were thinking in terms of trying to create units of self-governing towns where the farmer, in living in a small town, would be the defining character of the future of America. In that sense, it wasn't that this is a surprising evolution. It's already in people's minds, certainly by the 1780s, that the whole western part of the country is going to develop that way. The Louisiana Purchase, of course, is happening fairly soon thereafter. So you could say that people could have imagined it. As far as I can tell, they didn't. It would have been, as I said, very different from their experience. That's the best I can do.

5. Bank Risk Management

AT A GLANCE

Formalizes bank risk management as a choice of equity relative to liabilities and cash relative to loans, illustrated with default-risk curves for New York City banks from 1929 to 1940, which survived the Depression by shifting from loans to cash under depositor discipline. Presents evidence of market discipline from Chicago in 1932, Argentina in 1995, and Mexico in 1996. Argues the Great Depression was caused by contractionary Fed policy operating on a fragile unit banking system, resolved by Roosevelt's bank holiday and triage. Then examines the Florida land boom of the 1920s as a case of genuine uncertainty, conservative banks, and ex post fraud in the Manly-Anthony chain.

- 5.1** Welcome back. What I want to do right now is pick up on a theme that we've been developing and talking about—the banks—which is how banks manage their risk using a combination of their funding sources: that is, how much they lever themselves with debt—debt versus stockholder's equity. And then how banks manage their asset risks, partly by holding more cash assets when they want to reduce their risk. I've been mentioning this, and how banks respond to pressures by changing their risk management to reassure people.
- 5.2** But what I want to do in preparation for our further discussions, I want to formalize this a little bit so that we have some tools where when we are talking about the 1930s, which we're going to now immediately. Also, right after that, we're going to turn to talking about the Florida banking and land boom and bust that's happening in the 1920s. Then tomorrow, we're going to talk about the agricultural problems that are happening in banking in the 1920s in the US. You'll see that in all of those discussions, we're going to have a focus on how banks are responding in their risk management to what's going on around them.
- 5.3** It's really helpful to develop some simple tools, coming from financial economics, that help us understand how to think about bank risk and how banks manage their risk. The key things that I want to point out when we talk about what I'll call "old-fashioned disciplined banking" is that banks have both sides of the balance sheets as tools that they can manage when they're thinking about their risk. I also want to be clear that people are able to observe the bank's balance sheets—not necessarily on a continuous basis, but periodically.
- 5.4** Here's something that might surprise you. When examiners—let's say, national bank system examiners—coming from the [Office of the Comptroller of the Currency](#)—would come to examine national banks, one of the requirements in national banks was after the examiner had audited the books and said what the books were, the bank had to publish that in a newspaper. Then the bank had to clip the published newspaper article with the bank's balance sheet and send it to the examiner. Notice the interesting connection there. There's a connection between the fact that the examiner has the job of actually auditing the books.

- 5.5 Everybody can't go in and audit the bank's books. That's private information of the bank. But somebody has to audit the books. But then part of the examiner's requirement—the rules of requirement—is that the bank has to share that conclusion with the whole world. That is a really interesting combination, because what it says is the examiner understands that he has power, but that like Archimedes with the lever, his power is mainly coming from the fact that the truth that he is revealing is being revealed to the broader market.
- 5.6 And that market is where the real power is, because the market then responds to that information by creating consequences for the bank. So the real focus, when we think about bank regulation during this period, it's largely about auditing banks and making sure that the information about the banks is actually going to the market where the real consequences for the banks are being determined.
- 5.7 Banks have two different—I'm finally going to use the whiteboard over here—banks have two key variables. One is how much they lever themselves—so they can go to the stockholders and say, "Give me more stock." That doesn't happen very much. Or they can retain earnings so they can build up more stockholders' equity through what's called retained earnings. They can manage, to some extent, stockholders' equity directly by issuing more shares or by having more retained earnings, paying out less dividends. They do that sometimes, but this is not very frequently a variable that's changing. When they decide to make a loan, those loans often end up being funding the people who are borrowing as deposits. There is a bit of a correlation, not perfect, but a bit of a correlation between these two.
- 5.8 If a bank is suffering some deposit withdrawals, as depositors are saying, "Oh, there's a problem." The way that the bank deals with that is that it allows deposits to fall by not issuing new loans and instead accumulating more cash. So deposits are getting a little antsy. So what I do is I say, "Okay, the loans that are coming due, I'm going to just hold the cash." I'm going to use some of the cash to pay down the depositors who are coming in, and I'm going to use other of the cash just to accumulate more cash assets here.
- 5.9 Because my risk of default goes down if I have less deposits to equity and if I have more cash to loans. Why? Well, if I have less deposits, if you make fewer promises, you can't break them, right? So obviously, less deposits means less risk of default on deposits. And the more I'm holding riskless assets—reserves and treasury bills, which I'm calling cash—the less risk there is. So you can't have a decline in the value of it if it's cash. Cash is safe. So banks understood this. It's that simple. This isn't that complicated. This is what you manage.
- 5.10 And in finance theory, I'll show you a graph in just a minute that shows us that we can write down what we think of as **default risk** curves for different levels of **default risk** for banks that are simply the result of the combination of the ratio of equity to total liabilities and the ratio of loans to total assets.
- 5.11 As equity rises as a fraction of total liabilities, **default risk** goes down, and as loans rises as a fraction of total assets—or you could say loans relative to cash, either way—that increases the

default risk. Finance theory tells us, if you want to know what the **default risk** is, you have to tell me those two ratios. Of course, you also have to tell me what the riskiness is of each of the loans, but let's not get too technical. It's going to turn out that a very useful way to think about these levels of **default risk** in the banking system is if you just know the ratio of cash to total assets, and you just know the ratio of equity to total assets for a given level of lending that you're doing. Here that is.

- 5.12 What I want to do is use this language to start talking with you about all of these episodes, because we can observe this. We can observe this as it's evolving, as it's changing. And so we can see how the private market is reacting and how banks are managing their risks as problems unfold because we have the information about the bank's balance sheets.
- 5.13 Okay, so let me go through this and show you how powerful this can be. Here's the ratio of **equity to assets** on the vertical axis here. On the horizontal axis, I've done something a little bit different. I took the riskiness of loans and I multiplied that by the loan-to-asset ratio. I used the stock market as a way to estimate this.
- 5.14 What I'm doing over with $S_{sub A}$, it's the riskiness standard deviation of returns to assets is what $S_{sub A}$ means, but it's basically using the information over here and from the stock market to infer what's the total risk on the asset side. The main thing that banks are using to change that horizontal axis is the ratio of cash to loans. When you see this axis, you may be a little confused because there's a lot of finance theory and the **Black-Scholes** model and a lot of other things that are driving the bus here. But what's really causing the change in $S_{sub A}$ is largely banks' decisions about how much cash to hold relative to loans.
- 5.15 On this axis, E/A is mainly changing. Equity over total assets is mainly changing as deposits are moving around. Banks aren't moving equity that much—so the main source of variation. When you see moving from a point like X to Y and from Y to Z, those movements are mainly reflecting changes in deposits on this vertical axis and changes in cash relative to loans on the horizontal axis. But I didn't want to label it that way because that's not quite right.
- 5.16 We're now going to look at from about 1920 till about 1936. X is going to be about 1929—January 1929, before the **Great Depression** starts. What you see here is banks are holding on a market value basis. The value of stockholders' equity is about on average for New York City banks during this period, about 35% of the value of assets. That's very high compared to current standards, where that number would be maybe 10%. I'm telling you, that's a very high number relative to what you're used to today. About 35%. Then similarly, the riskiness of assets is indicated by an $S_{sub A}$ value of about 15%. But what's really relevant there is: that's accomplished with a cash to asset holding ratio of about 25%.
- 5.17 When the **Great Depression** hits, the shocks of 1929 through 1933 are acting to push the banks out in this direction. The shocks—why? Because when the recession hits, you know that you're going to suffer losses. Let me come over here. You're going to suffer losses on some of your loans. What that's going to do is cause a decline in your assets. But because your deposits don't

change, because they're senior claim on your assets, the main thing that's going to change here is equity over assets is going to fall.

- 5.18 Equity is the shock absorber. When loans fall, total assets fall in value, and the main thing on this side that's changing is, stockholders equity falls. Why? Because deposits are a senior claim on assets, and therefore, when there's a downward movement in assets because of loans falling in value, that doesn't cause deposits to decline. That only happens if the bank gets into serious trouble.
- 5.19 So what's going to happen when the recession hits? The first thing that's going to happen is because loans are declining, equity is going to decline, and that means equity over assets is going to decline. Second thing that's going to be happening is the recession, as people see this decline in loans, their perception of the riskiness of the individual loans is going up. That means that the $S_{sub A}$, the measure of risk, coming from market perceptions, is going to be higher.
- 5.20 Back to this graph. The first thing that's going to happen to you when you get hit by the Great Depression is your perceptions of your risk are going up, the risk of your loans, and perceptions of the value of your loans are going down. The reduction in the value of your loans is causing your equity-to-asset ratio to fall, and your overall perceptions of risk of assets to go up. That's pushing you in this direction.
- 5.21 Now, this first line here, you can see it says P equals one basis point. What that's telling you is that based on using the Black-Scholes model in the stock market—I'm not trying to say it's extremely accurate, but it's helpful, right?—as a reference point, we'll call that a one basis point, actuarially fair, physical default risk. That means that if you wanted to compensate a deposit for bearing that risk, you'd pay them the treasury bill plus one one-hundredth of a percentage point. Very low, right? We think of this as very low risk at point X.
- 5.22 But as I get pushed in this direction, look at this next curve coming out of the Black-Scholes model. It says, if I'm on any of the points on that curve, I'm at a 50 basis point, actuarially fair, physical default risk compensation. So as I get pushed out by loan losses that are associated with all sorts of perceptions of increased risk, notice that deposits are facing more default risk out here than they were here. That's the first point I want you to see.
- 5.23 But banks don't like that. Why don't banks like it? Because depositors don't like it. What happens when the banks get pushed in this direction? The depositors in the New York City banks start doing some interesting things. If you're at a relatively risky bank, they move their money. Where? To Canadian banks that have offices in New York. Interesting, huh? They knew that. They knew Canadian banks were not as risky. Or to postal savings accounts. Back in this period, the post office operated its own savings system. My mom, for example, had a postal savings account. She would take a dime to the postal savings, put it in, and she accumulated maybe, I don't know, \$20, \$30 over a long period of time. And you knew that that wasn't going to fail because it was guaranteed. So the postal savings—and it was backed by treasury bills, effectively.

- 5.24 So as you're being pushed this way, depositors are starting to say, "Well, that's starting to look a little risky." Now, don't think of this as runs. None of these New York City banks failed. None of them failed during the Great Depression. What they did instead was they adapted to the discipline coming from deposit withdrawals. None of them failed.
- 5.25 So as the shock of the Great Depression is hitting you, causing loans to fall and risk perceptions to increase, the banks are getting pushed out there. They can't stay out there at point Y. They can't stay out there because depositors are leaving the bank. And the banks don't want the depositors to keep leaving to go to postal savings or Canadian banks. So what do the banks do? They start managing the risk. And that eventually leads them to a point like this, point Z. How did they get from point Y to point Z? By reducing their loans and increasing their cash.
- 5.26 Notice they didn't really increase their equity-to-asset ratio. Why? Because it's a bad time to issue stock. It's the middle of the Great Depression. People don't think, If I go to issue stock, everyone's going to think, I must be really in bad shape if I'm trying to get people to give me stock right now. So they didn't issue stock. Instead, what they did was they decided to stop rolling over loans and to accumulate cash, and that brought them back to Z.
- 5.27 Now I'm going to show you this in a table of numbers so that you can see everything I just told you. You've seen it in the balance sheet, the basic idea. You've seen it, what the actual evolution was in the graph. Now, let's look at a table. If you look here, basically, loans are 3.3 times cash assets in 1929. But by the time you get to 1940, look at that, loans are only 0.3 of cash. What that means is cash is about 75% of your total assets. So the banks went in New York in 1929 from having 75% of their assets in loans to having 75% of their assets in cash. Because that was the way to convince depositors that they were managing the risk as the Great Depression shock got more and more severe.
- 5.28 I want to emphasize, none of these banks failed. You can see the P-value of their implied default risk going up on average to 33 basis points to 41 basis points, but then falling back because through risk management, they were able to communicate to depositors that the risk isn't something to worry about. But that was a dynamic process. And that's what we mean by market discipline. That's what we mean by bank risk management, responding to depositors' concerns when depositors are risk-intolerant.
- 5.29 What does it mean to be risk-intolerant? The depositors want this number to be zero or one or two. They don't want it to be 30. And if you try to convince them to take 30, they move to postal savings or they move to Canadian banks or to other New York banks that are doing a better job. That's what market discipline is.
- 5.30 The other thing that I found, and I've done a lot of work in different developing countries, advising people in bank regulation. One of the things that I found is very powerful is to show people market discipline by looking at the liability side of banks. What do I mean by the liability side? Looking at the flows of deposits. We're going to rely on this a lot when we talk about Florida and when we talk about deposit insurance in the '19-teens and 1920s.

- 5.31 Depositors leave banks. Believe it or not, if you looked at, let's say, Argentine banks during the 1990s, you'd be able to see **depositor discipline** operating through deposit withdrawals because the banks were not insured very much in the 1990s. **Deposit insurance** was not very strong. And so depositors had a strong incentive to withdraw their money whenever the banks looked like they were increasing in their risk. And you can see it in two aspects of the deposits. We're not going to show that here today, but I'm going to talk a little bit about it.
- 5.32 What you see is when the risk goes up, as this analysis would suggest, deposits flow out. It's not a run. It's just deposits start moving out. Every week, a few percent moves out. What else do you see? Those banks that had the perceived shock see a little bit of an increase in their interest rates, and in some cases, more than a little bit. Because some depositors are willing to stay, sometimes it's **wholesale depositors**— that is, bigger people are willing to take a little bit more risk. They're not necessarily completely **risk-intolerant**. But then what are they going to insist on? Compensation, right?
- 5.33 What we see is when risk goes up, the **risk-intolerant** depositors leave, and the more risk-tolerant depositors that stay, what do they do? They charge more money. If you want to know, is a banking system engaged in **market discipline**? That's what you do. You can look at that because guess what? These are things that we can observe, and they don't tend to have hanky-panky.
- 5.34 Now, a lot of government statistics in banking, like, Oh, how bad are the loans? Hanky-panky all over the place. Government supervisors are politically motivated to lie, lie, lie, lie about loan quality in today's world. Constantly, predictably, very severely. But they don't lie about the total deposits, typically, and they don't lie about the interest rate paid on deposits. So if you want to see **market discipline** operating, if you think it might be, that's the place you look. Let me try to show this to you a little bit what I'm talking about.
- 5.35 So let's look at, during the 1930s, banks in Chicago. Chicago has a crisis in June of 1932, where a bunch of banks fail. As you can see, 46 banks fail, but 62 banks in Chicago don't fail. After they're failing in the summer of 1932, we can ask some questions about comparing these two groups. The ones who failed in the summer of 1932, what interest rate were they paying in 1931 to their depositors? They were paying a full percentage point more interest to compensate for risk.
- 5.36 Okay, what was their ratio of **wholesale deposits** to total debt? Because I mentioned what happens? The deposits who are **risk-intolerant** leave, and then the depositors who aren't **risk-intolerant** stay but get paid more, right? Well, for the ones that are failing, 12% of their depositors in 1931 were wholesale kinds of depositors. Whereas for the banks that survived, only 2%. Why? Because these guys in the failure column lost a lot of the **risk-intolerant** deposits. So the **risk-intolerant** deposits left, the risk-tolerant deposits got paid more. Makes sense? That's **market discipline**.
- 5.37 On total, of course, the ones who were in trouble lost a lot more deposits in 1931. I want to emphasize, all three of these numbers are 1931. The failures happened in June 1932. Don't get

confused. This is long in advance of the failures. The market is seeing the riskiness of this group and distinguishing it from this group a year in advance, or more, of the actual failure experience. That is what **market discipline** is about.

- 5.38 I mentioned Argentina before. There were several banks in Argentina that failed in 1995. They were mainly banks that funded themselves with **wholesale deposits**, not **risk-intolerant** deposits. They failed in 1995. What interest rates were they paying in 1993, two years before their failures? The ones who failed were paying 13% interest. The ones who didn't fail were paying 9.5% interest. You see that even, and by the way, there were few failures, and Argentina did quite well in 1995. **Market discipline** functioned quite well in getting banks to manage their risk properly.
- 5.39 But what I'm trying to show you here is that the theory of **market discipline** that assumes that banks and markets are involved in a conversation about risk and risk management has a lot of support in history. We're going to see it again when we talk about Florida in the 1920s and when we talk about agricultural failures in the 1920s elsewhere.
- 5.40 I also worked in Mexico in 1996. What's really interesting to me about this experience is even though Mexico had given, as we'll discuss more tomorrow, had given 100% liability insurance for all liabilities of Mexican banks, even though they had 100% liability insurance, here's the catch. 1996 was a time where politics was very unpredictable. The **PAN** had just won, for the first time, the lower house in Mexico. The **PR**I had just lost, for the first time, the lower house. People knew that things were changing permanently in Mexico. The **PR**I dominance era was leaving.
- 5.41 There was a lot of sense that the deals that had been made with the banks were very unfair. Even though there was de jure, as a matter of law, 100% **deposit insurance** for the banks and liability insurance, there were some real concerns that that wasn't going to be honored. What did that mean? All of a sudden, **market discipline** came back into effect. Even though the banks were de jure, completely insured, de facto, they weren't. What we find is— I was working for the Mexican government at that time on various things, first as a **World Bank**, part of a **World Bank** mission, and later working for the government directly.
- 5.42 And so when I say **Banamex**, which is one of the Mexican banks, was marginally solvent, I think that's true. And when I say **Bank Serfin** was deeply insolvent, again, this is now my opinion, only my opinion. So you have to take it for given right now. But the official accounts, of course, were not revealing any of that. But it was officially not the case that this was deeply insolvent, although everyone in the government knew that it was true.
- 5.43 What I want you to notice is **Banamex** was paying 17% interest on its funds, and **Bank Serfin** was paying 29%. But all these funds were 100% insured by the government, but not necessarily de facto. So what you see is other evidence, also, after the crisis, people that were holding these deposits at **Bank Serfin**, they weren't holding checking accounts. They were holding mainly wholesale funding.

- 5.44 And so Bank Serfin, as you can see, after the crisis, a lot of banks in Mexico expanded their deposit network quite substantially. So you can see—and the reason for that is lending was not profitable, was not a good business. But if you could get depositors to give you deposits that you paid low interest on and you hold treasuries, you can make a big spread business on that. And so there was a big advantage to establishing new branches. And in Mexico, if you look at it, over this period, the number of branches increased by about 1,200. That's a big change, like a 20-some-percent increase in number of branches.
- 5.45 But Serfin, which was insolvent, had almost no increase in branches. Why? Because they couldn't attract. They knew they couldn't compete for checking accounts, for retail deposits. All they could do is get wholesale, speculative money, which they pay really high interest on. Again, showing you about risk-intolerant depositors. When risk is really high, you can't get their business.
- 5.46 Okay. Now, what I want to do is turn to the Great Depression. What I want to emphasize that makes the Great Depression in the US different from all the episodes we've been talking about—the National Banking Area crises, the other countries crises—is, from the US standpoint, this is a time of very large bank loss, but still, total cumulative negative net worth relative to GDP, about two and a half to three percent. It reflects a major and persistent shock to the economy called contractionary monetary policy by the Fed. That's what makes it so different.
- 5.47 The other thing that makes it different is that's happening in the context of a unit banking system. So you have a fragile banking system, and you have a major shock. And so that's what made this unusual. The shock continues, and it's not like the typical national banking era, little recession shock happening at a seasonal time of vulnerability leading to a temporary concern which gets resolved in a matter of weeks or sometimes a couple of months. No, no. This is starting in 1931, going through to March 1933. We have, and even before then, persistently negative monetary policy actions leading to these banks suffering huge losses.
- 5.48 Now, why does contractionary monetary policy lead to losses of banks? Because it makes GDP fall, it makes national income fall, and also it makes prices fall. So if you're a borrower and you don't have any income or you have a much reduced income, that means you won't be able to pay back your loan. Let me also point out that if prices fall, the value of your debt goes up. I owe \$100, and then there's a big deflation. Oh, wow. Bread now only costs 10 cents; used to cost a dollar. So we've had 90% deflation. Unfortunately, I still owe the same amount of money in nominal terms that I borrowed. How do I get that money? How do I get the money to pay what I owe? Incomes are falling because prices are falling.
- 5.49 So the Great Depression is all about a monetary contraction, and in a very vulnerable unit banking system. How does it end? When President Roosevelt gets elected, there's really just havoc everywhere. And what he does is he says, "I'm going to declare a holiday." It sounds like a fun time. It's not. He says, we're just going to—no more banks are paying out. What we're going to do right now is do a triage operation. We're going to have the examiners audit the banks and put them into three categories.

- 5.50 Category one, you're fine to reopen, and some of those banks reopened immediately. Category two, you're not fine to reopen, but you're a marginal case. We think that with a little bit of help, you can reopen, and you're going to be a candidate for a new program that we're creating called **Reconstruction Finance Corporation: Preferred Stock Assistance**. The government's going to give you a little bit of preferred stock, which is a liability in between deposits and stockholders' equity. It's junior to deposits, it's senior to stockholders' equity, and it's going to help you retain your deposits because it's junior to them. That means that the deposits are in a superior position to it. It can help bolster from a depositor's perspective, the riskiness of your bank. However, when you get that assistance, we're also coming onto the board of your bank. We're also going to be requiring you to come up with a plan for how you're going to manage your risk to get back into really the perfect situation.
- 5.51 Then there's the third category: hopeless. We're not wasting any assistance on you. You're too deeply in the tank, and we're shutting you down. They didn't decide all this at the same time. They looked at the banks. If the bank was clearly okay, they allowed it to reopen immediately. If it wasn't, then they had to decide whether to give it assistance or not. You get the idea. The whole thing took a matter of months, but some banks were able to open almost right away.
- 5.52 This was credible because the government was giving some assistance based on it, and because it was also happening in the context of a world in which people were familiar with a credible government examination process. It was, I think, a very successful case. By the way, if you're thinking right now—which now being June 2024—after we had this crisis just a year or two ago, in the US, that came from the fact that we had big shocks from interest rate increases to the solvency of many banks, finance professors who have been looking into this claim that there are about 600 banks right now in the US that are teetering on the brink of insolvency.
- 5.53 Why isn't the government doing something about it? Well, that's a good political question that you could ask. It's not forcing them to recapitalize. It's not forcing them to cut down their risks—for political reasons. Then when we get another **banking crisis**. We had **SVB**, and we had a few, three other banks that went down in 2023. But you might say, "But did we solve the problem?" No, we did not solve the problem. Because solving the problem is politically difficult. It requires you to actually do what **Roosevelt** did: establish a credible process for shutting down banks or forcing them to recapitalize. We didn't do that.
- 5.54 We're now sitting on a time bomb of about 600 banks, mainly small and a few medium-sized banks that look like they will explode. But I want to point out, when we think about the predictability of **banking crises** today, which is very distinct from this historical period we're talking about, we always come back to the politics of what decision we made about tolerating insolvency.
- 5.55 I've already mentioned that, here's a list of questions: what shocks precipitated the **Great Depression**? It's clearly **contractionary monetary policy**. You might say— I'll skip to this question—why did the **Fed** keep doing it? Why didn't they stop doing **contractionary monetary policy**? **Allan Meltzer** has shown in his History of the **Fed**, that the **Federal Reserve** officials were

completely confused and believed that they were not contracting **monetary policy** because they were looking at unhelpful signals of what the effects of their actions were. So they saw interest rates falling. They said, "See, that means we're not tightening things. Things are loosening."

- 5.56 But the interest rates were falling because the deflation was so great. The nominal interest rate is the sum of the real interest rate plus the rate of expected inflation. And so if the expected inflation gets to be deflation, nominal interest rates are falling. And **the Fed** looked at that and said, "Oh, that means that we're doing things that are easy. We're having an easy influence on the markets," which was completely wrong.
- 5.57 They also had another signal that they looked at called **net free reserves**. And **net free reserves** were showing them that banks were accumulating more reserves. And they said, "See? Banks are accumulating lots of reserves. That's an indication that policy is loose." But why were banks accumulating more reserves? Because they were managing their risks and because **the Fed** was trashing the economy, in order to keep their depositors, they had to have lots more reserves, like those New York City banks that went from 25% of their assets in reserves in 1929, January, to 75% in reserves by 1936.
- 5.58 The **Fed** simply misunderstood, and it saw this easing even as potentially a problem. So they kept pressing on the gas pedal, or you might say pressing on the brake, and didn't understand that they were causing incredible distress.
- 5.59 Was the depression the result of **panic** or fundamentals? I'll show you some evidence. It's clear that the banks that are failing are the relatively weak banks, on fundamental grounds. So bank failures are reflecting fundamental weakness. I showed you some evidence from the Chicago 1932 failures. It showed that they were observably weak even in 1931. But I have lots of other evidence on that. I'm going to go through it pretty quickly. So you'll have to trust me—based on the papers that we're going to discuss—I won't go into too much gory detail, but you'll see.
- 5.60 What explains the depth and **duration** of the depression? Why did it take so long to recover from it? Part of the explanation is this trashing of the banking system and the trashing of the balance sheets of companies. Because remember, if I borrowed \$100 and then there's a big deflation, that means in real terms, I now owe a lot more than \$100. So even if I'm able to continue to stay solvent and pay off my bank loan, I now am operating under the burden of a huge amount of debt, which might make me much less able to pursue new projects.
- 5.61 So the Fed's shock had lasting consequences through the way it destroyed the banking system and the way it destroyed the financial health of borrowers. **Irving Fisher** called it the "**debt-deflation**" explanation of the **Great Depression**. What effect did the Depression have on bank regulatory policy? So you might say, well, it should have at least gotten everybody to say, "let's get rid of **unit banking**." Now, I pointed out some states did relax a little bit their **branch banking** policies.
- 5.62 But actually, the national government, for the most part, came up with a completely false interpretation of the Depression. It said it was the fault of the New York City banks, and the

financial power, and all that stuff about the cities. And the main thing that the Federal Reserve Board in Washington, DC, took away from this was they got a new power that Congress asked them to stop bank consolidations, to keep banks small because big banks were very powerful and risky.

- 5.63 It's strange, but not that surprising when you see how the persistence of bad ideas is a political outcome. Even the Great Depression, which caused some relaxation in some of the state branching policies, basically led the Federal Reserve Board to have a new power to prevent mergers of banks.
- 5.64 Here's some graphs that give you a sense of how much personal income is declining during this period. This is from Friedman-Schwartz's classic 1963 book. Industrial production is falling, wholesale prices are falling. So this is what we mean. This is what a disaster looks like over a very long period of time. And when we look at—I mentioned Irving Fisher's debt-deflation discussion—and so you can see that corporations are seeing a big increase in their real debt as deflation is happening.
- 5.65 So even though the economy is going south starting in 1929, the debt that corporations are holding is going up in real terms. And of course, there's not just the crash in 1929, but a failure of stocks to recover, which is reflecting those troubled situations in terms of income falling, prices going up, and generally, corporate balance sheets being less healthy. If you look at pricing in the debt market, you can see, of course, that bond yields are also jumping during this period. You can see the effect on this debt-deflation and the increased size of debt burdens on corporations are reflected, of course, in big increases in credit risk spreads in the bond market.
- 5.66 If you were going to build a model of this to explain the contraction, or to describe the contraction of the money supply, here's Friedman-Schwartz's data showing the contraction of the money supply and showing that it was mainly driven by a contraction in deposits, that actually cash is going up a little bit during this period. But deposits are falling dramatically relative to currency and relative to bank reserves.
- 5.67 If you wanted to build a microeconomic model—which we did, Joe Mason and I, in our various papers, looking at the Great Depression—we found that some of the most useful tools for measuring the distress that banks were experiencing were local economic conditions such as building permits, other kinds of indicators of what activity's happening, or business failures, local business failures. And that when you did that, you find that you're really explaining the failures pretty well just based on these kinds of fundamentals.
- 5.68 This is a graph that allows you to see how there are some times, it's not uniform, but there are some times where liabilities of failed businesses—which is this graph here—is tracking pretty well the liabilities of the actual bank failures—the probability of failure of banks, which is this lower graph. So you can see they're moving together very closely. And that's the aggregate. And when you break it down and look locally, you get very good explanations of the locations where bank failures are happening and the particular banks, because we have individual bank data.

- 5.69 So we can sort out what's driving some of the banks to be able to survive, even in locations that have problems, whereas other banks aren't surviving. And what do you guess it is? So if you're in a place that's hit hard by problems in construction or problems in business failures, but you've been fairly conservative in holding lots of cash and holding lots of equity relative to assets, you find that you're more likely to survive. We combine the macroeconomic data with the microeconomic bank exposure to risk, and we find that we can really understand who's failing and why. Which, of course, again, reinforces how much of this could have been avoided in a nationwide **branch banking** system.
- 5.70 This is from the Chicago paper. I mentioned that we got the survivors and the **panic** failures, that you can see that the failures in 1932 are also the stock market is valuing their shares much lower, long before the failures here in 1931. The survivors are understood by the stock market to have much higher value assets. Not only can we explain it **ex-post** statistically, but we can observe that the market understood it. Of course, I've already shown you the market understood it in terms of flows of deposits being different between the two groups and total interest that it's paid on debt being different in the two groups. This is a slide you've basically seen before in a little less detail.
- 5.71 Now, what I wanted to show here is, so this shows the predicted failure, cumulative bank failures, predicted, and then based on a model that's just based on fundamentals. Then if I put in some dummy variables for a particular moment, I can see how far did the predicted deviate from the actual. You can see during what some people have called a **panic** period where things were irrational, what we find is not much deviation. We're explaining almost all of the explicable bank failure risk.
- 5.72 Whereas when you get to 1933, there it's much more of a massive shutdown of banks, even if they're not weak. So the model does a great job until the very last few months when things are so bad off that the market really can no longer distinguish, is not interested in distinguishing safe from unsafe banks. And that's when you get FDR's intervention with a **bank holiday** that basically says, "Okay, we're shutting everything down, and now we're going to come in and do our audit." Then the auditors will say, "Are you safe or not?" Put you into one of those three categories. That makes sense from the context of this graph.
- 5.73 This is also showing that a lot of people would say, "Oh, depositors can't distinguish." If a bank fails next to another bank, then everyone runs on the bank that's next to the failed bank. What we found is that's really not true, that adding the banks that are near you as an influence has a very minor additional explanatory power. People might learn a little bit from that. They might say, "Well, if that bank failed, that means that maybe there's a common shock for the other bank." I think that's true. But that wasn't really the dominant source of information in our model. It was mainly coming from observable macro-characteristics and observable microeconomic balance sheet characteristics of the banks. It's really allowing us to explain what's going on.

- 5.74 There's another element, which is that it was particularly important how you were getting your cash. New York City is at the top of the **pyramiding** of bank reserves. Imagine now you're a bank and you and your depositors think you have a lot of cash, but some of that cash is held at a New York City bank. And then the New York City bank says, "I'm sorry, but I'm suspending **convertibility**. I can't give you access to the cash." That is a major problem, right? And maybe some depositors even anticipate that problem and say, "You've got so many reserves as cash that are held in New York that I'm worried that New York, if it suspends **convertibility**, I better get my money out now."
- 5.75 So in addition to the fundamentals, there's some evidence that the **pyramiding of reserves**—again, a consequence of what? The **unit banking** system—adds to the jitteriness of depositors because they realize that there's some **liquidity** risk that comes from the interbank, the loss of access to interbank reserve holdings through the pyramid of reserves.
- 5.76 That is, if you're a bank, you're holding accounts in New York—suppose that I'm in, let's pick one, Indianapolis. If I'm a bank in Indianapolis, then the rural Indiana banks have accounts with me, and I have accounts with the New York City and Chicago banks. All of a sudden, those rural Indiana banks start saying, "he has a lot of his money in Chicago and New York. I better take my money out because I'm worried that my money is dependent on his being able to access Chicago and New York."
- 5.77 So we're able to show through empirical analysis of this, a couple of papers are, that this was actually a major magnifier of fundamental risks. As banks got more fundamental risk, the **pyramiding of reserves** actually magnified depositors' concerns about fundamental risk.
- 5.78 So what does that mean? That means there's yet another way to understand why **unit banking** was a destabilizing choice. Because **unit banks** had to operate their reserve balances through this **pyramiding** system where they had a lot of money in the major financial centers like Chicago and New York. And that added to the fragility through a new kind of **liquidity** risk problem of the **unit banks** operating the periphery.
- 5.79 When people have looked at the US, in comparison, let's say, to the UK, one of the things that they've emphasized is that it was the fact that the **monetary policy**— and the commitment to the **gold standard** was part of that, but it was really mainly the Fed's actions—that led the US to such a negative outcome in the **Great Depression**. Whereas the UK leaves the **gold standard** in September 1931 to be able to make possible more expansionary **monetary policy**. You can see they have a relatively mild experience in the **Great Depression** in terms of the contraction of industrial production. I want to emphasize, yes, the Depression was a global phenomenon, but not all countries suffered the same as the US because not all were pursuing the same kind of **monetary policy**.
- 5.80 I mentioned that the trashing of the banking system was a major contributor to the severity. It had this persistent effect, because once you've lost your banks, how do you provide credit locally? Then how do you get growth restarted? This is from another paper I wrote that looked

at trying to measure how the loss of local bank credit had long-term effects on the ability of places to start growing.

- 5.81 And what we found was that there were very major effects, persistent effects, through 1935 that came from—and I've decided I'm not going to review this, but just summarize it—that shocks to local banks in the early 1930s, because they led to such contractions in the supply of local credit at the state level, produced major persistence of decline of income for several more years. Again, another effect of the unit banking system is that it made this Great Depression local shocks much more persistent than they otherwise would have been.
- 5.82 We always try to ask after our discussion of all these crises, what lessons we expect might have been learned if you adopted, let's say, some evolutionary perspective and whether we learned them. The main point, of course, was, well, unit banking is a terrible system, but we knew that. We knew that a long time ago. We knew that in the 1830s. We knew it in the 1850s. We knew it after these six crises. We knew it after the agricultural problems that we're going to be describing.
- 5.83 It's strange, but maybe not surprising that we didn't learn to get rid of the unit banking system as a result of this crisis. Remember, it's not until the 1980s that we really see that progress, because it's the 1980s where the distress, like the Great Depression, is happening, but alongside changes in the political equilibrium that are making the decision possible. If you want to see learning, learning requires not just observing the economics, but having the political bargain shifting.
- 5.84 Okay, what I want to shift to now— although it's actually you'll see very closely related in terms of this discussion we're having about market discipline in particular—I want to shift now to something closer to home because we're filming this in Miami. I want to talk about the first major nationwide land boom in the US. When I say nationwide, I don't mean that Florida is the whole country. What I mean is that the Florida land boom was experienced by the whole country.
- 5.85 Everyone was aware of this new idea, which was: there's going to be a new garden of paradise where people will be able to retire and have beautiful homes in lovely weather. I know it's a little warm today, but overall, the weather is pretty good here. Lovely weather, enjoy their lives, and they're so abundant that maybe many people, maybe millions of people, will be able to move here almost immediately. We're going to talk about where that came from, how that happened.
- 5.86 But the first thing I want to say—and I'll come back to this in a minute—they were right. Here we are, right? It's not that they were wrong. Maybe they thought it was going to happen a little faster than it did. Maybe they didn't appreciate how getting the infrastructure set up to make all the things happen in Florida would take more time than it did. They thought it was all going to be lickety-split, and it wasn't lickety-split. But fundamentally, I just want us to start by recognizing that they weren't so wrong.

- 5.87 Okay, so when people look at this, we're going to do a **deep dive**. We're going to look into what banks were up to, what was happening in land markets, what were people thinking, to try to do as we always do with historical crises, try to understand from an **ex ante** standpoint what's going on. What does it look like when someone's not doing a **deep dive**? Sometimes making fun of the people who lived through the episode.
- 5.88 Here's **John Kenneth Galbraith**, not my favorite analyst of **financial crises**, who's using this as typical looking down his nose at these poor, unwashed masses who aren't as smart as he is. "The **Florida land boom** was the first indication of the mood of the '20s, the conviction that God intended the American class to be rich." he's saying it's the arrogance of the people of these unwashed masses to have believed that they were somehow destined for this beautiful paradise that is explaining why everything went wrong.
- 5.89 Now, of course, as I've already pointed out, they weren't wrong. They were right. This is a place where people who aren't even very wealthy can come and have a pretty amazing retirement. That is exactly what people were thinking in 1922, 1923, 1924, et cetera. Of course, it didn't work out very well. "Dead subdivisions lined the highways. Their pompous names half obliterated on crumbling stucco gates." By the way, they were names. We'll talk about that.
- 5.90 "Lonely streetlights stand guard over miles of cement sidewalks." Also quite accurate. In other words, when people were developing this, they didn't build a bunch of houses. What they did was they built sidewalks and streetlights and sold lots and then made people think, because it was reasonable to expect, maybe, that only after people started moving to Florida would the houses actually be built.
- 5.91 So try to imagine that what's really being constructed is really sidewalks and streetlights. By the way, this happened again. I think it was in the '60s. If you go to West Coast, a little bit north of here, there are whole areas where there are these enormous concrete things of streets being built and all that that never had any houses built on them. That's the way this developed, "where grass and palmetto trees take the place of homes that were to be." It's very poetic. In fact, that is quite true.
- 5.92 Now, let's get a little bit more scientific about this. As I say, it's the first nationwide land boom in the sense that people are purchasing homes here because Florida is a national attraction, and they're purchasing homes from all over the country. By the way, some of the people are buying homes sight unseen. They didn't even come to Florida.
- 5.93 Many of the people are buying homes simply by mail based on seeing advertisements in the newspaper. What do those advertisements look like? Well, they're not photographs because there's nothing to photograph. They're sketches, sketches that are pretty sketchy about what the future could be, might be. You're going to see some of those in just a minute.
- 5.94 What was driving this? What was driving this was, first of all, there was a railroad boom, as all who live here might recognize the name **Flagler**. **Flagler** builds a railroad that makes it, for the first time, really accessible to a lot of people to come to Florida. There's also lots of new

technology relating to land. What that new technology means is people can now, for the first time, fill in a massive scale swamps.

- 5.95 People can also build coastlines that they want. You don't like the coastline? Build another one. Extend it. This is no longer science fiction because we now have the technologies and the equipment to do this. This is really interesting because what it means is even the physical facts of Florida are not given. They're now to be determined technologically.
- 5.96 You might say that's an enormous positive, but it's also a huge uncertainty because how do you know what this thing called Florida that no one's ever experienced? How's that going to take shape going forward exactly? How much land is there going to be that's inhabitable, really? How much people in the aggregate really want to move to Florida?
- 5.97 That first question is about supply. The second question is about demand. I'm going to show you a graph in a minute that shows how confusing that is, that no one could have possibly known the answers to that. But that's coming from the vast improvements of technology, the establishment of the railroad so that people can be here, making it possible. Secondly, all these new technologies.
- 5.98 Now, there's also no data aggregator here. If you look at— we've tried to collect data on housing starts and other kinds of building permits and various things like that, pretty much it's Jacksonville, Miami, and Tampa. The rest of Florida is also developing, but there's no ability to do any data aggregation outside of those three urban areas.
- 5.99 Another thing to keep in mind is if you wanted to do a very technical analysis of the development of the Florida land market during this time, it would have been almost impossible. You could have had an opinion about Tampa, Miami, and Jacksonville, but you wouldn't have really been able to aggregate the fundamental conditions in any useful sense across the whole state.
- 5.100 Furthermore, you didn't really have, even within those locations, very useful data. If you were a developer, for example, you didn't know what other developers were doing. We all take these statistics that we currently have for granted. There weren't any. So, you're trying to figure out what's going on in Florida. How much should I pay for land? What I know is Flagler built a railroad. These new technologies exist. I'm seeing some ads in the paper. It sounds like this may be my only opportunity before the prices go through the roof to have a retirement home.
- 5.101 What would you do? I probably would have bought one. Because if you're a middle-income person, this is your chance, right? This is it. And maybe those prices are going to go up very high. I'm not saying I would have, but I'm saying before you make fun of people, think about it. All of these facts are real facts. It didn't have anything to do with being arrogant. It had to do with, wow, something's really going on in Florida.
- 5.102 Yes, it seems strange to buy a house in a place you've never visited based on advertisements and market sales tactics by, and who knows, to trust real estate developers. Of course, it's not a

good idea, generally. But you don't have a thousand choices. You don't have a lot of data. You make your choice. That's the perspective we want to try to bring to understanding the land market.

- 5.103** And then furthermore, in this paper, what we want to do is try to understand the bank's behavior. So, did banks go cuckoo? Or did banks continue to operate using sound principles like this? Those are the questions that motivated us. How do we think about this land market and how crazy it is or not crazy it was? How do we think about how crazy the people who funded the growth of the land market were? And why did things come apart and can we tell a story that takes us from **ex ante**, fairly reasonable actions, to the actual **ex post** outcomes that we observe?
- 5.104** What we find is, yes, that banks were funding a lot of the growth here through their deposits. But what's interesting is banks were maintaining very conservative risk management practices in doing that, with the exception of a few banks which did end up failing, who were heavily influenced by and, in fact, managed by a particular **chain** of banks. It was a part of a particular **chain** that were effectively involved in fraudulent activities with the developers themselves.
- 5.105** To make matters worse, the supervisors were complicit in the fraud. The government, both in the national banking supervision and in the local Florida supervision, seemed to have been complicit. The story that we end up telling is not a story of craziness. It's a story of, first of all, with respect to the land market, huge **ex ante** uncertainties that had to do with an unprecedented thing happening where people couldn't have possibly known whether to think that the prices that were being discussed were reasonable or not. No way to know.
- 5.106** Secondly, banks were actually quite conservative, with the exception of some conflicted banks who were operating corruptly and with the complicity of some government officials. That's our story. I have a few minutes in which I want to just go through a little bit of this to show you the scale of it and to give you a sense of how we reach those conclusions.
- 5.107** First of all, something like 20 million lots were sold in Florida. Lots. That's half of the US population. In the 1920s, half of the US population thought it was going to have a place in Florida and bought it, as far as we can tell, or at least put down down payments of some kind. Some land prices, this is anecdotal; we don't have an index of land prices. We don't have it. There's no data that would allow us to say what the prices of these transactions were. It doesn't exist. But anecdotally, in some locations, land prices rose during the boom by 1,000 times. Again, anecdotal. Let's just say we know land prices went up a lot during the boom.
- 5.108** There was about \$900 million in deposits that flowed into Florida from other banks outside. The growth of the Florida banking system during the boom was enormous. Of course, those deposits found their way into helping to fund some of the development that was going on. In the end, about 10% of Florida's banks ended up suspending or failing, mainly failing in 1926. Almost all of those were about half of the banks that were part of something called the **Manly-Anthony** bank **chain**. A **chain** means the same owners of multiple different banks.

- 5.109 The Manly-Anthony group was—Manly and Anthony were the main people running it. They had their group of stock investors. They did investments in a bunch of banks, and then it turns out they used those banks to take on a lot of risk to fund their own operations. They were able to get away with that largely because of the complicity of examiners whom the depositors were dependent on.
- 5.110 And so what we want to do is look at, well, were banks in general operating badly? And it's hard to fault the bankers or their depositors for the Florida boom and bust if they didn't end up failing and they didn't end up losing a lot. And if the losses were all pretty much concentrated in this one fraudulent group. And by the way, I want to be careful about using the word fraud. They fully believed in what they were doing. So when I say fraud, fraud was what happened after they had lost everything and they were trying to keep the game going a little bit longer.
- 5.111 It's not that they were committing fraud from the beginning. They put all of their money into this. They were completely convinced, and they knew more than anyone else. They were completely convinced that this was a real boom, and they invested in it fully. Then, as they're losing their shirt, that's when fraud kicks in. That's when you start trying to lie about what the conditions of your loans are or whether you're making real estate loans with your money or not, which was prohibited or limited.
- 5.112 There's a lot of that action happening, but it's only happening after they've lost everything. This is not a fraud in the sense that they're trickling, taking people in their own minds in advance. It's more of an ex-post thing. That's the way fraud generally works. People don't commit fraud in advance. They commit fraud when their backs to the wall.
- 5.113 Okay, so this gives you a little sense for Jacksonville, Miami, and Tampa of what we mean by building permits going up. You can get a sense of the peak of this, of course, is happening in 1925, and then it's collapsing pretty dramatically after 1925. But it's a several-year process. It's a little different patterns in the three different cities, but all three of them are experiencing a boom.
- 5.114 Aren't these great? These are newspaper ads for some of the great places. And by the way, I want to emphasize, some of these, I think, are still around, right? So the Flamingo, the Lincoln, the King Cole, the Nautilus. And some of these actually exist, these resort hotels. So this looks pretty impressive. At least this is real.
- 5.115 Now, over there, we've got this beautiful, colorful, watercolor, I guess it is, painting of how beautiful the beach is in Florida. This is by the Pennsylvania railway, and of course, they want people to come down on vacation. This is all about the initial appeal of Florida. But you can see how wonderfully exotic it is, like gondoliers here, and it gets better. So these are where we now have sketches of Miami Beach is calling you. Miami Beach is calling you. Miami Beach is calling you. Do you get the message? Miami Beach is calling you. This is your moment. Don't miss it.

- 5.116 You can see with these beautiful graphs, often with young women at the beach there in case you needed any further enticement and all sorts of wonderful things going on. It's a wonderland and would be completely unfamiliar to anyone.
- 5.117 This is not something that—so here you are slogging away in Ohio or something, and you see one of these ads because these ads are in all the nationwide newspapers, and you're thinking, "That seems like a pretty nice idea. I'm about two years from retirement. If I could get in on this, that could be pretty nice." Not crazy at all. If I went back in a time machine, I'd probably still do it. I couldn't resist.
- 5.118 And to emphasize that people aren't necessarily wrong, here's a picture of a particular location in Florida in 1925, where everyone was saying it was going to build out hugely. And, of course, as of 2017, it did. It wasn't like everything's a swamp that it's uninhabitable. No, this is a pretty good idea to build a resort there and to build all sorts of homes there. In fact, it did work.
- 5.119 I want to emphasize, we want to pay attention to what information exists, what people reasonably believe. As I pointed out, very little actionable information exists. Clearly, there were fantastical narratives of developers. There was a lot of investment, in fact, huge investment in marketing. This gives you a sense of things are really moving here in the 1920s.
- 5.120 So the Florida East Coast railway is showing a lot of growth. Now, that doesn't mean that the Florida East Coast railway can do everything. And of course, it's just one railway, and there's a lot of other infrastructure needed to make the Florida dream happen. But the point is, based on objective statistics, you would certainly conclude something very positive is happening in Florida.
- 5.121 There's a lot of investment in advertising. They're spending a huge amount of the developer's budgets on this advertising. They're getting people like William Jennings Bryan, the famous prosecutor, but also presidential candidate from the 1896 election that I mentioned before. He comes down to Coral Gables Venetian pool to give Bible classes on Sundays. He's still out there, by the way, if you want—no, he's not.
- 5.122 Then we had golf courses where you'll see people having caddies that are baby elephants, and even hitting golf balls off of the trunks of baby elephants that are holding the golf balls up. So when I say fantastical, that's what I mean. Giving people a sense that everything's different down here in Florida. A lot of this buying is happening through the mail in response to these ads.
- 5.123 The other thing that you'll see in a lot of these ads is the use of things like hills or heights because what is it that people are most worried about coming to Florida? Swamps. And so what they do is they have Boca Heights, Miami Beach Hills. What hills? What heights? There are no hills, there are no heights. But it sounds good. You can call your property anything you want, right? So that's one of the particularly amusing aspects of the naming that goes on. See? Lakewood terraces. Terraces, they're up high.

- 5.124 These are beautiful drawings, don't you think? Reasons for Villa Rica's values? The Atlantic Ocean. Ocean Boulevard—I've been to Ocean Boulevard. It's pretty nice. Florida Coastline, Waterway, Dixie Highway. Look at all these things that they're listening to. So I find these very convincing, I have to tell you. Procrastination is the thief of time. Because if you wait, all the good ones are going to be sold at high prices. And what are you going to be doing? Wishing you had bought when the chance was there. By the way, I will be passing out—if you want to sign up afterwards—I'll be passing out a list.
- 5.125 And now, look at this. This looks like a Gauguin painting, right? Here we are in Florida. And these natives, it looks like something right out of a Gauguin painting. That's what you're buying. You're buying the fantasy of walking into a Gauguin painting. And where is that happening? Boca Raton. Mizner Development Corporation. Was Mizner a crazy person? No, not at all.
- 5.126 So, how does it come apart? There's a terrible hurricane that makes people, first of all, wake up and say, "Wait a minute. I guess it's not all good news down there." And then another thing happens: this boat called the Prince Vladimir overturns in Miami Harbor, and they can't get it out. So all of the shipments of supplies are interrupted. They can't do everything.
- 5.127 The railroad has a moratorium on nonperishable goods for reasons I can't remember. And so, all of a sudden, they realize how vulnerable the infrastructure is and how limited it is. So this is a wake-up call, a combination of the hurricane, of things aren't perfect, and then this. And then that brings people to a more conservative way of thinking.
- 5.128 Now, I don't want to say that I can disprove in some sense that this is all the result of the Minsky-Kindleberger greed and fear cycle. I'm not saying I can, but I'm saying that what I think we want to do is not just look at this as some crazy greed and fear, but also recognize that the land market was very hard to judge at this time because this combination of new technologies and lack of aggregate data collection and lack of any experience with a land boom like this.
- 5.129 How would you have gauged how many Americans are willing to buy homes in Florida? First of all, half of America ended up apparently purchasing something, putting a down payment down on something in Florida. But half of America didn't move to Florida. How would you have gauged that? How would you have used that information, which wasn't available *ex ante*? We're not even sure it's true. It's anecdotal.
- 5.130 How would you have figured out, given that there was no price index to think about what Florida land prices were and what was reasonable? How would you think about how much coastline is there going to be? You don't know what the demand is really going to be. You don't know what the supply is it going to be. And so we drew this graph to show that based on reasonable beliefs, you could believe that this was the equilibrium with a very low price and a very elastic supply of inhabitable prime land.
- 5.131 So, if you think that demand is low and supply is very elastic using the new technologies to build out a lot of land, then the price might be down here. If you think that, really, you can't build out a lot of new land, land is limited, and demand is very high, then you think the price should be up

here. Well, which is it? I don't know. I don't think it's possible to know. Before we make fun of everyone the way John Kenneth Galbraith likes to do, let's be a little bit more humble about what was knowable, and let's recognize how new technologies really were changing the way we could think about that.

- 5.132 We've already talked a lot about risk management in banks, and we're just going to have yet another application of this. This is all stuff that I've already told you in the context of this model that we developed before. You can see here that there's a huge inflow of deposits. But that doesn't mean that people were acting in the banking market, were acting with some greed, followed by fear. In fact, what we can show is they weren't. What they were mainly doing was, deposits were going up, but alongside deposits, deposits, equity, and cash were growing, too.
- 5.133 The depositors were being fairly conservative in taking on this new risk. That's the first major point of our paper. You can see the allocations of deposits are also varying very much by region. So, deposit growth is very region-specific. The first ratio to talk about is loans over assets and then cash over deposits. Loans over assets, you can see, are actually declining during this period. What that's telling you is banks are building up a huge amount of cash. So, even though deposits are growing, the ratio of cash to loans is actually rising.
- 5.134 What about the ratio of equity to total liabilities? Well, it is rising a little bit. I'm sorry, it is declining a little bit, but not very much. If you put the two together—so what we do is we show that it can be very useful to just put equity and cash, add them together as a combination relative to total assets to get a sense of the overall effect of risk management. It's not a perfect way to go, but it's a first-pass useful way. We simulate in our paper to show that that's not a bad first step.
- 5.135 From that perspective, this thing we call loans to buffer, where buffer is the sum of equity and cash, that it's actually declining. In other words, the banks are being quite conservative. In fact, they don't fail, with the exception of the Manly-Anthony chain. About half of them do fail. The developers are heavily involved and influencing in the banks that ended up failing. We're finding that in those banks where the developers were involved, they're committing this ex post fraud.
- 5.136 They're borrowing and mischaracterizing the loans that they're taking. They're influencing the banks to charge low interest on the loans to the insiders. Of course, they're also engaged in getting the officials who are overseeing all this to basically be complicit in the fraud, too.
- 5.137 I think given that we're running low on time, I don't think there's a lot more to be said about this. These are just quantitative comparisons from this paper that I wrote with Matt Jaremski on this topic that came out in the Journal of Financial Intermediation, which is all about ways that we think about the non-chain banks versus the chain banks. The point is the non-chain banks in terms of their loans to assets, only seven of them out of the 187 suspended or closed in 1926, which is a fairly low number. That's not what we would call a banking crisis.
- 5.138 But in the chain system, 30 out of the 54 closed or suspended. But those had recognizable characteristics, which we call the fingerprints of fraud. That is, they were involved heavily with

the developers who were basically causing the banks to make loans that were inappropriate in their own interest. What we find is that the **chain banks** that were involved had more funds invested in particular categories of loans that clearly were going to the developers, and they were charging much lower interest rates on those loans.

- 5.139 What lessons would you expect to draw from all this? It's hard to say that there are clear lessons about mistakes people made. The banks in general had remained conservative. Failures were limited to a corrupt **chain** run by developers who lost their shirts as equity investors who bet everything on it and who became corrupt or fraudulent basically after they already had suffered those losses and then got some people, regulators, to be complicit with them.
- 5.140 How about the land market? Very hard to judge anyone's behavior as irrational. I admit that, on the one hand, you could say buying land through the mail sounds crazy. But of course, if the alternative is don't buy land because coming down here is very costly for you and you're thinking about potential opportunities, it's very hard to say.
- 5.141 Sometimes, in economics, we call this **Knightian uncertainty**. If you don't know where things are going and you don't have a very good way to do an analysis of it that's clear, it's hard to judge people for making the wrong decision. My wife definitely would not have let me buy the land by mail, but I would have probably tried.
- 5.142 Okay, I think we will leave it there. When we come back, we will turn to the stock market boom and bust, which is also happening during this period that leads to the discussion of the crash of 1929.

6. Crash and Depression

AT A GLANCE

Examines the 1929 stock market crash, separating it from the Great Depression and asking whether the boom was a bubble. Reviews evidence that price increases tracked citation-weighted patents (Nicholas), professional valuation models (Kabiri), and bond investors' optimism, against Rappaport and Whyte on credit conditions and the New York banks' own decision to delist because they thought their shares too high. Concludes the evidence is mixed: the run-up was largely warranted but perhaps slightly excessive. Emphasizes that the Federal Reserve deliberately caused the crash through tight money, and warns that central banks now embrace macro-prudential targeting of asset markets.

- 6.1 Welcome back. I want to continue now with a discussion of one of the most written-about financial events in the US, which is the **stock market crash** of 1929. By the way, on a personal note, when I got married, I had the foresight to pick October 29th, which is the date of the crash in 1929, to pick that as my wedding date, so I'd be sure never to forget my anniversary. Because if you're a financial historian, this is one of the dates that you know pretty well.
- 6.2 Before we start with an analysis of the crash, I think it's important to also just mention, we've talked about the **Great Depression** already today. You'll notice I never mentioned the **stock market crash** of 1929. The reason is that financial historians don't think there's much of a connection between the **stock market crash** of 1929 and the **Great Depression**. Actually, the stock market recovered in 1930, not fully, because there were continuing negative macroeconomic shocks in 1931 and beyond.
- 6.3 But generally, the **Great Depression** is something that we think of as, really, 1930 to 1933. 1929, the stock market crashed. There was a recession that began in the fall of 1929. The recession was not really something that is in any direct sense connected to these big shocks we've been talking about in the **Great Depression**. In any case, the **stock market crash** doesn't seem to have been a causal or contributing factor to the Depression. The Depression is a story about **monetary policy**, and its severity is a story about bank failures, and the crash is really a preceding interesting financial event, but not one that's really related to the **Great Depression**.
- 6.4 A lot of people who were writing about the crash in the 1940s or 1950s or early 1960s, without doing quantitative analysis of the **Great Depression** or the crash, mistakenly connected the two. I would dare say that if you took high school in the United States, you probably thought there was a connection between them, because a lot of non-economic or financial historians writing in textbooks make that connection. But I would like to tell you, very, very few people believe there's any connection.
- 6.5 There is some indication of a slight connection between the crash and effects on declines in consumer sentiment and consumption. But that is not the story of the **Great Depression**. So yes, maybe small connection, generally not a very big one. We want to study the crash, really for its

own sake, as an event, what we can learn from it as a financial boom and bust. I was studying in the 1980s, one of the things people noticed is, oh, there seems to be a remarkable similarity between the pattern of stock price movements in the 1920s and the crash on the one hand, and the 1980s. If you remember, we had a big stock market boom and bust.

- 6.6 And if you looked at it on percentage growth terms, the two track each other interestingly for much of the period leading up to the boom and the immediate bust. I don't know what to make of that. That's just something that people pointed out. And some people, I guess, use that as a way to think about it from a **chartism** standpoint, about whether markets have their own inherent dynamics of getting **greed and fear** or God knows what. Maybe there's connection. But as we're going to see, quantitative research has really approached this very differently and has a very different way to think about the stock market movements that are happening during this period.
- 6.7 My own view on this is not to make too much of this connection between the '20s and the 1980s. I think they just are similar, not identical, and they're very different kinds of processes in any case. OK, there's a vast literature, serious literature, as well as a qualitative soft literature that's looked at this. Some people regard it, of course, you know what **John Kenneth Galbraith** is going to say. It was self-evident that this must be an irrational boom and bust cycle. I'm sure he had many insulting things to say about the stock market investors, just like the Florida land speculators.
- 6.8 But some other people have done what I would call serious empirical research and viewed this as possibly, at least in some places, a **bubble**. **Rappaport and Whyte**, we'll talk about their research, they've claimed that there's a lot of indication that there was a **bubble** component to the stock market increase. A **bubble** means an unwarranted increase. As you'll see, I've also written some papers about New York banks during 1928-1929, and argued that there there was a sense in which there was a **bubble** in some of the New York bank stock prices.
- 6.9 There's some people who have reason to believe, based on empirical research, that there was an element, maybe just at the very end, of a kind of excess of stock price. But as we're going to see, not everybody agrees with that view. In fact, I agree with, in a sense, both views. I find, I think we have, my ultimate conclusion for you is going to be, there's a lot of mixed evidence on this question, that there's some evidence that's somewhat convincing, including my own, that some stocks may have experienced unwarranted price increases, but there's a lot of evidence that is in the other direction. So let's talk a little bit about that.
- 6.10 In that respect, I want to contrast this episode very much with what I said about the **South Sea bubble**. So if you remember in the **South Sea bubble**, smart investors were actually profiting from the **bubble** and got out before the crash. It was inexperienced investors who were driving up, at the very end, driving up the price and experienced most of the losses. That's not apparently true in the 1920s, as we'll see.

- 6.11 Furthermore, to contrast it with the South Sea bubble, the reason that experts knew that the South Sea bubble stock was overvalued is that looking at the franchise, they could see that there was no value other than the bonds that were being held, and those bonds could be valued. The implicit huge franchise value had no connection to any, let's say, intellectual property or licenses or monopoly rights that were particularly valuable. In contrast, what we're going to show you is that we can actually identify in stocks why some stocks went up while other stocks didn't during the boom in the 1920s. It's very much related to observables that have to do with technological improvements.
- 6.12 OK, so this is going to be a case where we're going to say that there's some ambiguity. Now, the thing that there is no ambiguity about is what precipitated the crash. The thing there's no ambiguity about was, the Federal Reserve did it. The Federal Reserve did it on purpose. The Federal Reserve believed that its own policies had been very loose in 1929, and that that had led the stock market to excessively increase. It wanted to restrict credit through call loans being made through stock brokers. It wanted to restrict those loans by making monetary policy tight so that they could pull back on credit to the stock markets because they wanted to actually cause the crash.
- 6.13 I want to emphasize—why the crash happened when it did, I would say, is not that controversial. The Federal Reserve felt that it had been too easy. By the way, I already mentioned that the Federal Reserve had a very poor understanding of its own policies' effects during this period. If you recall, the Federal Reserve continued to tighten throughout the Great Depression because they thought they were being easy. Similarly, in retrospect, people don't think the Federal Reserve was being particularly easy just prior to the crash.
- 6.14 So the Fed's inference that the stock prices were high was based on no real analysis, either of itself, no credible analysis, either of itself or of the stock markets. Nonetheless, that doesn't prevent monetary authorities from doing things. They concluded the stock market was rising because they were being too easy and they needed to stop it. And boy, did they. So if the Federal Reserve tells everyone, "We don't like the stock market being so high," do you think they're going to be able to make people sell stock? Yes.
- 6.15 And so that, I think, is very important to recognize, that there's no, there is some controversy about how warranted the stock price increase was. There's no controversy about how incompetent the Federal Reserve was, in both its evaluation of itself and its willingness to actually go out and target the market and make it very clear that they wanted to rein in speculation and rein in this excessive price increase.
- 6.16 Now, the literature, though, has found lots of reason to, as I said, there's a lot of evidence that this is not the South Sea bubble, that the stocks that are going up have a lot of franchise value relating to intellectual property, and that the investors who are most interested in these, the models that have informed investors that are trying to value stocks, are very consistent with the price increases. In other words, this is an amateur hour. Let's talk about that.

- 6.17 Tom Nicholas' work, Tom is a professor of finance at the Harvard Business School, and he did an analysis of different firms' stock price run-up and asking, can you distinguish across the firms that the firms that had the most valuable technology, new ideas, were the ones that the market was rewarding with higher stock prices? The answer is, oh, yeah. That's how he did it. When people want to measure the value of your technological idea, they don't just measure how many patents you have. What they do is, they weight those patents by how much they are cited in the future. We'll talk a little bit about that evidence.
- 6.18 It's absolutely clear that, in the aggregate, the 1920s are an unprecedented and maybe never repeated episode of technological advance across many sectors involving a lot of new ideas, and that those ideas were embodied in patents that got cited and are still being cited today. An unusual proclivity for creating ideas that had dramatic future consequences in reality, not just in expectation. Then what Tom shows is that, if you measure the patents that are possessed by each company weighted by their importance based on future citations, that lines up very well with which companies had relatively high price movements.
- 6.19 This connects to another point, which is, the price movements during this period, the increases, were not uniform across the stocks. It wasn't as if the whole stock market was going up. It wasn't. It was only some firms that were going up. Other firms weren't going up at all. Which firms were going up? The ones that had huge citation-weighted patent scores. So you might say, well, then why am I saying this is controversial? It's like Nicholas' evidence is pretty strong. That's true.
- 6.20 But Nicholas' evidence really has mainly to do with cross-sectional differences. It still doesn't say, could the average of those have been, of all stocks, be 5% too high? I think what it does say is it couldn't have been 20% too high. But that doesn't say that there wasn't a little bit of overpricing. When I say, I think the evidence is mixed, I want to be clear. I'm not saying that we can have a view of this like the South Sea bubble. We can't, based on the evidence. But we can't conclusively show there wasn't a little bit of excess pricing on average going on.
- 6.21 But we know that, cross-sectionally, we can explain differences, and that, as you'll see, those differences are quite big that we can explain. Another piece of research by Ali Kabiri goes back and looks at, what were professionals who were doing valuation of firms, what were their models telling them the warranted prices were of these firms? And what he shows there is, they're telling them that the market price is warranted, so that the actual investors who are looking at it based on the performance of the firms at the time are driving the bus of the market price.
- 6.22 So what does that tell us? It tells us that the market price run-ups aren't due, as they were sometimes in the South Sea bubble, to unsophisticated investors coming in at the very end and pushing the prices up even more based on some momentum beliefs, like that poem that we read yesterday. That's not the story. The prices that we're seeing in the market are prices that are coming out of the valuation models that investors are taking into account ex ante.

- 6.23 We know that we can understand the cross-section of returns based on intellectual property, and we know that we can understand the overall each of these prices as consistent with what professionals believe, based on the fundamentals of the firms at the time. Now, the contrary evidence, I'll point to a couple of things. Rappaport and Whyte, what they basically show is that some of the average movement in the prices is connected to how loose credit is. I think that that shows some of the ambiguity here.
- 6.24 There's no reason to believe that credit was excessively loose leading up to the Fed's purposeful tightening. But there is clearly some indication that there's a connection between the supply of credit and the stock market price. And so that leads Rappaport and Whyte to say, well, some of these prices are driven by easy money, just in the sense of access to credit, if not due to an overall easy money posture by the Fed. So that's why I think we have to reserve judgment on whether we can say that there's no connection between credit markets, which is distinct from fundamentals of firm valuation, and the stock market.
- 6.25 Rappaport and Whyte show that there is an aggregate connection there. I just want to emphasize, you can believe both. You can believe that the relative prices are following intellectual property, but that the average price is a little bit goosed up relative to what it should be. I mentioned I had also done some work on this. Let me tell you a little bit about that. New York City banks during the 1920s are also innovating. They're getting involved in trust management. They're getting involved in underwriting. They're operating internationally.
- 6.26 They're expanding their bank operations in a lot of ways, especially banks like Citigroup, at that time called Citibank or First City or whatever. I can't remember what it was called. Citibank and all the other large New York commercial banks delisted from the New York Stock Exchange in the beginning of 1928. The reason was that they believed that their share prices were too high. So how many times have you ever heard a CEO publicly say that their stock price is too high, and then say that they think the reason is because there's people who are doing small lot, small-size purchases, they're driving up the price, and so they're going to delist?
- 6.27 So all the New York City banks met on the same day with their stockholders in early 1928 to vote to delist from the New York Stock Exchange, and they didn't go back for decades. They delisted. They found out almost immediately, though, that their diagnosis was wrong because when they delisted from the New York Stock Exchange, all of the sales that continued that weren't New York Stock Exchange single-share transactions, that is, no longer small lot shares on the New York Stock Exchange, had the same price. In fact, their stock price kept going up.
- 6.28 So, it's funny, right? I mean, who would know better than the CEO what their stock price should be? And they delisted because they believed their stock price was crazy high. To me, I take that very seriously as evidence that at least their share prices were probably unwarrantedly high. But I don't have any other basis for making the argument than that. I want to show you all the evidence that we have and the sense in which I regard it as a little bit mixed.

- 6.29 Now, one thing to bear in mind here is whole classes of shares didn't go up during this period. What I'm showing you here is, if you look at what was one of the big growth sectors, well, actually, it was utilities, because you might think of utilities as boring stocks, but not in the 1920s. And that railroads, as a group, didn't participate much in the big growth. There was no new big technological change in railroads. It was industrials that were mainly driving the bus and utilities. And then within those, as I said, individual stocks differed quite a bit.
- 6.30 I should also mention that some of the best economists at the time were completely convinced about the stock price being warranted. Irving Fisher is perhaps the best-known example, and based exactly on technological change. There's a lot of good arguments that technological change is driving, let's say, 80, 90% of what we're seeing, and also that investors are not behaving in any mechanical sense. They're not just buying everything. They're buying things very much discriminating based on where the intellectual property is.
- 6.31 I mentioned before that innovation in the 1920s is just extraordinary. Here are some examples. The number of patents— there were about 20,000 patents granted to firms between 1920 and 1929. 21% of those are cited in patents that were granted between 1976 and 2002. That's an extraordinarily high percentage. What it's telling you is, a lot of these patents are fundamental patents that have to get cited for a very long time in new patents. 31.8% of the patents of the greatest inventor of the time, Thomas Edison, between 1910 and 1930, are cited between 1976 and 2002.
- 6.32 This gives you a sense, we're talking about people that were making extraordinary progress. Alex Field wrote a book in 2003 all about how progressive and important the patenting in the 1920s was, and concluded that, despite the Great Depression, diffusion of the technologies that came about during the 1920s was at its height in the 1930s. So even though the 1930s were a horrible time to do investing, notwithstanding all of the negative macroeconomic shocks, it was still one of the most progressive periods for diffusion of technology because they had this amazing stock of new technologies that they had to diffuse.
- 6.33 Who are some of these incredibly innovative companies? Well, again, you've heard their names, but at the time that they were developing—Kodak, General Electric, DuPont, General Motors— these companies were the big innovators of the time. They were the Teslas of their time. They were the ones creating these patents. I'll just give you a little bit of a sense. So if you graph market value over book value versus the patent stock, which is the solid line, market value over book value is the dashed line. On the right-hand side, we have it citation weighted. On the left-hand side, we have just the simple patent stock.
- 6.34 What you can see is that, over time, patent stock and market value are trending together in these progressive firms. You can see this for General Motors, for Westinghouse, and for others. Furthermore, from a fundamentalist perspective, I've already mentioned market prices recovered in 1930. That would maybe indicate that the Fed's targeting of the stocks, which precipitated the stock decline in the fall of 1929, had more of a temporary influence than a permanent influence.

- 6.35 I've already mentioned, the Fed was probably incorrect in its own view that it had precipitated this as a result of loose policy. But you can see pretty dramatic increases in interest rates as a result of the Fed's actions. The point is, if a central bank tells you, "I don't like what just happened here, and I'm going to contract monetary policy to make that change," generally speaking, they're going to be able to succeed.
- 6.36 Some more evidence that's consistent with the view that there was a fundamental driving the price increase is that bond investors were also quite optimistic about these companies, so that long-term increases in leverage were happening. If you thought it's not like the South Sea, but again, unlike the South Sea bubble where you just have unsophisticated people participating in this one market, the stock market, well, the people participating in the bond market are all sophisticated investors, and they are allowing these high-growth firms to increase their leverage substantially.
- 6.37 Well, what does that mean? If they're allowing leverage to go up, they must be thinking that asset risk is declining and that asset growth possibilities are increasing. This gives you a little bit of a sense of how much long-term debt is rising during this period, reflecting that view that corporate leverage tolerances were rising because of reflecting those same long-term opportunities. There's a literature here, this is inside baseball finance literature, about whether closed-end fund premium in investing in stocks, differences between closed-end funds and the actual underlying shares, reflected something that's consistent with this being a bubble.
- 6.38 I would I'd say, the reason this is inconclusive is, you can't really distinguish between the closed-end funds being pessimistic and people being pessimistic about the managers of the closed-end funds. I'm sorry, the closed-end funds doing a poorer job or the closed-end funds having poorer investments. There's some literature coming at this from using something called the closed-end fund premium, but I think not very convincing.
- 6.39 There are more formal statistical tests of the relationship between dividends and stock prices, and those tests tend to be able to not reject the view that stock prices were warranted. I just said not reject the view. But on the other hand, they're low power tests based on time series, and they're not really able to conclusively accept it. My view is, there's some mixed evidence. There's some things that are hard to square with a full perfect fundamentals view, but that, overall, I think that we'd have to agree that a lot of what's going on is reflecting very much positive long-term technological improvements that are reflected in some firms.
- 6.40 Another thing to bear in mind is that, another explanation for why sometimes, when there are these waves of optimism, justified probably by a lot of new technological beliefs, why there might be a little bit of excess in the market, is because people with different views, like suppose that we're talking about Westinghouse or General Electric or General Motors. You might say, "Oh, here's my view of how great that technology is going to be." Maybe I have a slightly contrary view and I say, "I think you're a little too optimistic." How do we vote in the market?

- 6.41 Well, if you're an optimist, you vote by buying the shares. If you're a pessimist, you have to short the shares. But just from a pure transactional standpoint, buying the shares is not costly, but shorting the shares is. You have to borrow the shares. And so there's actually, markets tend to have a sort of implicit bias in favor of optimism during these waves of optimism because the cost of being a short seller can often, just from a transaction cost standpoint, is higher. So my overall take is, this was—the run-up in shares when it happened and where it happened was warranted. The extent of it was probably a little bit excessive.
- 6.42 But the biggest lesson that we really can learn from this is that central banks are very risky things to have in your financial system. The Fed really had a very poor understanding of its own effect on markets during this period, but a very bold willingness to use discretionary actions to target a whole stock market and to successfully cause a major crash. You might say, well, I guess we've learned for central banks not to do that. Sorry, but since 2008, globally, there's now a whole new commitment to what's called macro-prudential regulation by central banks, so that central banks will be more aggressive to try to identify asset bubbles and target them.
- 6.43 Whether that means central banks are able to do so wisely, I hugely doubt. But I'm just pointing out that our democracy, after the 2008 crisis and democracies all over the world came to the conclusion that they want central banks to be doing highly discretionary activities, including macro-prudential policy, even though there's some disastrous examples of central banks unwisely targeting stock markets. Does that mean that we should have learned differently? I'm not sure.
- 6.44 But I think what's interesting, again, from the overall perspective of adaptation I've tried to bring to this class, that I think that currently, at least, this seems to be central bank policy discretion, including targeting asset markets with macro-prudential policies, seems to be very politically favored.

7. Subsidies and Bailouts

AT A GLANCE

Defines risk subsidies, contrasts them with lump-sum subsidies, and explains why politicians prefer invisible credit guarantees and why prudential regulation predictably fails. Walks through Calomiris and Jaremski's study of eight state deposit insurance systems (1908-1920s), showing insured banks gained deposits, held less cash and equity, faced no depositor discipline, and collapsed after wartime crop prices fell. Recounts how Roosevelt opposed federal deposit insurance yet signed it through log rolling with Henry Steagall. Covers the farm credit system and Mark Carey's evidence on the 1970s farmland bubble, then the 1934 New Deal mortgage programs and the thrift collapse of the 1980s.

- 7.1** Welcome back, everyone. Today, we're going to take a deeper look into the financial crises of the 20th century worldwide. And because that's our topic, we are going to focus on the most relevant factor in those crises, which is what we call risk subsidies. So what we're going to talk about today from the perspective of many different crises is, what is a risk subsidy? Where does it come from? Why do people enact them? How do they work? Then how did those evolve during the 20th century in the United States and around the world?
- 7.2** Before I begin, I want to just define what a risk subsidy means because it's a very general category. We're going to be talking about it from a variety of different specific versions. But you know what a subsidy is? If you're trying to help someone in some way, you direct resources to them with a subsidy. Let's say that I wanted to help you buy a house. I could decide to subsidize your down payment. For example, in Australia, and this is also true to a lesser extent in the US, first-time home buyers can enjoy a government subsidy that helps them on a matching basis, helps them buy their first house.
- 7.3** You are a first-time home buyer. The government's going to help you buy your first house. That's not a risk subsidy. In fact, if anything, it's an anti-risk subsidy because it's actually helping you when you go to borrow the remaining amount to have less that you'd have to borrow. It also means that the cost to you of your house is going to be lower, but not because the government is bearing any risk. The government's just giving you money.
- 7.4** But there's another way the government can help you buy a house, and that is to encourage you to take more risk by making the cost to you of taking risk lower. So that would be, for example, if the government guaranteed your mortgage or guaranteed even part of your mortgage and said, if you borrow 100%, like let's say in the Veterans Administration in the US, if you're a veteran, you borrow 100%, that's OK, and we guarantee your mortgage. And so that means that the bank or other provider who's lending you the money for your mortgage is going to lend that money to you at a much lower interest rate because they're not pricing any risk.
- 7.5** So the government, in that extreme case, is subsidizing your mortgage risk. Now, those are two different ways to skin that cat. One of them is just a lump-sum subsidy. The other one is a risk

subsidy. And those are two very different policies. The **risk subsidy** works only if you're willing to take more risk. It helps you, but in a very risky way. And so, that's really an important theme for us as we're thinking through all the applications that we're going to talk about today, that what the 20th century is all about is the government discovering the idea that it's very convenient for governments to give subsidies in the form of credit guarantees, of **risk subsidies**.

- 7.6** We've already studied one example of this from the late 19th century, which was the Argentine case of the **cedulas**, which were exactly a **risk subsidy** for wealthy landowners in Argentina in the late 19th century that reduced the interest cost to them as the way the government passed through its subsidy. There's a broad political question, of course, which is, given that there's more than one way to skin a cat, why do you sometimes skin the cat with **risk subsidies** like the ones we're going to talk about all of today, and sometimes you just give cash? Let me give you a hint.
- 7.7** First of all, how many of you have ever thought about the value the government spends on **risk subsidies**? I'm going to guess the answer is, very few of you. **Risk subsidies** are not typically accounted for in the government's spending accounts. They're invisible. That's great because that means that governments can engage in political bargains, decide to throw benefits to some people, and those benefits are not very visible. Those are exactly the kinds of government transfers that politicians love. Make somebody better off without showing everyone else that you're spending the money on them.
- 7.8** Now, that's not true of all governments at all times. But as we'll see in the 20th century, this has become especially important. It's going to start off with subsidies early in the 20th century to agriculture landowners we'll see in the US, because you'll remember, **rural populism** is a dominant political movement in the US in the early 20th century. It's going to transition to subsidization of mortgages, not just in the US, but worldwide, which we mentioned even early in this course on the first day.
- 7.9** It's also going to show up as **risk subsidies** to banks. That is part of the overall **game of bank bargains** that we've talked about, which is governments and banks form coalitions where banks do things for governments, from the very beginning, **Bank of England**, if you remember. But different manifestations of this happened throughout history. Banks do things for governments, chartered banks do, and governments do things for banks. One of the things that governments do for banks is create some sort of guarantee of their situation.
- 7.10** You'll remember that **John Law**, back in the early 18th century, was doing this. This is not a new idea, but it's been increasingly implemented as part of the government bank bargain. Of course, that is, as the research in economics has shown, it's that last **risk subsidy** for banks that helps us understand why banks have been taking so much risk and why we've had so many **banking crises** in the period since 1970. The answer is, governments have really discovered their willingness to use the tool of subsidizing banks as part of this government bank bargain through **risk subsidies**.

- 7.11 The only way you can get the subsidy if you're a bank is to take risk. Then you might say, well, wait a minute. Governments are supposed to be, in addition to offering bank subsidies, they're also supposed to be regulating banks in prudential regulation to prevent banks from taking risk or too much risk. But now you see that there's going to be another theme in this discussion, which is, wait a minute. If I tell you you get a subsidy, but the subsidy is only there if you take risk, and then if I regulated you to prevent you from taking risk, that would undermine the whole point of the subsidy.
- 7.12 Now you're going to learn how to start predicting things. When you start understanding how **political equilibria** work, you're going to be able to predict that if governments worldwide decide that they want to use the tool of **risk subsidies** in their partnerships with banks, they're certainly also going to decide not to be very strict on prudential regulation because that would just undo the subsidy. So now you start understanding, why is it that prudential regulation seems to always be failing? Why are our regulators and supervisors, whose job it is to make banks limit their risk, why is it that they seem like they're always not doing that job?
- 7.13 As recently as the 2023 experience in the US, as recently as many different **financial crises**, over 100 since 1970, and the answer is, once you start understanding that **risk subsidies** are a purposeful tool of state, then this is not puzzling. In fact, this is just part of the equilibrium. That is an insight that I'm hoping to develop through our discussion today. What we're going to do first then is talk about **deposit insurance**, which is one of the main manifestations of this partnership between governments and banks, through government provision of **deposit insurance**.
- 7.14 We're going to talk about how that expanded throughout the world in the 1980s and 1990s, especially. We're going to start with an early example of it, which is studied by a lot of papers, and I've written myself, I think, four papers on this topic from this period. Most recent paper is one that **Matt Jaremski** and I wrote that was published in the Journal of Finance in 2019. We're going to start off with a fairly technical academic discussion that's going to take you through this paper.
- 7.15 By the way, one of the things about this course that I guess you've noticed by now is that I tend to teach, even though I look at this as an undergraduate-level course, I teach it with very high-level academic research based on published articles in top journals. The reason for that is because I don't think anything else is convincing. I don't think professors should play the role in our society of giving opinions that aren't backed by clear evidence, that only their authority and their opinion is supposed to be taken by students seriously.
- 7.16 If you're a student and you're struggling through some of the more difficult analytical aspects of this course, please think of it that way. Think of it as, it may be that you have to struggle a little bit with some analysis, but you don't have to take my word for anything. I hope that you will have sat through this course with realizing that you don't have to take my word for anything. Everything I'm talking about is clearly indicated by empirical research, historical narratives that have stood the test of time, as well as statistical analysis, and none of it is a matter of opinion.

- 7.17 In our first topic, **deposit insurance** and its development, I also want to emphasize that there's a very large literature on this topic. We're only going to talk about two articles at length. The first one is about the early experience in the US in the 19-teens and 1920s. Then the last article we'll turn to at the end of the day is another article that I wrote with Sophia Chen at the **IMF**, which is about **deposit insurance** growth and its effects on risk throughout the world during recent years. Let's get into our first discussion then.
- 7.18 As I've already explained to you, **deposit insurance** spread throughout the world. There have been a lot of studies of it and its effects. Of course, you might think, in fact, if you were talking to most economists who don't study **deposit insurance**, and you ask them, "Tell us, what's the point of **deposit insurance**?" They might say, "Well, it's to reduce the riskiness of the banking system." Because if we have **deposit insurance**, then depositors are more relaxed.
- 7.19 Of course, many people wrongly believe that depositors' jitteriness is the cause of systemic risk in the banking system. If we can just calm those depositors down, then we don't have bank runs and we don't have bank failures. But then there's another point of view, which is, wait a minute. Maybe depositors' jitteriness is a reflection of problems in banks. By exerting discipline, by being observant, some class of active depositors—not all depositors invest in information. Of course, they don't have to. Just a small group can.
- 7.20 But that small group, by investing in information, noticing when banks are weak, and being willing to go in and start withdrawing funds from the banks, they can help banks restore their risk management equilibrium, as we've discussed. And this table here, which is a bank balance sheet, shows that that happens by banks who face that **market discipline**, choosing to increase cash or reduce their deposits relative to equity, which puts them back into a low-risk situation.
- 7.21 And from that perspective, maybe what **deposit insurance** does is, it makes banks a little too safe from deposit discipline, and therefore knowing that they can take risk with impunity. And maybe even that's the point of creating **deposit insurance**. By the way, it assuredly, as a matter of historical narrative, as I'll show you today, of course it was the point of creating **deposit insurance**, to allow banks to take risk that's safe from deposit or discipline. That was the reason **deposit insurance** was created in the cases that we're going to talk about this morning.
- 7.22 From that perspective, you might say, well, then, I guess **deposit insurance**, I would predict, is going to cause more banking system collapses because banks are going to take more risk and no longer manage their risks prudently. And that would be correct. So the entire literature, the entire empirical literature on **deposit insurance**, there aren't any exceptions that I'm aware of, of papers, and there are scores of papers in this literature, has reached that conclusion—that **deposit insurance** makes the banking system much less safe.
- 7.23 Then, of course, you might say, well, wait, that's surprising. If deposit insurance's purpose should be to make banking systems safer, then if all the research has pointed in the direction of the opposite, why don't we get rid of **deposit insurance**? And of course, that's an interesting question. I wonder how, if I had stood up here and said, "By the way, I'm here to convince you

- to get rid of **deposit insurance**," if those were the first words that came out of my mouth, how you would have reacted to that.
- 7.24 Maybe you would have said, "Wait, **deposit insurance**, I'm used to that. That's part of my life. All my deposits in US banks are insured by the Federal **Deposit Insurance** Corporation. Who is this character? That's been in place for almost a century now. This is what we're used to, and it feels good to have government **deposit insurance**. I like it." So once you start doing that kind of introspection, you'll see why, once you enact **deposit insurance**, it's very hard to get rid of it, and partly because people don't really understand **risk subsidies** and how they work.
- 7.25 I'm not Don Quixote. I'm not out here with my sword and weird hat to tell you I'm on a campaign to get rid of **deposit insurance**. I'm a little smarter than that. Unfortunately, but that helps us understand what I was saying early in the course about the difference between economically efficient or economically beneficial on the one hand, and **adaptive**, socially **adaptive**. **Deposit insurance** has been adopted without being undone, except in very few cases. Estonia, Argentina. There are a couple of countries in some periods who got rid of their **deposit insurance**, but pretty much **deposit insurance** doesn't get reversed.
- 7.26 That teaches us that this is somehow a socially, politically **adaptive** policy, even though it's very destructive economically. OK, so in this first paper, we're going to talk about this historical episode of eight states that adopt **deposit insurance** in the early 20th century in the US. What's the point? What are we trying to do? We first want to establish what the effect of **deposit insurance** is on bank risk-taking. We also want to look at the mechanism to see if we can see why **deposit insurance** is having the effect that it's having. The historical periods can be very useful because they allow us to see things that we can't always see in our current period.
- 7.27 Let's talk a little bit about causality in empirical research. As I mentioned, all of the literature in financial economics indicates that once countries take on **deposit insurance**, their banking systems become much riskier. But one of the problems that we have in that literature is we lack a kind of airtight causal **identification** story. What does that mean? Suppose that it were true that countries adopt **deposit insurance** because they're entering a very risky moment or that risk exogenously unrelated to **deposit insurance** that's relevant for the banking system goes up.
- 7.28 Suppose that that event tends to be correlated with governments adopting **deposit insurance**. That then could mean that it's the risky moment that's causing both **deposit insurance** and banking system problems subsequently. Not that it might be, therefore, that it's not **deposit insurance** itself that's causing the risky problems in banks, but **deposit insurance** adoption or expansion is just a symptom of an increase in **exogenous** risk. I'm not telling you that's true. In the last paper that we're going to talk about today, we directly investigate that and show that that is not the driver of the result.
- 7.29 But before that paper was written, we didn't have a paper that had done that. One of the advantages of this historical study that we published in 2019, before I worked on that other paper, one of the advantages of this is, it provides airtight **identification** because we can do

comparisons that are very specific and that allow us to put aside the possibility that something about the risk that's changing in the economic environment is what's driving the risk. Let me explain how.

- 7.30 In the cases we're going to look at, when the state develops **deposit insurance**, there are other banks in the same state that aren't covered by the **deposit insurance**. And there are other banks in neighboring states also that aren't covered by it. What we can do is, we could say, well, if there were a change in the riskiness of the overall environment, that would affect all the banks that are in that environment. But what we find is, the risk problems are specific only to the insured banks. That provides a kind of airtight causal **identification** that you wouldn't get in today's **deposit insurance** systems.
- 7.31 And why? Because when **deposit insurance** is adopted, it applies pretty much to all banks. But historically, in the cases we're going to talk about, and that's where we're going to get all of our statistical power, is comparing what the effect of **deposit insurance** is on the banks that are covered by **deposit insurance** in comparison, in contrast to the banks that aren't covered by it. That's what makes it powerful.
- 7.32 What we're going to do is, we're going to make that comparison of insured banks compared to the banks that aren't insured in their state. But we're not just going to look at the behavior to show that, yes, it's true, that they're the ones who are taking on more risk, and of course, that these systems are collapsing. We're going to look at the specific mechanism to try to understand, not only do they take on more risk—so let me remind you, what we're going to be looking at are these two ratios that we developed yesterday.
- 7.33 Do the banks that are insured, do they choose to hold less cash relative to assets? Do they choose to maintain less equity relative to their liabilities? I'm going to show you that the answer is yes to both of those. But we're going to go beyond that. We're going to go beyond it to say, is it also true that the depositors in those banks are less attentive to the risk of the banks than depositors in the banks that aren't insured? Remember, we're able to do that because being attentive to a risk means, when the risk goes up, what do you do as a depositor? You withdraw some funds.
- 7.34 We're going to then look to see, when risks go up in the uninsured banks, do depositors withdraw the funds? The answer is they do. When risks go up in the insured banks, same kind of risk going up, does that cause the depositors to withdraw their funds? The answer is no. In this paper, we're going to both show you that, within the same locations, the insured banks are behaving very differently. But we're going to show you that the reason they can get away with that behavior of taking on more risk is that depositors of the insured banks behave differently.
- 7.35 That's what allows the tolerance for the high risk. But we're going to go beyond that. We're also going to ask the question, how does this affect bank strategies beyond just this analysis? For example, when these political movements to establish **deposit insurance** in these states succeed, what happens is, many of the people who are pushing for the movement take

advantage of the **deposit insurance** in new strategic ways. Some of the people—let me give you an example.

- 7.36 Some of the people in the western part of Nebraska, for example, which is wheat-growing land, decide that this is a good moment to grow crops in those new locations which weren't really under cultivation yet because world prices for wheat are high. They create new banks, which would have been very hard to create. They're inexperienced bankers. They're taking big risks. They're just starting up these new enterprises. They're able to enter into these very risky new banks with impunity.
- 7.37 Why? Because depositors are willing to provide the funds. Why? Because the funds are guaranteed. And that's not a coincidence. That's not some random fact that happens to be true. It's, in fact, part of the lobbying for **deposit insurance** is to make it possible for those people to do that. They were part of the political advocacy, and they're going to be part of the story. What we're going to find is, the most extreme cases of the effects of **deposit insurance** are going to be for new entrants into banks that are occurring in locations where these particularly risky strategies are happening that have to do with crops whose prices go up a lot in World War I.
- 7.38 Some people say, "Wow, now the land is worth cultivating." And so they want to establish banks to help cultivate in these new areas that are profitable because prices are high. OK. Is that a smart thing for them to do? No. The history of war throughout time that we've been able to observe, whether it's the Crimean War in the 1850s, or the Franco-Prussian War, or the US Civil War, you name it, what happens to grain prices during wars? They go up. Why? Because the number of people cultivating is reduced, and the number of people consuming hasn't been reduced so much. So demand is still pretty much what it is. Supply is contracted.
- 7.39 What's going to happen once the war is done? Prices are going to fall. Now, that's not very hard. That's not very complicated. Some people aren't very smart. Normally, the market prevents people who aren't smart from being able to organize banks, because usually, to be able to get funding for your organized enterprise, you have to demonstrate that you're doing something that's reasonable and smart. But if you have **deposit insurance**, that's no problem. You no longer have to demonstrate that your business plan makes sense.
- 7.40 That's a big part, too, of what **deposit insurance**, or generally **risk subsidies**, do, that they allow people who are taking foolish risks to still be able to take those. So it's not just smart risks that we're subsidizing, it's also foolish risks. I've already basically previewed for you all the conclusions, but I do want to just take you through it quickly so that you can see how economists use statistical analysis to convincingly make points. Because I'm hoping that some of you who are sitting through this lecture will find financial history and the study of **financial crises** interesting and might even decide to undertake research careers, academic careers.
- 7.41 That's not a requirement. That won't be on the test. But nonetheless, I want to show you how we learn to think. The first thing we're going to demonstrate is that deposits, once **deposit insurance** is enacted, within the same locations, deposits shift from uninsured banks to insured

banks. The way the insured banks do that is, they offer a little bit higher interest. I say to you, "I'll pay you a little bit more interest. I don't have to pay much. Also, you know that my deposit is insured." That is attracting some deposits.

- 7.42 As they're doing that, though, because they're insulated from **depositor discipline**, they tend to adopt riskier strategies. Strategies, as I'll show you, that involve, to maintain that growth, less cash ratio and less equity ratio. I'm also going to show you that the thing that enables them to do that is that depositors in insured banks don't react to increases in risk, in contrast to the way depositors in uninsured banks react. Then finally, I'm going to show you that this is happening exceptionally in areas where these World War I price effects on crops were the greatest, were the highest.
- 7.43 And it's happening with young banks, new entrants that are coming in in these high price run-up areas. And then we'll also look at the collapse of these banking systems. But it won't surprise you to hear that the uninsured banks don't collapse and the insured banks do collapse. Once the economic environment changes, the World War I price effects are done, these banks collapse along with those changes.
- 7.44 There are going to be three core hypotheses that we're going to be investigating to get a little bit formal here. First, we're going to investigate whether insured banks compete away deposits from uninsured banks. Secondly, we're going to investigate the hypothesis of whether insured banks take more risk. And third, we're going to investigate the hypothesis of whether depositors don't engage in discipline to punish banks for increasing their risk if those banks are insured but do otherwise.
- 7.45 Remember that one of the things that allows us to do this is that all of these states are **unit banking** states. They're all going to be agricultural states. You know that that means that the bank has one location. We can talk about the economic environment of the bank very clearly because we know exactly where the bank is located. We can talk about what crops are produced in that county or in that place because we have a lot of statistical evidence that allows us to identify that.
- 7.46 **Unit banking** was horrible for American banking history, but it was wonderful for banking research because we have lots of observations, many different banks, and we know their geographical locations, and we have a lot of data about those environments, and there's a lot of variation across those locations. It's wonderful. Bad for you, good for me. These are the states that are adopting this. One of the things that's really interesting about this is, there are lots of them. They're all being pushed by, as part of this **unit banking** advocacy because **unit banking** creates risk.
- 7.47 And so the people who are unit bankers also want to now create a **risk subsidy** that helps them manage that risk better. And so they're all advocating it. What's interesting is, many of their nearby states almost also advocated it. That's very helpful to us because, from a statistical analysis perspective, we really want it to be the case that the comparison groups we're going to

be using are going to be very similar, otherwise similar. Sometimes our comparison group, most of the time, is going to be national banks or in some cases, even state banks who weren't members of the insured system within the state.

- 7.48 But we'll also sometimes use neighboring states, as you'll see, those right on the border, right across the border from the subject insured banks. It's relevant to also note that all of those states also considered adopting this legislation, and it was only narrowly defeated. They're quite similar in many other regards. This is a list of the eight states that adopted it. It's important to bear in mind that they are quite similar to their neighbors who are going to be our **control group** in some of the analysis.
- 7.49 As I said, in some of the analysis, the **control group**, we call this the **treatment effect**. **Deposit insurance** is the **treatment effect**, and we need a **control group** who aren't affected by the treatment. That's either going to be uninsured banks within the state, or it's going to be, when we're looking at nearby next-door states, all the banks in those states that are uninsured. We get the same results for everything.
- 7.50 What are the insured banks? The ones that are going to be mainly analyzed in most of our analysis based on data availability are going to be the ones that are in this black strip that goes straight down. So first of all, most of the states that we're talking about are just part of this one strip. And the neighboring states' comparison groups are going to be the gray states on both sides. We're also going to, in some of the analysis, look at Washington state and at Mississippi, which also are adopting it, but not in all the analysis based on data availability.
- 7.51 And I won't get into all of the data availability issues or which states are in which sample. I'm just going to show you the results. Any of you who wants to read about it, you can go to the original paper to see all those details. OK, so let's start with just looking at a simple graph of the aggregate. Most of the analysis we're going to be relying on is bank-specific econometric regression analysis, but in making those comparisons to test directly those three hypotheses that I've already listed. But I also want to just start with some very basic graphs.
- 7.52 If you look here, you can see that the non-deposit insurance states, the neighboring states, are the blue line. The **deposit insurance** states that are part of that axis coming straight down are going to be, which are adopting relatively early, they're going to be in the red line, and then some other insurance states that are adopting later are the green line. What you can see is that, looking at the entire state, whether it's insured or uninsured banks, because in each of these insured states, there are also uninsured banks, you can see that from this perspective, the insured and the uninsured states look pretty similar.
- 7.53 And that's all I'm really trying to show you in this graph. Let's just compare the blue and the red line. You can see that overall economic growth, which is driving deposits in both states, sets of states, look pretty similar. Now, if we look at state bank deposits relative to national bank deposits, there you start seeing some difference. The blue line of non-deposit insurance states,

after **deposit insurance** is adopted, start growing. The blue line is growing slower. The red line is growing faster.

- 7.54 So then the aggregate, the adoption of **deposit insurance** causes a shift within the state to banks that are enjoying **deposit insurance**. I'm just showing this to you graphically. I'm going to show it to you in a statistical analysis that's much more powerful in a minute. But just to show you, nothing up my sleeve, right? The result is showing up just in aggregate graphs. You can see it quite visibly.
- 7.55 Now, you might also say, well, how do we know that there isn't something else going on in these states that just happens to be driving some of this? So we have the ability to do what we call **placebo tests** in statistical analysis, which is, wait a minute, it isn't just the passage of **deposit insurance** that's reflecting some political event. It's the actual enactment of **deposit insurance** that's really mattering. So it's not just like, suppose there were a particularly risky environment that caused them to enact **deposit insurance**. And suppose that we're just seeing the effect of that risk, not of **deposit insurance** itself.
- 7.56 Then we would expect this behavior that we've just documented. We would expect that to be reflected from the moment of the passage of the law. But if it's the **deposit insurance** itself, it's the implementation of the law that should be the moment that things begin. And in two of the states, Nebraska and South Dakota, they passed **deposit insurance** in 1909, but didn't actually implement it until 1911 and 1916, respectively. What we can do is show that, that's what this graph shows, that it's not the moment when the law is passed, that the effect of the shifting of deposits is happening; it's after the system is actually implemented.
- 7.57 In one case, that's seven years later. In another case, it's two years later. It's pretty clear that this graphical pattern is reflecting the actual implementation, not something about the moment in which the legislation was passed. Here are the variables we're going to be looking at, which I've already mentioned. The first two are related to the first hypothesis, the second two are related to the second hypothesis. That is, first I'm going to show you now, not just an aggregate, but when we look at individual banks, we're going to see that, compared to banks located in their same states, or in some cases, we'll see neighboring states, that they did grow their deposits more.
- 7.58 Then secondly, the last two variables, we're going to show that they maintained a less prudent balance sheet structure, in that they systematically held less equity relative to deposits and less cash assets relative to loans. Let's look at that. The key statistical analysis is going to isolate the relative effect of the insured banks after controlling for the economic environment where they're located, not just the state, but their specific counties, because other banks are located in those counties, too, that aren't part of the insured system.
- 7.59 The key variable of interest, which is shown in red here, is, after making all of those controls, what's the effect of insurance per se on the incremental activity of the bank relative to its comparison group? I won't explain all that in more detail. I'll just show you. Also, we have a lot

of county-level controls that were in that regression that I just mentioned. The key variable of interest in this regression is this one. Because this is the one that, after controlling for all these other variables, is looking at what's the effect of being a member of the **deposit insurance** system in a state where **deposit insurance** has been passed?

- 7.60** The answer is, all things equal, you have about 30% more deposits and about similar amount, about 25% more loans. So **deposit insurance** had a huge effect on your competitive ability to attract deposits and to make loans. That's the first thing that we show. Now, we can also ask, what's the effect on your risk management? Again, this is the key variable. What we're seeing here is that **deposit insurance** reduces your cash relative to assets, and it reduces your equity capital relative to assets.
- 7.61** And all of the results I'm telling you are highly statistically significant, but not just statistically significant, but economically large. So these are very large, even though that may seem like a small number. It causes your cash relative to assets to decline by 3.5%. Well, if that number is on average 10%, then 3.5% is a pretty big change. And that's roughly what we're seeing on both sides. These are very large differences. And now we can ask the other question, to what extent is this different in some areas?
- 7.62** So I want to emphasize, I've already talked about what World War I did, but I've already mentioned to you new entrants, which would look like small banks or young banks, which tended to locate themselves in these risky counties where crop prices had gone up a lot as a result of World War I. We're going to look for each of those three characteristics. Are young banks especially explaining this? Are small banks especially explaining this? Are these counties explaining it? Let me show you what we're talking about.
- 7.63** When we talk about these particular crops that everyone knows about, that went up during World War I, that's the 12 crops that are going to help us identify counties in which the crops really changed a lot. If you're looking at total agricultural output in these different counties, you can see that it didn't really change that much during World War I because there are only a few commodities that are really very sensitive to the global effect of World War I. But that's these things. So we're going to look for counties that were very intensive in those 12 crops versus counties that weren't.
- 7.64** What we find is, these results are very strong. Again, it's going to be this effect here. Are you a member bank in a **deposit insurance** state? And are you also a small bank? You can see that the effect is to basically double the normal effect. This is the effect for banks that aren't small. They grow their deposits by about 21%, but for small banks, they grow their deposits by about 40%. It's a much larger effect that you get that by summing these two coefficients. The same is true if it's a young bank, the same is true if it's a small bank, because after all, small banks tended to be young banks. And so small young banks tended to locate in these particular kinds of counties.
- 7.65** So whether you condition on the size of the bank, how young the bank is, or whether it's in one of those counties that's particularly intensive in the 12 crops I mentioned, you get the same

result. And that's because those are pretty much the same banks. What about their risk management? Same story. So the risk management effects on cash holdings and on equity are, again, much greater for those. Here we have it again. If you're not a small bank, it's 2.7% decline in your cash ratio. But if you're a small bank, it's 4% decline, the sum of those two.

- 7.66 I won't belabor the point by going through all the coefficients, but the point is, what we're finding, what helps us understand that this is a real effect is, we're finding that the banks that we expect to be relatively affected by the **treatment effect** are the ones that are relatively affected by it. Now, we also do two other things. We just take advantage of variation in the data. One thing is to look at border areas. Again, you'll remember that these are the insured states, these are not insured states. We can also, instead of just doing comparisons within the states of the insured and the uninsured banks, we can also do comparisons of the insured banks with the banks that operate in the adjacent states, which are all uninsured.
- 7.67 We're going to get the same results. In the interest of time, I'm just going to show you quickly that we get similar results. So whether you make your comparison with the uninsured banks within the state or with uninsured banks that are just on the other side of the state line, which are, of course, more similar in terms of their economic environment because they're very nearby, you get the same result. That's very reassuring.
- 7.68 In the case of Kansas, **deposit insurance** was optional. So we have some banks that we can now just look at state banks, excluding national banks, only at state banks within. This is a tricky comparison because, of course, the choice of **deposit insurance** is optional. You might say, well, we don't really understand why you wouldn't choose **deposit insurance** if it's a subsidy. We don't expect the results to be as large, but we do still expect to find something, and we do. The results are similar direction and statistically significant, but as we expect, a little bit smaller because of this **endogenous** choice.
- 7.69 OK, now we get to the question of the mechanism. Can we see that depositors are behaving differently? The answer is yes. We're going to have three measures of risk that depositors could react to because they are visible. One is, on the balance sheet, as I already told you, depositors get to see balance sheets. So how about loans of our assets? Do depositors, all things equal, if a bank's **loans to assets** go up, do depositors in uninsured banks withdraw funds because risk has gone up, while banks in insured banks don't? The answer is going to be yes.
- 7.70 All things equal, when banks' capital to assets goes up, do banks in uninsured bank, do depositors in uninsured banks tend to send more deposits to that bank because it's less risky? Well, insured banks don't do that. Depositors at insured banks don't do that. The answer is yes. When banks start to show that they're experiencing some losses on loans relating to real estate, that's how they get something called **real estate owned**. The way you, as a banker, it may sound good. "Hey, I got more **real estate owned**." No, it's not good. What it means is, I got more **real estate owned** because I foreclosed.

- 7.71 So when we see more **real estate owned**, we're expecting depositors to say, this is a riskier, this bank has a riskier lending strategy, all things equal. We expect withdrawals if they're uninsured banks. But in insured banks, we expect the depositors to ignore that and like they ignored the other information. And so that's what we find. This is just showing it on average. I think what I'll do is just show it to you for the most interesting period. We explain why this year is maybe the most interesting one because World War I is fully operational, and you haven't experienced any of the price decline or the end of the war yet.
- 7.72 What you can see here is the effect of, what we're expecting to see is that for the uninsured banks, a very big effect that's then undone by the effect of insurance. What we're looking for is to see these two coefficients of opposite sign and maybe of a similar magnitude. What we see is, yes, it's true that if you have more **loans to assets** in an uninsured bank, you'll lose deposits. But if you're in an insured bank, that cancels out the effect. There's statistically a zero effect by combining these two.
- 7.73 Similarly, if you increase your equity ratio for an uninsured bank, you get more deposits. For an insured bank, it cancels out that effect. If you have more **real estate owned** because of more foreclosures, in an uninsured bank, that leads to deposits leaving. But in an insured bank, it offsets the effect. You can see very clearly how powerful this mechanism that we've been talking about is. We tested it just now, not just in normal banking environments like what we were talking about yesterday, but in these agricultural areas, too, between these two types of banks. We found depositors really understand how to react to bank risk when they have an incentive to do so. But when they don't have an incentive, they don't react.
- 7.74 Now we're going to look a little bit at the declines of these systems. I'm not even going to spend the time on the regressions because the graphs tell the story so clearly. You can see that if we look at, within these states, national banks who aren't part of the insured system, that's this solid line. So after World War I's over, you see maybe a little bit of decline in the national banking system, but not very much. In fact, if anything, maybe an increase. Why? Because as this system, these are the insured banks, is falling apart, the deposits come back to the national banks.
- 7.75 These are state banks in the non-deposit insurance comparison groups. So you can see, yes, you might say after World War I, there's a little bit of decline in this red line. But these banking systems are still quite viable. They're continuing. National banking system, if anything, is growing a little bit. What's happening to the member bank systems? Death and destruction. Disappearance. And so that's the end. The denouement of these systems happens pretty quickly.
- 7.76 I could show it to you in loans, too. I think it's interesting to note that this is happening in all eight of the **deposit insurance** systems. It's observable. It all happens within a couple decades of the establishment of these systems. All of this, this is statistical evidence that's fairly convincing, but the narratives that were saying this were known all along. So these conclusions that we've

drawn from our statistical analysis, about what were the political origins of this, were to create a subsidy for unit bankers that would allow them to pursue high risks, which ended really badly.

- 7.77 If you doubt whether this was all known at the time, here's what, during the 1932 election, here's what Franklin Roosevelt wrote in his letter to the New York Sun, which was published. He wanted to make it clear in this letter that he would not be advocating deposit insurance by the federal government. Why? He said it would "lead to laxity in bank management and carelessness on the part of both banker and depositor."
- 7.78 Let's just slow down for a minute here. Franklin Roosevelt, it's as if he knew all of our results. He knew that the laxity of the banker, in terms of choosing these kinds of decisions to have very high-risk strategies, was reflecting the laxity of the depositor. Isn't that remarkable? And then he said, because he realized the destruction that happened in those systems and how much loss there was, that if the federal government were on the hook for deposit insurance, he says, "I believe that it would be an impossible drain on the federal treasury." So he's also recognizing the fiscal cost of this.
- 7.79 And of course, as we talk about this in the post-1970 period, what did I tell you? The cost of these protected banking system collapses, in the hundred collapses that happened worldwide, had a median negative net worth, bailout cost, of about 16% of GDP, which is a huge number. He forecasted also that the consequences would be greater severity of loss that would have to be borne by whoever had issued this deposit insurance guarantee. So this is pretty interesting. It shows that the experience of these eight states was well known at the time.
- 7.80 Now, here's what's really also interesting. Franklin Roosevelt was actually pretty smart about banking and made this a key part of his platform going into the election. And yet, once he gets elected, over his own objections, by the way, the objections also of the Secretary of the Treasury, the objections of the Federal Reserve, and the objections of the American Bankers Association, which represents mainly larger banks in urban areas; over the objections of all of those, including himself and Henry Steagall, who was in charge in the Senate of regulatory policy — all of those people were against it.
- 7.81 But he went for it. Why? Because he had to make a political compromise. There were things he wanted to achieve that also Henry Steagall wanted to achieve. The only way that they could move forward on the things they wanted to achieve is by compromising with guess who? The unit banking lobby, which was represented in those discussions. Oh, I said it wrong. Carter Glass was the one in the Senate. I'm sorry, I got the two names backwards. Carter Glass was the one in the Senate who objected. Henry Steagall in the House from Alabama was the advocate of unit bankers in this whole discussion.
- 7.82 And so federal deposit insurance is created. It goes into effect at the beginning of 1934 because of what we call log rolling. Log rolling just means, if you want one thing, you have to give another. And so this thing that's considered one of the great surviving legacies of the New Deal, we'll talk about some more of those legacies. But the thing that's considered to be one of the

most important surviving legacies of the **New Deal**, in fact, was something that **Franklin Roosevelt** was very clearly opposed to, and which many others were opposed to because they were well aware of these facts.

- 7.83 So at the end of today, we're going to talk about regression results from a study of mine with **Sophia Chen** that's looking at the current era, since 1970, that shows all these same findings. But what's so interesting to me is, we didn't enact **deposit insurance** in the US as a national policy because we lacked the information about what it was going to do. We had that information. We didn't have the statistical version of it, but we had very clear understanding of it, based on having lived through it, and we ignored it.
- 7.84 Part of the story was **Henry Steagall** said, "We'll just do it, it'll just be temporary, and we'll do a small amount of it." But then why was that so smart? Because once you put it in, it's very hard to get rid of it. Then you broaden the constituency for it. So he knew what he was doing. OK, let's move on to talk about a couple of other things that are happening during the same era. One of them has to do with agricultural land **credit subsidies**, specifically. Now, of course, you could say, well, **deposit insurance** was an agricultural land **credit subsidy**. You just showed us that.
- 7.85 Yes, that's true. But there also were things called **land banks** that were explicitly designed to make loans, guaranteed loans on land in agricultural areas. Then we're going to talk about residential mortgage subsidies, because those are also starting off in a big way in 1934 as part of **New Deal** programs, federal mortgage subsidies. Let's start with the first one. Given how important the rural populist coalition is to banking policy, it won't surprise you that the first federal **GSEs**, **government-sponsored enterprises**, are **federal land banks** established in 1916.
- 7.86 You might say, wow, that's interesting. 1916, we were just talking about 1916 as this kind of hot moment when there's a lot of pressure to expand credit so that people could take advantage of these very risky hot new areas, right? That's an interesting coincidence. Maybe not a coincidence. The **farm credit system** structures itself through cooperatives that are decentralized. The incentives of these cooperatives to make sure that they're making smart loans are very weak. Who gets the profits? Well, they're cooperative.
- 7.87 The profits are held by the lending institution as its capital. The institution, there isn't a shareholder who's minding the store the way there is in a normal bank. So governance in these kinds of cooperatives is very poor. And they also need the political support in order to continue. They need the political support of a wide range. So if I told you that we're establishing a government-sponsored entity that has government guarantees and that needs political support and is running as a cooperative, how would you expect its risk management to work?
- 7.88 Would you expect them to be, the governance to lead to really smart risk management or to more politicized, weak risk management, trying to get more risky members and not really feeling the incentive? No one individual really feels a strong incentive to make sure that the risk is being managed because there's no individual who's really exposed to loss very much. That's a

problem that is still true in the **farm credit system**. So the big event that happens in this set of intermediaries is the collapse in the 1980s. And what this reflects is the agricultural boom that's happening in the 1970s.

- 7.89 So I'm going to show you what's happening to farmland prices. So the method that we're going to use here is coming from my friend **Mark Carey**, who spent most of his career at the **Federal Reserve Board**. He's an economist who was trained at University of California, Berkeley. He had this great insight in this paper, which is, it's very hard when you're analyzing most big booms in finance to say whether the price boom was warranted. But in the case of agricultural land, we can do statistical analyses that will do a pretty good job of saying what the warranted price is.
- 7.90 Here's why. Suppose I told you that I had a model of the time series properties of the price of wheat. So I know when wheat prices go up, they tend to mean revert over some period of time. By the way, that's true in real terms. So we're taking out the effects of inflation. When a particular agricultural price goes up, it might have some temporary permanence, but then it recedes, because there's going to be supply reactions all over the globe. If the price of wheat goes up, more people start cultivating wheat somewhere. Prices don't tend to persist very much.
- 7.91 You can show that pretty well in your statistical analysis. Then you can do, if that's true, then what that implies, and **Mark Carey** will show that it is true, if that's true, then that implies that the value of farmland shouldn't change very much just because the crop that you're cultivating, the price goes up. Now, the value of the farmland could change a lot if you develop more productive methods for farming. But not just because there's a blip in the price.
- 7.92 He uses this analysis to be able to say what the warranted prices would be of agricultural land, and then show that, the first thing he's going to show is, agricultural land during the 1970s experienced what we call a **bubble**. That is, an unwarranted price increase. Well, sometimes the definition of **bubble** is different. It could just be a price increase. And we've used that, in this course, we've used it more generally. But let's just be clear, an unwarranted price increase. He's able to show that that's true.
- 7.93 That's interesting. It's maybe the first time anyone has been able to show so conclusively, in my mind, the effect of a land price increase and to show that it's the result of an unwarranted increase. Because it's hard to do. If we were doing residential properties, how would we decide whether the increase in the price was warranted or unwarranted? But if it's agricultural properties, we know that the purpose of that property is to grow wheat. As long as that's the highest and best use of the property, then we can analyze what the cash flows are from wheat.
- 7.94 If there isn't a change in productivity, we can say how much the price, the temporary price change, reflects a permanent price change that should be imputed into future cash flows and therefore into a present value of those cash flows, which would be an increased price. The conclusion of his research is, not much. So the price increase. So the first thing he's going to show us is the price increase was very large and not warranted, the price of land.

- 7.95 But then he's going to show us that, during the **bubble**, the **farm credit system** lenders increasingly were the only source of funding. So what's happening is, as the **bubble** is happening, other lenders, commercial banks, not farm **land banks**, drop out of the market. They say, "This doesn't make sense. We can't lend against this land just based on the fact that the land price has gone up because we don't think that's sustainable." So bankers who have their own equity at stake, unlike the **farm credit system**, Mark Carey shows, dropped out of the market.
- 7.96 I did some work on the **farm credit system** back in the 1980s and 1990s, visiting many of the **farm credit system** banks and talking with people about what they had done because there was a widespread recognition that they had lent very unwisely. What was clear was that they adopted a procedure that said, "All you have to do to get a loan from us is bring us an appraisal. The appraisal should just be based on the most recent transaction that's relevant in your area." In other words, during the land price **bubble**, all you had to do was show that the price of land has been going up.
- 7.97 The banks that were competing with the **farm credit system** had much more rigorous standards, and it wasn't good enough to get a loan for a certain amount to just show that prices in your area had been going up. That's why they dropped out of the market. This was, I think, one of the most convincing studies on the effect of credit **risk subsidies**. When governments subsidize credit risk, they tend to create very unwise kinds of lending.
- 7.98 We have, again, just as we did in our study of the **deposit insurance**, we have this very nice comparison between the non-farm credit lenders, the non-farm credit system lenders, commercial banks located in these ag areas, what they did during the **bubble**, what they did was stop lending. And what the **farm credit system** lenders did, which is increase their lending share. OK, so let's take a look at this. First, just to give you a sense, here's a graph of US farmland prices. Just in raw terms, you can see that there's a big increase in prices.
- 7.99 That can be a little bit misleading, though, because what's happening during the '70s, as you probably remember or maybe have heard, is that there was also a general inflation. Let's adjust those prices for inflation. Now we can see, this is really the **bubble** that's happening starting in about 1972. We can see there's a big increase in agricultural farmland prices, which is peaking in 1982 and then declining, and it pretty much goes back to where it was. Why is that? Well, because commodity prices are jumping, but long-term value of farmland isn't changing.
- 7.100 But if you were just using appraisals based on this to justify your lending, what would you do? You'd be making huge amount more lending during this period. Now, let's look at different lenders. This is what tells the whole story. When we look at the **farm credit system**, once the **bubble** gets going by about 1977, now you're starting to see that you can have appraisals based on prices that are going way up, so that their market share goes from about 30-some percent of the market. By the end of the **bubble**, 1981, guess what? 100% of the market of new loans.

- 7.101 So the **farm credit system**, because they're making decisions just based on these inflated land prices, they're 100% of the market by the end of the **bubble**. But of course, the land prices were themselves reflecting who? The **farm credit system**. It was a very circular process. Because they were willing to lend just on the base of any recent run-up, if there's a run-up in price, they lend against it, which fuels the **bubble**. They're the ones who are causing the **bubble** to really unfold over time, and they become the ones who make 100% of the loans in it.
- 7.102 Now, one thing you might ask is, why wasn't some smart person who saw that this wasn't sustainable, why wasn't some smart person out there betting against the increase in farmland? How would you do that? Nobody is raising their hand. The answer is, you can't do it. In the stock market, you could do you do it by shorting the stocks. But stocks are like the stock of General Motors is a homogeneous asset with lots of shares. How do you short farmland?
- 7.103 Well, you'd have to imagine a short transaction against all the individual parcels of farmland, but there is no market for shorting farmland. The only people who express their opinions about the value of land, that's expressed by buying and selling. That's how it's expressed. You can't short it. You can sell it. But then if you don't own it, how do you sell it? You can't. And so during a **bubble**, smart people sell the land. They can't keep expressing their opinions once they're out of the market and have sold the land. So it's only the optimists who can express their opinion.
- 7.104 So this is reflecting the opinions of optimists, the run-up in price, which has fueled, that optimism has fuel, which is the government-subsidized credit. What we learn from this is, if you want to see how to really fuel land bubbles, then you provide government-subsidized credit to land purchasers. We're going to see, after our break today, we're going to come back to talk more about the 2008 crisis, which is, of course, exactly the same logic, fueling through subsidies to residential mortgages a **bubble** in residential mortgage land.
- 7.105 You might also wonder, after the 1980 crisis, what happened—the crisis of the 1980s. There were lots of changes in the **farm credit system**, but they really didn't address the fundamental governance problem. They did require the **farm credit system** to increase the capital ratios. So these were required to go way up. And there hasn't been, to my knowledge, at least, I haven't tracked it very closely, a collapse of these institutions since they collapsed about 30 or 40 years ago. But do we expect this to happen again? I would say probably yes.
- 7.106 Why? Because the incentive problems of the combination of the subsidy and the cooperative structure is maintained. I don't think we've fixed that problem, although it's been dormant for a few decades. The last topic I want to cover before our break is the **thrift** collapse of the 1980s. Well, this is another example of a **credit subsidy**, in this case, a **credit subsidy** to residential mortgages, that led to a major event, very costly financial collapse, during that same period. I think this isn't very well understood.
- 7.107 Prior to 1934, the way that people got mortgages on their homes was completely different from what happened after 1934. Let's first talk a little bit about the old system. The old system prior to 1934 was that you, the typical mortgage would have very small number of years of maturity,

maybe one year, maybe three years, maybe five years. The reason for that is, it would be very hard, it's very hard anyone to make a loan contract, any lender, to make a loan contract with a long-term fixed rate of interest, because that exposes you to **interest rate risk**.

- 7.108** If you don't have some natural match for the long-term exposure, that risk is very hard to manage. Now, insurance companies were sometimes in this market, and because they had longer-term exposures on both sides, they could make relatively long mortgages, but they didn't make **30-year fixed-rate mortgages**. Nobody did. The **30-year fixed-rate mortgage** is not something that the market has ever produced because it creates risks that are very hard to manage. And so typically, **thrifts** prior to 1934, what were they? The word **thrift** means they're involved in the housing business.
- 7.109** Well, they were things that were either insurance companies or building and loan associations. Building and loan associations are not **savings and loans**. That's a different thing. There's some wonderful new research about this. We don't have time to go into it in great detail. But building and loan associations and banks, to the extent that they provided mortgages, and insurance companies tended to provide short-term mortgages that didn't have a lot of long-term **interest rate risk**.
- 7.110** I want to teach you the concept of **interest rate risk** in a simple way. Suppose that you had any asset with a fixed interest rate. Let's say it could be a bond or it could be a mortgage. Let's talk about it as a bond. Let's say that you had a 30-year treasury bond. So we'll just take credit risk out of the equation. Now, the same logic is going to apply to mortgages or other assets that have some risk. A 30-year bond, and let's say that you're getting about 5% coupon on this 30-year bond. Then let's say that interest rates go up by one percentage point in the market.
- 7.111** So at the time that you bought the 30-year bond, the yield to maturity on the 30-year bond and the coupon were both 5%. And let's say a year later, interest rates for 30-year bonds go up to 6%. How much have you lost? I'm embarrassed I'm going to do this on videotape, and I'm doing it based on a rough calculation, but I'm going to say you've lost about 15% of the value of the bond, just based on what we call **duration** analysis. But I'm guessing what the **duration** is. I'm guessing it's about 15 on the 30-year bond.
- 7.112** Think about that. Then if interest rates went up an additional 1% to 7%, another 15% loss. That's what I mean by **interest rate risk**. That because something has a very long maturity, if there's a change in interest rates, it tends to produce a dramatic change in the value, even of a riskless instrument like a treasury bond. We call that **duration risk** or **interest rate risk**, and it has to do with long-term **instruments**.
- 7.113** The market simply couldn't manage risks like that. Supply and demand in the market prior to 1934 never had **30-year fixed-rate mortgages**. But you could see that it could be very popular to create **30-year fixed-rate mortgages**, particularly if there were an additional **risk subsidy** involved in those mortgages. In 1934, the **New Deal** under **Roosevelt** establishes three programs

that are targeted toward reducing the interest rate on mortgages and allowing people to have very long maturing mortgages. What were those?

- 7.114** First, it was the creation of federal **thrifts**, so **savings and loans** that enjoyed the protection of **deposit insurance**. As long as you were willing to accept a charter that put you squarely into the business of lending residential mortgages, so that's going to be your whole portfolio, residential mortgages, you can have **deposit insurance**. The argument was, well, now we don't have a lot of **market discipline**, and that's good because it frees these **savings and loans** to be able to do **30-year fixed-rate mortgages**. They don't have to worry about depositors disciplining them.
- 7.115** Did that get rid of the **interest rate risk** of doing a **30-year fixed-rate mortgage**? No. All it did was subsidize it. Because now in the state of the world where everything's fine, you, the stockholder of the **thrift**, get to keep the money. But in the state of the world where interest rates jump up, what happens? You become maybe insolvent, but maybe you don't experience a run. Why? Because the depositors don't have to run. Why would they? In that state of the world, maybe you can buy time, maybe you'll hope interest rates come down, maybe a lot of things.
- 7.116** Now you understand what was the initial shock that led to the savings loan collapse that we experienced in the 1980s. Interest rates, because of the high inflation episode and the Fed's fighting inflation, finally after 1979, interest rates spiked. And when interest rates spiked, **savings and loans** lost a huge amount of their value. And then what happened was that the regulator saw that the savings loans lobbied. They said, "Look, we've lost a lot of our value. We need new sources of revenue," which meant new risk-taking opportunities. Why?
- 7.117** Well, if I've lost everything, I've got nothing more to lose, right? So then I lobby the government to give me new opportunities to take risks. I'm not kidding. So now I can lend against shopping centers. I can invest in junk bonds for the first time. And so we're going to do that because we've got nothing to lose. And so the losses in **thrifts** reflected first the big shock of the increases in interest rates. And then secondly, the regulatory choice to allow increased risk activities that deepened the losses of the **thrifts**. A lot of them experienced both of those losses.
- 7.118** Then you might say, well, why didn't the government resolve this problem? Because most of the insolvency was there by the early 1980s and visible, and everyone knew it. I was there doing research. I can tell you, we all knew it. Why didn't the government resolve them? Because this is not politically something that anyone wanted to do. If you talk to people in the George H. W. Bush administration, what they'll tell you is, the reason that **FIRREA**, that's a regulation that was passed, not a disease, a regulation was passed in February of 1989, I believe it was.
- 7.119** Why was that the case? Guess what had just happened? The presidential election. So nobody wanted to take on this very politically unpopular problem in the early 1980s, and certainly not in 1987 or 1988, when it was really obvious prior to the election. So they waited until after the election. As I've told you, this is a general pattern. Resolving unpleasant things that you ultimately feel like you have to deal with is something that's done after elections. And that's not

just a story about Latin America or East Asia. That's a story about very much also the United States.

- 7.120 That wasn't the only **New Deal** program. There was also the creation of this thing called **Fannie Mae** that we're going to talk more about later today, and the creation of the **Federal Housing Administration**. **Fannie Mae** was there to create a secondary market for normal mortgages. In other words, if the mortgage originators had a place to then sell their mortgages to somebody who was purchasing the mortgages, constructing mortgage portfolios and funding those in the secondary market.
- 7.121 **FHA** was designed for people who couldn't qualify for mortgages from normal originators like **savings and loans**, who then therefore got a special **credit subsidy** from the **FHA**. There were three **New Deal** programs, one designed for people who couldn't qualify for normal mortgages. The other, that was the **FHA**. Then there was the **thrift** charter with **deposit insurance** that was there to subsidize normal mortgages. Then an additional subsidy for normal mortgages was **Fannie Mae** that created a secondary market for purchasing them at subsidized prices, at higher prices.
- 7.122 The combination of **deposit insurance** and **Fannie Mae** subsidized normal mortgages, and the **FHA** subsidized the risk for people who couldn't get normal mortgages. The resolution costs of the savings loan crisis, when it hit, were perceived to be as great as \$300 billion. Things improved in the market. As they're selling off all of the foreclosed assets, **ex-post**, the cost was only about \$150. But I would say the true economic cost is better thought of as 300 because that's what we were all on the bill for.
- 7.123 Do we know that people took advantage of the **deposit insurance** and incentives to take on risks in these **savings and loans**? And the answer is, we do. And so my friend **Eli Brewer**, who was recently retired from the **Federal Reserve Bank of Chicago**, wrote maybe the most convincing article showing that the **thrifts** that were initially positioned to have the biggest losses, which they didn't fix by raising their capital—so **thrifts** that were very severely insolvent or near-insolvent, behaved differently from **thrifts** after, let's say, 1982 that remained very well capitalized.
- 7.124 So the **thrifts** that were well capitalized, once they had suffered these initial losses from high interest rates, they didn't have any incentive to take on the new risky activities that were allowed under the 1982 law. But the **thrifts** that were deeply underwater, they were the ones that went crazy with lending against very risky assets. What his analysis really showed was, that second round set of losses in the **thrift** industry was very much understandable in terms of the incentives of **thrifts** that had already lost their equity to gamble for what we call **resurrection**.
- 7.125 Whereas if you hadn't lost your equity, you didn't have that incentive to gamble. This is all now part of what I would call the vocabulary of the economic literature on **risk subsidies**, how they create systemic risk in the financial system, and how incentives operate through the effect of these **risk subsidies**, both incentives of depositors and incentives of lenders.

8. Need for Reform

AT A GLANCE

Introduces the two gorillas of the post-1970 crisis era, government protection of banks and subsidization of mortgage risk, and shows they are married. Analyzes the Mexican crisis of 1994 (bank privatization with 100% liability insurance, fiscal expansion, sterilization, an overvalued crawling peg) and the Korean crisis of 1997 (chaebols implicitly guaranteed by the government), both predictable twin crises. Explains the US subprime crisis through the GSE Act, merger-related commitments to activist groups, deteriorating mortgage standards, phony capital measures, and loose Fed policy, plus Spain's Cajas. Presents global evidence that generous deposit insurance raises risk and mortgage lending, and closes with questions on geopolitics and fiscal dominance.

- 8.1 Welcome back. In this last 90-minute session, we're going to talk about the most recent events relating to **financial crises** worldwide. And of course, our motivation, we've already mentioned, comes from how unusual the period since 1970 is globally in terms of **financial crises**. And when I say **financial crises**, as you'll see, it's not just **banking crises**. There are some events that aren't just **banking crises**.
- 8.2 But the thing that's unusual is the frequency and severity of **banking crises** has really been the defining characteristic of this period. We don't have another period in history. The **Great Depression** doesn't even come close to the current era, even if you just take those few years around the **Great Depression**. As you saw in the US case, maybe 3% of GDP, negative net worth of failed banks. But in our current era since 1980, it's a median loss rate, for the hundred **banking crises** that happen, a median negative net worth of failed banks of 16% of GDP.
- 8.3 To compare it to, I think, a very comparable era, so if you take 1980 to 2013, and you compare it to the period 1874 to 1913, which in many respects is a very interesting comparison. But the current era, 1980 to 2013, has **banking crises** 10 times as frequently, and on average, they're five times as severe. That's a pretty dramatic fact. How do we understand this fact? I think that there are, well, if we're talking generally, I like to talk about the **two gorillas** in the room.
- 8.4 We have this expression, the **gorilla in the room**. Often, the reason I use that expression is, **gorillas** are very destructive. Don't bring a **gorilla** into your room unless you want to see a real mess. What about these **two gorillas**? Why do I talk about it as **two gorillas**? Because there are two different influences that are causing the destruction. But also because the expression, **gorilla in the room**, is often used to talk about something that people know is a problem, but purposely don't talk about it.
- 8.5 Oh, the **gorilla in the room** is, she doesn't get along with her mother-in-law. That's the **gorilla in the room**. Nobody's talking about that. But that's why we're having all this tension here tonight. The **gorillas** in the room, there are two of them. They are the subsidization of banking risk through the protection of banks. **Deposit insurance** is the most important element of that. But

then there's also additional protections. When banking systems go bust, banks get bailed out beyond what the commitment in **deposit insurance** was.

- 8.6 Remember, we had in the US **TARP** in 2009, which was a program that provided **preferred stock assistance** and even common equity assistance to a couple of the largest banks to help them deal with the problem. In other words, the government not only guaranteed their deposits, but also helped them to economically recapitalize with preferred stock and even equity in a couple of cases. The first **gorilla in the room** is bank protection by government.
- 8.7 The second **gorilla in the room** has been residential real estate subsidization through risk. That is, **mortgage risk** subsidization. We're going to talk about that primarily from the perspective of, the example we'll talk about of that is the US **subprime crisis** of 2008. When we talk about **banking crisis**, banking protection risk, we'll talk about the **subprime crisis** of 2008, too. But we'll also extend that to our discussion of Mexico in 1994, to Korea in 1997.
- 8.8 Then what we're going to do is, take these **two gorillas** and analyze them using data from all over the world during this period, to see if we can quantitatively show, as we did in our analysis of **deposit insurance** in the early 20th century in the US that was passed by some states, see if we can show convincingly that the **treatment effect** of **deposit insurance** is leading to purposeful risk-taking in banks related to these kinds of variables. But we're going to find out something more. Here's the big shocker. Well, actually, it's not a shocker for you. The **two gorillas** are married.
- 8.9 So that's not very encouraging. That means we're going to have some more **gorillas** maybe, right? But the **two gorillas** are married. Now, we already saw one example of this. What was it? The chartering of **thrifts** in 1934 with **deposit insurance**, because the subsidization, you wouldn't have gotten **30-year fixed-rate mortgages** to be something anyone was willing to do, which was a big residential real estate subsidy, if you hadn't also given those **thrift** institutions **deposit insurance** that insulated them from the discipline of the market.
- 8.10 That's an example of where the **two gorillas** are married. They're related. Doing something with **deposit insurance** can encourage the banks to be part of the government's plan for real estate subsidization. That example turns out to be a very telling one because we're going to see this happening all over the world in the last four decades. That's the message that we're going to get when we think about these **two gorillas** and about the more recent period.
- 8.11 That the two most important sources of systemic risk are these **two gorillas** of the protection of banks, the subsidization of bank risk by governments as part of their partnerships with banks to produce political outcomes, and the subsidization of real estate risk. Then we're going to see these two feed on each other. There's going to be another element that's going to come up in these first two cases I'm going to talk about, which is the Mexican crisis of 1994 and the Korean crisis of 1997.
- 8.12 That other element is, sometimes in some countries, let's call them relatively autocratic countries with government partnerships between large conglomerates and government. In

Mexico, that means the so-called "**grupos**" that are running, not just running banks during this period, but are also running major industrial conglomerates. Those **six families** who control those are the six prominent and most politically influential families in Mexico. In Korea, you had something similar. In Indonesia, you had something similar.

- 8.13** In Thailand, you had something similar, where the government, even though it's democratically elected in all these countries, is relatively autocratic in terms of the way things are actually run, and that there are close partnerships between the people that are running the government and a small group of industrial conglomerates, whether they're called **grupos** or they're called **chaebols** in Korea.
- 8.14** What we're going to see, the reason that this is this third element, it's also a **gorilla**, but it's not as pervasive. It's true in some countries. I wouldn't say that's true in, let's say, the US or in Germany or in other countries that have **financial crises**. It was, however, true in Iceland as well, by the way. There's this other element, which is sometimes the government partnership with banks is reflecting an additional partnership that has to do with a particular group of very powerful industrial conglomerates.
- 8.15** When that's happening, we have an additional element of credit risk subsidization that's happening through the subsidization of the funding costs of those large conglomerates. When we talk about the Mexican and the Korean collapses as examples of that, you're going to see very much how that subsidization is reflected in very large collapse costs. What I want to do is give you a quick review of the events in the Mexican and Korean crisis with this understanding that, in these cases, you might say that these influences were more important than they are in other countries, where it's really a story about the **two gorillas**.
- 8.16** After we go through the Mexican and Korean case, we're going to turn to talk about two cases where it's really all about the subsidization of residential real estate risk. That's going to be the US **subprime crisis** and the Spanish crisis of 2008. Then we'll conclude with this general nationwide analysis of the statistical evidence that these **two gorillas** are so important and that these **two gorillas** are married. That's our plan.
- 8.17** Let's talk about Mexico. As I said on the first day, one of the things if you've been studying, I've been a professor working in this area for more than 40 years. Of course, I was reading in the early 1980s, I was reading about the Chilean crisis as it was unfolding. What's really uncanny, I won't talk through all the elements of that was, but those of us who experienced that couldn't believe how similar the Mexican crisis was to the Chilean crisis. It had all the elements.
- 8.18** The **grupos**, who had very deep involvement with the banking system, and also this fixed exchange rate, which ends up collapsing, and also these very large bank negative net worths that are reflecting how overextended and protected the banks were that made them take on huge risks. Often, and those risks were very much related to the **grupos** that they were directly involved in. Those same sorts of themes come out again in Mexico in the early 1990s, and it was really surprising to many of us.

- 8.19 I've been working many years with teams of people at the World Bank and later at the IMF, and the World Bank had a great team working on systemic problems in banking systems. And so all of us really got to be very close friends and were involved in all of these crises together. There's an interesting little cultural aspect to that. The people who were involved in that went on to play central roles later at places like the IMF, this World Bank group under Jerry Caprio. Then those people who were involved in that as their training ground went on to play very important roles at the IMF, at the ECB, and at the BIS.
- 8.20 It really was this—we were all learning together about the crises that were unfolding, Scandinavia in the early 1990s, Latin America, also East Asia. This was really an interesting worldwide phenomenon. People were collecting up a lot of cases. Now, when you start with the discussion in Mexico, you have to go back to something we've already mentioned, which is the collapse of the Mexican banking system through nationalization in the early 1980s because the government was so desperate for fiscal resources.
- 8.21 Now you might wonder, well, wait a minute, how do you have a private banking system collapse in 1994 if you had the banks all nationalized in the 1980s? The answer, that's a tricky one. Even you might find it hard to believe how clever governments can be. After you steal the banks from your friends, remember, what's the definition of a friend in emerging market banking? The last one you stab in the back, right? But you did stab them in the back, right? So after you stab them in the back, now in 1990, what's your new plan?
- 8.22 You want to sell their banks back to them, so you can make more revenue. It's a great plan. First, I steal the banks, and then I sell them back. Wait a minute. How does that work? First of all, I've also trashed the banks in the meantime by basically not making good loans, stealing all that, whatever was there. So hold on. How can governments pull this off? This is not easy. First, I steal the banks from my friends who now know they can't trust me. Well, they always knew they can't trust me under certain circumstances. Then I'm going to trash the banks by using them to make revenues for myself.
- 8.23 And then I'm going to try to get more revenue by selling the banks back to the same people that I stole them from. How in the world are you going to convince them to buy the banks back? And the answer, in the case of Mexico, is brilliant. The banks are trashed. They're not worth very much. So what am I going to do? I'm going to tell the people who are buying the banks—and I'm pretty much going to allocate the banks to my friends as I normally would, not have a real auction. Don't think like this is a real auction. I'm going to allocate it to my friends and the six families.
- 8.24 But the way I'm going to do that is, I'm going to tell them, "When you buy the bank, no money down." What you can do is, the money that you need to pay for the bank, you can get from the bank lending to you. You have, don't have to put any real capital into the banks. That's number one. But that's not going to solve the problem. OK, what's going to make this something worth my time? We're going to have 100% liability insurance on the bank. So any liabilities the bank undertakes, the government guarantees 100%, not just deposits, any liability.

- 8.25 So now you're telling me, very little downside to you, all upside, 100% **deposit insurance**. So what is this deal really? It's the government, actually, that's desperate for resources, getting resources from who? Not getting them from the people who are buying the bank, getting them from future taxpayers. Because what I'm doing is, I'm setting up a very risky system that's probably going to collapse in the future because it's not properly capitalized and with lots of incentives for risk-taking.
- 8.26 But I know that by making this commitment to **deposit insurance**, it's credible. And so where am I getting the money? Where's the money coming from? It's coming from the protection of the government, which means by the taxpayers, when you-know-what hits the fan. So it's very smart. It's borrowing from the future by offering this contingent liability protection right now. It's a brilliant plan. Notice that that is going to work, and you're going to get even very deeply insolvent institutions like **Banca Serfin** are going to be purchased and going to operate now as private banks.
- 8.27 The other background here is going to be that the Mexican central bank and government are going to maintain what's called a **floating peg**, and I'll show you some graphs that will illustrate this. But think of it as, I'm sorry, call it **crawling peg**. It's an exchange rate. It's not absolutely fixed relative to the dollar, but it's fixed based on a formula that can gradually change, having to do with the difference between Mexican inflation and US inflation. But you can think of it as a fixed exchange rate. That's going to be good enough.
- 8.28 What's going to happen, that's really, what's happening here is the **PRI** is in desperate circumstances to preserve its power. As we know, after this crisis, the **PRI** is pretty much destroyed by it. It loses the lower house in Mexico, and it really never restores its power. You get the rise of the **PAN** initially, and now, of course, the Moreno party. This was a last-ditch effort to preserve its power using these kinds of tools of extreme, trying to rebuild those alliances with the **grupos** and at the same time using the **deposit insurance**, in this case, 100% liability insurance, as a way to try to grab resources from the future as part of its fiscal measures.
- 8.29 What we're going to see in the Mexican case is that the government, because of its desperate fiscal situation, borrows hugely and also prints lots of money, and ultimately it experiences a complete collapse of the banking system and of the exchange rate in a pattern which is now very familiar, the so-called **twin crises**. I'll come back to talk about that in just a minute.
- 8.30 The Korean experience is different. Unlike the Mexican experience, we don't see any big expansion of the normal fiscal accounts in Korea or any of the other East Asian countries, nor do we see any obvious, irresponsible **monetary policy**. This is not driven by monetary and fiscal expansions that are inconsistent with the financial system's health or the exchange rate as they were in Mexico. That made it somewhat puzzling to people initially. Why is it that these three East Asian economies, Korea, Indonesia, Thailand, all collapsed in 1997?

- 8.31 But what's interesting in these cases is that that **chaebol** connection or **grupo** connection is still present. The way it works is that the **chaebols** have access to subsidized government credit to take on risks with government funding through what? Protection of the banks that are lending to them, but also implicit protection that the government is providing to the international bond issues by the **chaebols**. And there's clear evidence that I will show you in a minute of the fact that the markets understood that the Korean government was protecting these **chaebols**.
- 8.32 And so the **chaebols** could issue, not just get domestic loans from the banks that the government was protecting, but also be able to even borrow in the international bond market at very far reduced yields because everyone understood, if a problem happened, what would be likely to be the outcome? The Korean government would go to the **IMF**, borrow dollars, hand them to the **chaebols** who would pay off their dollar-denominated debts in the international bond market.
- 8.33 By the way, that is exactly what happened. But the market isn't surprised by that. Everyone knows that. That's why the **chaebols** were able to borrow incredible amounts that are completely inconsistent with the debt ratings that their debts had, as we'll see. Their debts should have been rated like junk, but weren't rated like junk because they were implicitly guaranteed by the Korean government. Markets understand things that are not necessarily the news that's being published in the newspaper.
- 8.34 Those **gorillas**, we can talk about our **two gorillas**, and we understand what this means. In this case, it's that third **gorilla**, the relationship between the conglomerates and the government, that the markets understood. Another interesting thing to note about both the Mexican and the Korean cases is that neither of them is a boom. Remember, we started off this course talking about this **Minsky-Kindleberger** theory. But wait a minute. The **Minsky-Kindleberger** theory is that a boom leads to, driven by greed, leads to a collapse driven by fear. But we didn't have that.
- 8.35 Actually, both of these are happening in economies that are experiencing mild growth followed by recessions, but not anything like a boom. Korea doesn't experience a boom. In fact, there's a decline going on in Korea, a growth rate decline that's happening for many years prior to the crisis. So all of a sudden, when we start thinking about crises, we realize there are big limitations from this one-size-fits-all notion of asset price going way up and then collapsing. That's not general. That happens sometimes. We saw it in the **South Sea bubble**, maybe. But that doesn't mean it's happening always.
- 8.36 And these crises don't tend to have that kind of feel to them. I want to emphasize that what's new here in some sense about this whole era is that crises aren't just a stock market collapse, or a banking collapse, or an exchange rate collapse. But we're seeing, especially these things called **twin crises**, where there's a big change in exchange rate—you'll see how big when I get to the examples— where an exchange rate peg that some government has set is collapsing, collapsing by an amount that's greater than most of the things we've ever seen, simultaneous with the banking system collapsing.

- 8.37 You might say, what's the connection? Why should these things be connected? There's a few different connections, but the most important one is that if the banking system is protected by the government, and if the losses, when the banking system shows how much loss it has, are, let's say, 20 or 30, or we'll see 40% of GDP, or even larger in the case of Indonesia, then once those losses become realized and the government has to pay for them, that often puts the government's debt-to-GDP ratio so high that there's no choice but to print money.
- 8.38 And so the collapse of the exchange rate is often clearly aligned with the collapse of the banking system because of the fiscal consequences of the banking system. Remember, back in Australia in 1893, which I mentioned in this slide, there was a big collapse of the banking system with no change in the exchange rate. And the reason was, there was no fiscal consequence. But in Argentina in 1890, we had the first really big **twin crisis**. Why? Because when the banking system collapsed, the government had guarantee costs through the cedula offerings that it had to make good on.
- 8.39 There's a reason why **twin crises**, which we've seen only very occasionally in the past, we see in this era since 1980 quite frequently, especially in **emerging markets**. OK, let's talk a little bit more about the details here. As I mentioned, in the Mexican case, there's a lot of evidence, which I'll show you graphically, for fiscal expansion. The other thing to bear in mind is, this is an election year in Mexico, 1994. And so the banks are also being pressed by the government very much to expand, guess what, residential mortgage lending and other consumer lending as part of trying to build the popularity of the **PRI** going into the 1994 election.
- 8.40 So the banks are being asked to extend themselves. Of course, they have very little capital; in some cases, very negative capital. It becomes apparent that the economy is not growing like gangbusters. In fact, at the end of 1993, the Mexican government recognizes you're entering into a recession. But what's interesting is, we had what's called a clear **overvaluation** problem of the Mexican exchange rate, certainly by mid to late 1993.
- 8.41 In fact, the late great economist, **Rudiger Dornbusch**, who was a professor at MIT, publicly, to try to encourage the Mexican government to resolve this before the crisis hit, in 1993, more than a year prior to the crisis hitting, publicly said, "The Mexican exchange rate is **overvalued**, unsustainable," because what's happening? Well, when you set an exchange rate for it to be sustainable, the exchange rate is the price of your money compared to the dollar, right? So you have to make sure that the supply of your money is credibly not going to grow like crazy because then you can't maintain the exchange rate.
- 8.42 You can't maintain a high price for something if you're flooding the market with the supply of it. And so, fiscal policy, why is that relevant? Well, because if I'm issuing a lot of government bonds that I know I won't be able to pay back, that's putting pressure on you to have to issue money in the future, right? So if I issue unsustainably high deficits, that means it's very likely the government's going to be in what's called a **fiscal dominant** situation, which means they'll have to print money to repay that debt because they can't just keep borrowing.

- 8.43 Secondly, as we'll see, they were involved in a **monetary policy** called **sterilization**, which meant that, as the central bank was experiencing capital outflows from Mexico, they **sterilized** the effect to make sure that the money supply didn't shrink as a result of that. That means that, when you experience outflows, you increase your purchases of—the central bank increases its purchases of domestic government bonds. I'll show you a graph of that, too, that shows you that **sterilization** was 100% during this period.
- 8.44 You do that to try to preserve the current health of the domestic economy. That's why you're doing this expansionary fiscal and **monetary policy**. Make yourself popular. And after all, it's an election year, right? But that turns out to not be consistent with being able to maintain a peg. And then the banks, of course, are very insolvent. The government's bearing an additional fiscal drag related to the eventual fiscal cost. It'll turn out to be about 25% of GDP to bail out these banks. Everybody knows that those banks are insolvent. The market knows it. The market knows.
- 8.45 And so what you're doing is putting yourself in a position of maintaining an **overvalued** exchange rate, exchange rate that's not sustainable, that's not credible, and a banking system that everyone knows is insolvent. And so this combination isn't going to work out very well. In Korea, as I said, it's a very different situation. There's not a government issuing lots of money or lots of debt. Instead, what's happening is, the government is building up its own contingent liability, its own guarantee bill, coming from the fact that its allies, the **chaebols**, are losing their edge.
- 8.46 They did very well in the 1970s in their growth, but in the 1980s, not so well. We start seeing that the people who have control over being able to get access to credit through their political connections, both access to credit in the domestic banking system and the international bond market, aren't really producing very well. They're not producing profitably. In fact, small firms in Korea, small manufacturing firms, are being denied access to credit because these big guys have the political connections. But that means the banks are funding, and the international bond market are funding, unprofitable enterprises.
- 8.47 That's where the government's fiscal cost is going to come from. It's going to be about 30% of GDP. But it's not coming from the fact that they're issuing their own government debts. It's coming from the fact that they are subsidizing the credit risks of unprofitable, bloated conglomerate enterprises. Very different kinds of stories, but with a similar story, which is, there's going to be a **twin crisis** in both cases. One, because the Korean exchange rate, in real terms, as we'll see, has to depreciate because of the declining productivity of Korea. That's something we'll explain in just a minute.
- 8.48 It's called the **Balassa-Samuelson** effect. But also because, once they have to pay the costs of subsidizing the **chaebols** after the collapse happens, there's going to be a big fiscal cost, about 30% of GDP that's going to also put added pressure on the **nominal exchange rate** to depreciate. In terms of the timing in Mexico, it's also worth mentioning that everyone's waiting after President **Zedillo** is elected. The reason it happens sort of the end of 1994, beginning of 1995, is,

people inside Mexico are waiting to see what is Zedillo, who, of course, was a PhD economist, is a PhD economist, I think, still alive, and a pretty good economist.

- 8.49 So is he going to announce, as part of his new presidency, some major reforms that might make the exchange rate sustainable and they might try to rebuild the banking system? But that's a pretty big political lift, right? So what becomes clear, I sometimes call this the dog that didn't bark in reference to this famous Sherlock Holmes mystery, is, he gives the speech that doesn't say that. And when his speech doesn't say that, that's, I think, a triggering event inside Mexico, that people realize the new administration isn't going to fix the problem.
- 8.50 And the run on the peso begins inside Mexico because that's where the political information about things are really first visible. Let me just show you some graphs relating to this. We can see the big buildup in government debt happening, government domestic debt happening here. You can see during 1993, a huge increase in the amount of government debt. Reserves at the central bank are falling. I'll show you the sterilization, and we'll discuss that a little bit.
- 8.51 But this last bit, the collapse of reserves happening toward the end of 1994, was actually not disclosed to the market. So what both the Thai government did and the Mexican government did during their crises in the '90s was, lie to the market about the balance sheet of the central bank. There's a lot of lying. They'll also lie about the condition of the banks. That's very normal. This was a little bit more unusual, not properly revealing what was going on in reserve flows.
- 8.52 There was also, associated with the same impecunious situation for the government, big increases in printing money. You can see also, this graph shows a very close relationship between the printing of money and the inflation that's happening in Mexico. Clearly, the inflation is happening as a result of bond issuance and a government bond issuance and money issuance that's making people very nervous. This inflation is making people very nervous about the exchange rate, because if you print a lot of money and you cause domestic inflation, how do you keep your exchange rate fixed?
- 8.53 Remember, Mexico has no capital controls at this time. You can't have it both ways. You can't be printing lots of money, issuing lots of debt that makes people see more printing coming, and maintain this exchange rate. I mentioned sterilization. Let me just explain why that's so important. In central banking, we have something called the rules of the game for exchange rate management. And what the rules of the game say are, when you experience a capital outflow, when your reserves start to leave—so central banks, let's back up, central banks have to maintain currency pegs by having reserves.
- 8.54 So when people come in and say, "Give me the pesos back and give me dollars. Here are some pesos back, give me dollars," and then they leave the country with that, maybe. So when the central bank is losing reserves, it's because people are asking for it to redeem outstanding pesos with dollars. That's what it means to set an exchange rate. Now, the rules of the game in central banking is that when you see people coming in and starting to present you with pesos and ask

- for dollars, what are you supposed to do? What you're supposed to do is, contract the money supply. That's called the **rules of the game**.
- 8.55** You raise interest rates so that you express the fact that you're really going to protect this. You reduce the total supply of money, in other words, because you're seeing some example that people are losing faith. When people are losing faith, what do you do? You cut the supply of money so you can restore faith. Those are called the **rules of the game** in central banking for maintaining a fixed exchange rate.
- 8.56** **Sterilization** is the opposite. **Sterilization** is, when the money flows out, that has, through the central bank's balance sheet, mechanically, that has a slight negative effect on the money supply. So what the central bank does is offset that effect by, for every dollar that flows out of reserves, it buys a dollar of domestic treasury bonds to make sure that the money supply isn't affected by the outflow. But that means that what you're doing is the opposite of restoring confidence, right?
- 8.57** When you look at this graph, you see what looks like a mirror image a pond here of these two graphs. That's not mechanical. That's not something—that is showing you that reserve outflows are matched dollar for dollar by purchases of domestic bonds. So the government is clearly pursuing a **sterilization** policy. That's what that mirror image is telling you. So when you're doing this—issuing lots of debt, printing lots of money, and running a **sterilization** policy—what signal are you sending to the market about your exchange rate? Not a very good one. You're losing reserves and you're showing no real commitment to keeping them.
- 8.58** How **overvalued** is the exchange rate? You may remember I mentioned **Rudiger Dornbusch** somewhere in here saying, "Look, guys, this exchange rate is not maintainable." What he was basically saying is this line is the **real exchange rate**. And what he was saying is that the **real exchange rate**, which is the multiplication of the **nominal exchange rate** multiplied by the relative price levels in the two countries. What really is going on in this graph that's making this go up is that Mexican inflation is going way up relative to US inflation, even though the exchange rate, which is this graph down here, isn't going up very much.
- 8.59** When you see the **nominal exchange rate** is not moving very much and the **real exchange rate** is going up, what that really means is all this inflationary policy with monetary and fiscal policy going on is driving up Mexican prices and making them very unsustainable in their exchange rate policy. It wasn't rocket science for people like **Rudiger Dornbusch** or others to point to this and say, "Hey, you got to control your **monetary policy** and your fiscal policy or else you're going to have a big exchange rate collapse."
- 8.60** How big was that exchange rate collapse? Well, pretty big. The peso goes from a number like three and a half pesos to the dollar—see how rigid it is because you're maintaining it—but once it explodes, it basically goes to six or seven and even higher. That's what we call sometimes a **maxi devaluation**. It just means very big, sudden devaluation that happens in fixed exchange rate regimes when things are not sustainable.

- 8.61 But what I want to really emphasize about this example is that it was very observable that you were going to have a collapse. This is not about greed turning to fear. This is about unsustainable policies that are clearly observable and being undertaken by a government, despite the fact that everybody knows that they're not going to end well. Then finally, when a new regime comes into office, the hope that there's going to be some reform announced doesn't happen, and then you get the crisis.
- 8.62 We don't have really a hard time understanding why this crisis happened or even when it happened. It's quite clear. There are a lot of examples like this. When we're thinking about crises, especially in **emerging market** countries that have this **twin crisis** feeling, we shouldn't be too surprised that this is a pretty normal story.
- 8.63 OK, moving on to Korea. I want to make the same point. If you were following the news, both the Financial Times and the Economist were both forecasting many months before the East Asian crisis, when Thailand first and then followed by Korea and Indonesia had their **twin crises**: collapses of their banking system and their **nominal exchange rate**. That both the Financial Times and The Economist ran articles saying this was going to happen.
- 8.64 The Economist—it was really extraordinary—in The Economist, they have this middle section, which features their big story the week that they want to talk about. And so the middle section, back in April of 1997, about six months prior to the crisis, the title of it was East Asian—I forget exactly what it was—it was East Asian **twin crises** that are about to happen. Now, think about this. By the way, the Economist never did this before and has never done it since, as far as I know.
- 8.65 They ran this very long article in the middle that said, based on our understanding, same things that we're talking about in this course. "There are three countries in East Asia that are going to have **twin crises**. They are Thailand, South Korea, and Indonesia. When they happen, we're expecting you're going to have a similar exchange rate collapse to what you saw in Mexico." They even estimated what the cost in the banking system would be of the government bailouts. They got them pretty much right. They said, "You're going to have a Mexican-style, maybe worse, **twin crisis**, and it's going to be in these three countries." Notice, they didn't just say vaguely, "there's going to be a **twin crisis**." They said how severe the costs were going to be, and they said which three countries. And they nailed it. They nailed it more than six months in advance.
- 8.66 When I say things are sometimes predictable, I don't just mean, I think we have a vague the idea. I mean, someone like **Rudi Dornbusch** can be looking at the Mexican exchange rate and saying, "That's not going to work." Or somebody like The Economist or the Financial Times can be looking at the situation in three East Asian economies and say, "We've seen this play before. This isn't going to work out well." The academic literature, as early as 1992, I think it was, that the economist **Alwyn Young** was writing about the declining productivity that was happening in the East Asian countries.

- 8.67 And of course, what is that declining productivity? Why do you tolerate unprofitable enterprises? Because they're politically connected. So you just keep saying, look, nobody wants to fire people. Nobody wants to tell an enterprise, you're done, go away. Nobody wants to do that. But if you're not willing to do that—the market, of course, does that all the time—if you're not willing to do that and then support better ideas and pro-growth policies and all the rest of it, you end up with this kind of a bill because what you're doing is you are simply subsidizing a sinking ship. You're also making your economy much less productive, which is going to cause your exchange rate to have to decline. That's the **Balassa-Samuelson** story.
- 8.68 As these things are getting worse and worse, also, those East Asian banks are borrowing more and more short term from other banks in dollars. Why? Because dollars are cheaper. If people are worried that your exchange rate is going to collapse, then you borrow much more expensive denominated in your own currency than you do in dollars.
- 8.69 Adding to the losses was the fact that the banks who were underwater from having subsidized these losing enterprises themselves took on new currency risk. So once the depreciation happens, they have even bigger losses coming from their dollar borrowings. So the dollar borrowings are cheaper initially, but then when there's a depreciation, they're much more expensive. And in fact, they add to the magnitude of the depreciation. So that's how bad these strategies are. It's a little bit like the **savings and loan** crisis that I mentioned of **resurrection** strategies, right? Betting on **resurrection**. That's what these foreign currency bets were.
- 8.70 By the way, the Mexican banks engaged in swaps orchestrated by **Goldman Sachs**, and others, of dollar exchange rates identically as they were heading into their crisis. They also effectively borrowed more in dollars because it was cheaper, which magnified their losses after the fact.
- 8.71 When the crisis came, just as I said, the economists forecasted these, the actual cleanup costs for Thailand, Indonesia, and Korea were pretty large. These are a few different estimates of the **ex-post** costs, Thailand and Indonesia being the largest of them. This is as a fraction of GDP.
- 8.72 Other studies have documented that, just as **Alwyn Young** had pointed out with a productivity growth decline, that there was a lot of unprofitable investing going on in these East Asian economies because the conglomerates were protected by the government through access to credit. It wasn't just domestic credit, it was also international credit. I'll just give you a little sense of what that looks like.
- 8.73 If you look at the case of Korea here, let's say as early as 1992, their corporations who are involved in international borrowing have leverage ratios that are defined at about 200. They're borrowing at very good investment-grade ratings. But if you were to look at what would that imply if you were in the corporate bond market in the US, if you weren't protected, that would be even at a number as high as 65—and 200 is a lot higher than 6—even in a number like 65, that's very bad junk. This is a junk bond. This is a much worse junk bond.
- 8.74 If you're looking at this and you say, "Well, how do they manage to borrow at single A ratings?" The answer is because the people lending them the money are understanding that those

corporations are effectively guaranteed by the Korean government. That's the only way to make this connection. This is an article by Harvey and Roper that's making that point, which is published in a Brookings book.

- 8.75 **Twin crises** are reflecting all of these kinds of **risk subsidies** that are betting the whole country and leading to all sorts of destruction, what would you expect to learn from all this? Well, you might say, "I'm not sure what I expect to learn." By the end of this class, you probably don't answer that question too quickly because you might say—well—
- 8.76 Now it is true that Korea undertook pretty dramatic governance reforms as a result of this. There was a lot of reform going on in Korea. But this same pattern is very much a pattern that continues in **emerging market** countries. Why? Because it reflects the **political equilibrium**. You might say, "well, why don't they have a different politics?" Well, maybe they can't choose their politics the way you choose which ice cream you're going to have. Or the way many times it feels like our policy discussions are imagining that it's just like a bunch of economists sit down and decide what the good policy is, and then, of course, the government wants to do what's good. That just isn't true. I've been doing this for 40 years. That isn't even close to true.
- 8.77 It is sometimes true that reformers are elected and they try to do something on a margin that's a little better than before, and they might listen to some good economists or somebody who's smart, a good businessman. But it doesn't mean that generally, that's the way things happen. Generally, decision making is very constrained by what's practical. If you want to do something very different, you just don't get power.
- 8.78 You can't really think about the evolution of financial lessons as following some economic optimization problem. It's incredibly naive. It's worse than naive. It's misleading. I would say purposely misleading that people try to think about government policy that way because who benefits from thinking that way? The government officials. They're always trying to do the right thing. They're always good guys. No, that's not the case. That's a myth they want to create.
- 8.79 It isn't just **emerging markets**, as we'll see when we talk about the US, which we're going to do right now. We have a very similar story there, where after the US **subprime crisis**. Nobody, none of the things that caused the US **subprime crisis** were admitted by Congress or by the administrations or by any of the regulatory agencies. And instead, the story that you heard about the US **subprime crisis** is, "oh, there was just this greed followed by fear, right?" But wait a minute.
- 8.80 Why did it happen when and where it happened? Can we really not understand some fundamental forces that were pushing this? And the answer is, we don't want to understand those forces. We don't want to admit them because that's very embarrassing. And guess what they are, of course, They're going to be housing risk, subsidization, and also **monetary policy** mistakes. Nobody likes to show you, "I was doing a huge subsidization, invisibly, it was all my fault." No one likes to say, "I was running really stupid **monetary policy**. Sorry." Those are the

two things that were the drivers here. But it's purposely being misdiagnosed because you don't want responsibility for what you did.

- 8.81** Your first clue that there's something a little screwy going on in the banking system— we talked yesterday about the consolidation wave that's happening in banking. Remember, post-1980, we're getting these big banks. How is that being accomplished? Through a wave of bank mergers, we end up with a few banks in the US now that are huge. JPMorgan Chase. Well, that used to be Chemical Bank, Manhattan, Chase—all these banks that grew together, took over each other. Then finally, we end up with just JPMorgan Chase, which is—there are about 50 banks in JPMorgan Chase that were acquired over a very short period. The same is true of all these large banks.
- 8.82** There was actually a process. Once you allowed nationwide branching to happen, starting in the 1980s, but continuing through the 1990s, you had to then actually create these nationwide branching banks. The people who are standing on the sidelines who are seeing this being allowed, remember, it was allowed through state actions in the 1980s and finally federal law in 1994. As they're seeing all this happening, do you think that they're thinking, "How do I get a benefit from this predictable merger wave?" Well, of course they are. I can tell you how that story unfolds.
- 8.83** There's a requirement in the US: in order to do a merger, you have to be approved by the Federal Reserve Board of Governors. In order to be approved, you have to demonstrate certain things about the competitiveness of the market after the merger happens. That's all confined to the deposit market, which is a complete joke. There's a complete joke requirement about, show me that things are going to still be competitive after the merger happens. It's all confined to the deposit market. And many mergers clearly violated competitiveness requirements.
- 8.84** For example, the merger of Bank Boston and Fleet Bank in New England in the late 1990s created a huge temporary monopoly in the middle market lending market in New England, which everyone knew was going to happen. But that's not part of the Fed's mandate. They only look at the deposit market.
- 8.85** But then there's another requirement: you have to demonstrate that you're a good citizen. Now, what's wonderful about this, it's a wonderfully Orwellian or Kafkaesque process, that—how do you show that you're a good citizen? In the early 1990s, some staffer, who shall remain nameless in the Senate Banking Committee, really is the one who came up with this insight.
- 8.86** If banks have to show that they're good citizens before they merge, that means that we could actually have banks pay to have people testify that they're good citizens. Furthermore, we could have them show that they're good citizens even better if we made the cost to them of doing favorable things for these people who are testifying if we made that cost lower. Believe it or not, this was all very much thought of in advance and took the shape of something called the new regulation in 1992, just at the time all this nationwide merger stuff was really heating up, was called the GSE Act.

- 8.87 The GSE Act said, "We're going to create requirements for Fannie Mae—" Remember? Created in 1934. Fannie Mae and Freddie Mac, the other entity that was subsequently created that's very similar to Fannie Mae—"to purchase mortgages in low-income and center-city areas." They'll have to purchase, as a fraction of their total purchases, a certain amount of those mortgages, a percentage that will grow over time. A big insight that that Senate staffer had was that if we do this, we'll be able to get the negotiations between the activist groups and the banks that want to merge, we'll get that to favor the activist groups more.
- 8.88 Why? Because the banks will be able to meet their requirement more easily because Fannie Mae and Freddie Mac will have to buy the mortgages that they originate. This is the deal. This is how thoughtful this is. I know that you two want to merge. You're going to be the acquirer of her bank. I want to tell you that you have to originate a lot of really crappy mortgages as a condition for my allowing you to merge. You're not going to like that. But then I'm going to tell you the good news is— a lot of those crappy mortgages, I'm forcing him to buy from you so we spread the cost. That's the 1992 GSE Act. That was the vision of it.
- 8.89 The goal was to make the commitments that you would be able to make to the activist groups who would come and testify on your behalf that you're a good citizen. That would allow you to make bigger commitments to them. Now, what I'm telling you might seem almost crazy. What do I mean, commitments? I mean signed contracts that have a 10-year duration. That are telling how much money you're going to give the activist group for coming today to testify. You say, "Wait, that sounds like extortion." This is not just government that knows this extortion is happening, it's government that's planning the extortion. The extortion is executed through a government agency, the Federal Reserve Board of Governors, giving its approval. It's very aware of the extortion.
- 8.90 The payments take two forms: subsidized mortgages, that is, you have to issue mortgages at low interest rates to poor people, and very, very risky mortgages, by the way. And also, by the way, my organization needs some cash. Give me some cash. So 90% of the payments made in these contracts that went for typically 10 years, 90% is in subsidized mortgages, 10% is in cash. And that's how my organization gets followers. Because if you don't join my organization, let's say I'm Jesse Jackson, how do you join my organization? Well, you have to do things for me. You have to show up at demonstrations. If you show up at demonstrations and you do that, you get to be one of the people who gets the subsidized mortgage.
- 8.91 Now, I'm not making any of this up. There's something called the NCRC that was so proud of its strategies about how to maximize the negotiations of the activist groups with the banks that it published a book about how to do it, which you can get online. All of this is discussed in detail in my book, *Fragile by Design*, with Steve Haber. That's what this slide is about.
- 8.92 This slide is about George Butts, who's the head of an organization called Acorn, coming to—this is a quote from one of the Federal Reserve Board hearings, "We support the Nation's Bank acquisition of Bank America because they will make credit work for low and moderate income people, and they will work with the community institutions." In other words, they're giving me a

lot of money. That's how I know you're a good citizen. You're giving me money. That might sound funny, but that actually works at the Federal Reserve Board hearings.

- 8.93 These agreements are secret, but the amounts aren't. Let me show you. Notice the bank merger wave is starting, and then we get the GSE Act. These commitments that are being negotiated between the merging banks that actually merge during this period are almost a billion dollars. But if you look at all the commitments that are part of this strategy, it comes out to more than four—I'm sorry, I said billion, I meant to say trillion. It comes out to \$4.5 trillion.
- 8.94 From 1992 until 2007, when the crisis hits, that's how the merger wave is executed. There's four and a half trillion dollars of commitments being made to community activist groups by merging banks to get their permission to do it. Now, I'm going to first ask all of you who are watching by video and all here, how many of you were even aware that this was happening? The answer is you weren't. That's what's so beautiful about it. This works so perfectly as a political device because the GSE Act makes this number so large, in this invisible way, that nobody could have known that it was happening.
- 8.95 The GSE Act gives HUD, the government agency in charge of housing policy, the authority to increase the mandates on the GSEs over time about how much, what percentage of their mortgages have to be to low income and to inner city. And so what you can see is that those requirements are pretty large, 50%. So the total is growing to 50%, 60% by the end of the crisis, but they're growing over time. This is leading up to the financial crisis.
- 8.96 The GSEs, Fannie Mae and Freddie Mac are being required to do a huge amount of buying of crappy mortgages. Of course, they're not calling them crappy mortgages. In fact, they are sued by the SEC for lying—Fannie Mae and Freddie Mac—about the quality of their mortgages. They did a settlement with the SEC based on their accounting fraud related to that. All sorts of interesting stories about where fraud and risk are really government policies. They're serving these two gorillas.
- 8.97 OK, then we could look at what are some indications of how much the mortgage credit is deteriorating. This is the percentage of originated mortgages being made with down payments that are basically zero, 3% or less. So prior to 1990, this is not something that happened in the US. But after 1992, this becomes a very big—just before the crisis, 40% of mortgages being originated have down payments of 3% or less. By the way, that's not the worst of it.
- 8.98 Starting in 2004, there was no longer any limit on the need for documentation in mortgages by Fannie Mae and Freddie Mac. So you could just say whatever your income is, you didn't have to document it. Why? Because we know from evidence, and I testified on this before Congress, we know that they couldn't have met those GSE objectives that I just showed you. If they didn't keep deteriorating the credit quality.
- 8.99 We have emails from the risk managers of Freddie Mac and Fannie Mae saying, "Hey, you can't deteriorate the credit. We're going to go broke. And by the way, these people who we're lending the mortgages to, they're not going to be able to repay these mortgages. These are too

risky. You're not helping them by only helping them through **risk subsidies**. You're going to destroy lives as well as destroy us." That guy got fired because he was sending these things through emails precisely so he could create a record, and that got him fired. And then he went on to teach junior high school mathematics. And none of you have ever heard of him.

- 8.100** So that's why this works. I just want you to understand, have we really reformed this? Of course not. Are we likely to reform it? I would say no. Then another part of this, of course, is that the banks, and of course, the **GSEs**, Freddie and Fannie, are not being held to anything like rigorous prudential standards. We have, of course, all this nonsense, the Basel Committee standards, and all this—thousands of pages of prudential regulation. But that's really designed to make it very obscure. I'm going to show you how prudential regulation purposely fails. But this is a pretty good indication.
- 8.101** By the time the crisis is hitting, let's just look at simple things in these banks' balance sheets relating to things that we've been talking about. Real estate lending over total lending—and this isn't including any of the off-balance sheet stuff that they sold to the **GSEs**—is growing dramatically. And cash, assets, cash plus treasuries, are at an all-time low. Banks are taking on a lot more risk. Nobody cares because if there's a political deal afoot, you can't then subvert it with prudential regulation. That's not part of the equilibrium. But you can create prudential regulations that seem to be very tough, even though they're not doing anything.
- 8.102** I would say to you, and you can read much more about this in **Fragile by Design**, that the **banking crisis** was not a surprise. I'm going to show you some evidence for that very quickly, which is those of us who were looking at banks' market prices so that we could see what the market's opinion was of how much real equity was left in the banks, that many months, like a year and a half prior to the crisis, people started seeing that the banks' exposures to this through a combination of their securitization policies and their on balance sheet exposures were leading to something that was very much a forecastable disaster. That's visible if you're looking at bank equity prices. But the regulatory measures specifically don't take any of that into account, and that's very much on purpose.
- 8.103** If you looked at the banks that weren't intervened, that is, these are the banks that never really got into serious trouble during the crisis, they all kept their market value of equity—that stockholder's equity over assets measure—above 5%. So that's pretty good. And they never really required any assistance during the crisis. But the banks that were in deep trouble, **Citibank** is a great example of one. Let's see, I think they are probably here.
- 8.104** You can see that **Citibank**, initially, back in April of '06, they had a 13% ratio of **equity to assets**. But by the time they get, even here we are in February of '08, long before the crisis really hits, it's fallen to about 6%, it's fallen by half. And then by the time we get right onto the crisis, you might say, "Why does the crisis happen in September of '08?" Because the market knows, starts realizing what the losses are in these banks that are going to come.

- 8.105 The equity price of the banks is reflecting that and showing us that these banks are insolvent, unlike the ones I showed you on the prior slide. That's why then those banks are no longer able to roll over their short-term paper that's maturing because the market has come to a particular view on them. Of course, that's what cues the crisis and then the bailout.
- 8.106 If you were looking more generally at the banking system, you'd see there's a huge disparity: a variation between the market value of equity and the book value of equity relating to regulations. So one of the things that most economists like me who've been involved in banking and bank regulation have been advocating for a long time is to reform prudential regulations to allow the measures of capital to be more real, to reflect underlying value. That has been a completely unsuccessful mission.
- 8.107 The **Financial Economists Roundtable**, which I chaired the drafting committee for their response to the 2023 crisis—we gave very specific instruction in this direction. But also **Mark Flannery**, formerly from the University of Florida, who's retired recently, also has written a lot about this. **Dick Haring** at the Wharton School has written about it. We've all been writing this problem that the regulations are based on phony accounts that are purposely phony and unnecessarily so.
- 8.108 By the way, I was Chief Economist at the national bank regulator, the **Office of the Comptroller of the Currency**. I tried to make this point while I was Chief Economist at the number one regulator of national banks by showing all of my senior management that I was working with that we could use information to have a much more accurate perception of what banks' true condition was. But there was no interest—there was really no possibility for making any of that happen. I'm only going to show you a little bit of this.
- 8.109 You could see when you look at market value of equity over book value of equity, **JPMorgan Chase** looks great. When you look at the traditional measure that they focus on, it's 6.9% ratio. When you look at the ratio of market value book, it's 1.5, very high. So this number tells you they're in great shape. This number tells you they're not in such great shape. But this number is a phony number. This number is a very helpful one.
- 8.110 And when you looked at **Citibank**, which has for decades been the obviously weak, poor quality bank in our system, it looks like it's doing better than **JPMorgan Chase** on the phony number. But then when you look at the actual number, **Citibank** has, and has had for a long time, a market value that is indicating value destruction, that this is not a well-run bank, and it's a highly risky bank.
- 8.111 The point I'm trying to make, not from just an outsider academic perspective, but also from an inside perspective, is the problems are transparent. The problems are known. The solutions are also pretty clear. The willingness to do anything is absent. That's not a surprise, right? Because willingness to do reflects political agreements, the bargain that we live under.
- 8.112 Reflect also on your own responsibility as a citizen in the sense that how much time have you invested in learning any of this? I hope this class will be a wake-up call for that. But on the other hand, can I really expect citizens who are going about their lives to invest a huge amount of time

in becoming experts on the banking system? We're stuck in democracies because we delegate a lot of this, but we delegate it to people who are making lots of deals and not always in our interest. In fact, typically not. I'm going to skip through that.

- 8.113 Now, the other piece was, which I mentioned, is **monetary policy**. **Monetary policy** was also quite bad. Here's an interesting fact. What we've learned from the academic research on **monetary policy** is that when **monetary policy** loosens, it tends to reduce the price of risk in the economy. By the way, that's not something that you should say, "Well, that's just some general theorem." No, it's not. It's a fact, though. If the central bank decides to do unpredictable loose **monetary policy**, that's going to contribute to risk-taking by reducing the market price of risk.
- 8.114 The second major contributor to the **subprime crisis** in the US was that, was that the Fed decided in the years 2002 through 2006 to run a very unusually loose **monetary policy**, and to deviate, as I'll show you, from its own implicit rules in **monetary policy** that they developed over the previous couple of decades. Just to give you a quick graph of this, if we looked at the **Real Fed Funds Rate**, which is a rough indicator of how loose **monetary policy** is—when you see that being negative, that's during the high inflation period of the 1970s, and then for many consecutive years, only 2002 to 2006. That's a rough indicator.
- 8.115 Another thing that we could do is look at, are we following the rules that the Fed itself seemed to be following prior to 2002, which is sometimes called the **Taylor Rule**. We can compare the actual interest rate that's being targeted by the Fed by the interest rates that the Fed's prior behavior, that was clearly happening here, would have led us to expect. You can see starting in 2002, the Fed deviates from its own **Taylor Rule** of **monetary policy**. And by the time we're here in 2004, 2005, interest rates are 2% too low.
- 8.116 2% too low—that's a lot of Fed policy, right? Interest rates typically move in 25 basis points, a quarter of a percentage point increments. So to be 2% too low means that the Fed was having a huge effect on the underpricing of risk. Now, this is a graph you saw on the first day. Look how much the **pricing of risk**, whether we're looking at **VIX** or at bond spreads, is declining from 2002 to 2006. That same **pricing of risk** is contributing to the **financial crisis** by making everyone more willing to take on risk in the financial system.
- 8.117 Now, if we look at the Spanish **banking crisis**—I'm only going to talk about it very briefly in the interest of time—but just to say there are actually very similar forces at work. When Spain joins with the Eurozone, that has an immediate big effect on bond interest rates in Spain, and especially reduces interest rate spreads. So all Spanish bonds interest rates fall because joining the Eurozone is taken by the bond market as an indication of a more credible government and a more credible private sector. So they have that same influence of a **monetary policy** interest rate effect, which is causing credit spreads to contract.
- 8.118 But then, of course, the big story is they have these things called **Cajas de Ahorros**, which are savings institutions that are run by local Spanish provinces. They're political institutions that are

guided by the regulations of local political establishment. They are very much interested in expanding mortgage credit during this period.

- 8.119 The studies of Spain, I'll just reference them broadly, have basically said all of the over-exposure to risk in the Spanish financial system was due to the politically motivated expansion in many, but not all, of the areas in Spain by the **Cajas** that were controlled by the local political governments. The particulars of the politics are always very specific, but the **two gorillas** are going to be almost always present, and sometimes that third **gorilla** having to do with the **grupos** or **chaebols**.
- 8.120 Now, the final thing I want to do, in about 15 minutes that we have left, is to talk about the statistical evidence about the **two gorillas** more broadly. What we want to do is try to address the whole global database on this— we might call it pandemic of **banking crisis** since 1980—and try to understand the mechanisms and especially the connections between these **two gorillas**.
- 8.121 This is a graph you saw on the first day, which is showing you that there's a lot of **deposit insurance** policy change that's happening during the 1980s and 1990s. The first question to ask is, "Why is that happening?" Recent research has shown that one of the big contributors to this is that all of these international government organizations were actually pushing for **deposit insurance** to be adopted by countries. Yes, the **World Bank**, the **IMF**, and the **EU** were pushing very hard for countries to adopt **deposit insurance**.
- 8.122 What was going on here? I think that it's hard to be too specific about it, but here's what I would say. It was coming from **myopia**. If countries' banking systems were weak and you wanted to be able to have them in a healthy condition for some reason next year so that you didn't have an embarrassing **banking crisis** that you had to handle, you then say, "Well, **deposit insurance** would be very helpful." Even though you know that **deposit insurance** long term might be a very bad idea, if you have a specific thing like, "I want to get countries to be able to join the EU," or, "I'm the **IMF**; I'm currently handling the East Asian **financial crisis**. I really want other countries not to have a **financial crisis** right now because I don't have the resources to handle it." Then you push very hard.
- 8.123 By the way, 1999, the **IMF** makes **deposit insurance** recommended for the first time. Why? Because it's bereft of resources. Most of these policies, I would by these multilateral institutions, are part of a **myopia**, having to do with their own goals and their own short term perspectives. I think **myopia** has been a theme, right? We've been talking about how a lot of this is reflecting election concerns, short-term things, or EU accession, or **IMF** managing its problems. I think that that's a really important way to think about why countries make—especially democracies—make bad long term decisions because they have **myopic** incentives.
- 8.124 We talked about housing lending. I showed you a different version of this graph for a slightly different group of countries and defined slightly differently. But you get the point. The point is the second **gorilla** is also substantially growing over time. We have both increased protection of banks and increased subsidization of **mortgage risk**.

- 8.125 What I want to do now is show you how one can try to approach teasing out the causal role of **deposit insurance** in making **banking crises** happen. Then—here's the surprise—also in making more real estate lending happen. What we're going to do is we're going to look at changes in the generosity of **deposit insurance**. For econometric reasons I won't go into, we're going to completely ignore going from zero to some **deposit insurance**. We're only going to look at countries' decisions to increase the generosity of **deposit insurance** that they already have.
- 8.126 We're going to look at that and say—we're going to try to model it, and we're going to have lots of things that model it, including the local economy and all the rest of it. But we're going to have a set of things that we regard as **instruments**. That means things that we think are credibly unrelated to the domestic circumstances of the economy. What are those going to be? Changes in multilateral policies, changes in the EU's policies for the whole world, changes in the IMF's policies for the whole world. When you do that, you have an **exogenous** instrument, like a lever to move the world.
- 8.127 From a causality standpoint, you can say, "When clearly **exogenous** things happen to a country that make it increase **deposit insurance** generosity, how does that affect banking sector instability and how does that affect the amount of mortgage lending going on in a banking system?" What I'm going to show you is—the answer is— those multilateral actions had very positive effects on bank risk and on mortgage lending. That's really the conclusion of this.
- 8.128 First of all, when **deposit insurance** becomes more generous, this first coefficient tells you that the ratio of **loans to assets** goes up, and it goes up by a lot. Notice, what are our two core measures of risk? Loans over assets and equity over assets. I'm going to show you that when you get more generous **deposit insurance**—just like I showed you in the US state's experiences in the 1920s, remember, '19-teens and 1920s, We're going to see the same thing here. More **deposit insurance** makes banking systems hold less **cash to assets** and hold less **equity to assets**. Here's the other thing: it makes them lend more on real estate relative to total loans.
- 8.129 Loans are going up relative to assets. Mortgage loans, which are household loans in this database, are also going up relative to loans. We can sometimes measure that using a different database—we get the same result. Mortgages are very much going up more as a result of more **deposit insurance**. Just like that US **thrift** policy in 1934 that we showed. Debt is going up relative to assets—that means equity is going down. We can also mention these things relative to GDP as another way to measure how big the effects are, and we find that they're very big.
- 8.130 The final question is: what does it do to the likelihood that you're going to have a **banking crisis** and to the severity of the **banking crisis** that you end up having? The answer is, **deposit insurance** has a positive effect both on the likelihood of the crisis and on the severity of it. If you wanted to say, "What does **deposit insurance** do?" Making it more generous makes you more likely to have a crisis, more I feel like I have a severe one, because it makes banks purposely choose to do things that are systemically risky, like lend more relative to assets, hold less equity relative to assets, and lend in this very systemically risky thing, which is mortgage credit.

- 8.131 I think I've already reviewed these conclusions, but just the point is: **deposit insurance** doesn't make our banking system safer. It makes it much less safe and it even promotes the other major destabilizing influence, which is real estate lending. What lesson should we expect that we're learning from this? Well, you might say, "What we should be doing is reforming **deposit insurance**—also reforming prudential regulation, but maybe reducing **deposit insurance** rather than increasing it." But it's very unlikely that we're going to do that.
- 8.132 There was a slide in here which I skipped over having to do with the 2023 crisis. In this case, you had interest rates that ran up from 2021 to 2023 that caused the banks that were involved in Mickey Mouse arithmetic accounting—that were showing their assets as valued—unchanging—even though the interest rate increase, as we talked about before, created **interest rate risk** for them and caused them to collapse.
- 8.133 You're not even going to believe this. Not only did the government not change that rule, that rule still is in effect. But there are about 600 banks that are similarly positioned that needed to have supervisory interventions to force them to recapitalize so they could become stable. And that hasn't been done either.
- 8.134 It's very hard when you come to what lessons do we expect to learn? Well, by this point, I hope you're with me here. It seems unlikely that we're going to see a change in any of this. Until this becomes a major electoral issue, I don't see how it's going to change, and I don't see it becoming a major electoral issue anytime soon.
- 8.135 We now arrived at the last set of points I want to make in the class. What I've done now is returned to an earlier slide—a slightly different version of an earlier slide that we had on the first day—which is trying to address the question of what's a useful approach to categorizing and to thinking about **financial crises**.
- 8.136 In this course, what we've done has shown that you don't want to have one story about **financial crises**, but you want to have accountable number of stories—not that many here on this slide. Each crisis tends to have a one or two or maybe three of these—there's an obvious contributing influence. It's very helpful to think the way we think about zoology, that we group animals into species, genus, and we recognize that some animals are more similar to other animals than they are to still other animals. That's how **financial crises** feel to me.
- 8.137 The major things we want to think about are domestic political economies creating **risk subsidies**, on purpose through **deposit insurance** or **GSEs**, for mortgages or other such things. Geopolitics that's pushing in the direction of competition. This is especially important. We saw in the **Mississippi bubble**, the **South Sea bubble**, where it was all about sovereign competition, and in those cases, mainly between Britain and France.
- 8.138 But **emerging markets**, many of the crises that we saw in the '80s and the '90s, I would say, had to do with how competitive the global market is. It's pushing countries to want to make progress and to try to find ways to redesign their systems. Sometimes those lead to very positive

outcomes, but sometimes they lead to taking some risk as part of that competitive strategy. That's not going away.

- 8.139 Then we have some crises like the Florida crisis that we talked about in the 1920s or the stock market collapse. Now, of course, the stock market collapse is quite controversial. Maybe that wasn't even an excessive run-up in price to begin with. But in the Florida collapse, I think we would say there likely was an excessive run-up in price. But that was part of a learning, having to do with innovations that were going on. That it'd be very hard for us to be able to think how we would have avoided. It wasn't all about **greed and fear**. It was about fundamental uncertainties and novelties that were opening up for the first time, and it needed be analyzed based on almost no data availability.
- 8.140 We also have this fraud showing its hand in a lot of things. To some extent—I'm not saying the government fraud can't be avoided—but some of the fraud that we saw in Florida, for example, just reflected the fact that we have a private economy based on individual actions and decentralized decision making. Some of that fraud that occurred in the **Manly-Anthony chain** reflected that between the borrowers and the bankers, the fact that they have a relationship, they have private information. Now, of course, it was made much worse by the complicity of the regulators who apparently were bribed. But that's fairly rare. But nonetheless, I'm saying, I don't think we're going to avoid those kinds of magnifiers anytime soon.
- 8.141 Then finally, many of our examples had **monetary policy** errors— 2002 to 2005, 1929, 1930 to 1933, all of the US **Federal Reserve**. That comes from the fact that we've decided to have a **fiat money** regime based largely on discretionary policy. Again, I can see real major advantages from reforming that to be more systematic and transparent. But I don't see that point of view, which is my point of view, being widely represented among politicians or even economists. When I say that the world has adapted itself toward these kinds of things— continuing, and therefore, we shouldn't expect **financial crises** to end—I'm not saying that it's a good thing. I'm just saying that this is the way we think about the world. With that, I've run out of time, and I'll open it up to any final questions any of you have. Yes.
- 8.142 Professor, as I understand, historically, let's say, historically, the pillars of the **financial crisis** are land and housing. Let's say that's—
- 8.143 Well, not only. Sometimes.
- 8.144 I would add a third. It would be geopolitics. It's—
- 8.145 Well, it's on my list here.
- 8.146 Yeah, they're correlated with the land and the housing, the markets, expansion, all that stuff. Those three would be the origin of a **financial crisis**.
- 8.147 You can make your own list. It's a free world. I made my list, and here it is. But I agree with you that geopolitics is very important, not just through competition, but also we talked a little bit about the role of the **IMF** in supporting the Korean government's ability to pay the **chaebols** by

making dollars available to them. Who's making decisions at the IMF? It's the G7 finance ministers that run institution— pretty much as their slush fund—with full discretion. And so absolutely, I didn't get into it, but I agree with you, geopolitics is often expressed through these multilateral institutions, especially the IMF.

- 8.148** That was also true in the 1980s. My first job as an economist was to work on the Brazil desk at the IMF in 1981, the summer of 1981, just as a summer intern. Yes, I am that old. I've been around working since then. It was absolutely clear to me that the IMF was completely complicit in the accounting fraud that was going on in the US with respect to its own banks and their positions in the LDC debt at that time, and that this was an open secret. It wasn't a secret at all.
- 8.149** Why did they do that? Because they were told to do it by the people who ran that organization who were not the economists, but they were the G7 finance ministers. Still true today. Absolutely. Geopolitics is very important.
- 8.150** One of the themes or one of the major points that we've learned during this course is: know thyself. With that in mind, if we look back at the 2008 crisis, and if we continue this discussion about geopolitics, it's clear that we were in a very different climate than we are today. It could be argued that in 2008, we were still in a unipolar world where right now it might be splitting geopolitically.
- 8.151** If we take a look at, A, the geopolitical landscape of today, B, the fact that the myopic view of politics affects how the sovereign responds to the whole structure of the financial system—we have elections coming up. Then C, the fact that this multipolar world is creating other institutions other than the traditional liberal ones like the IMF and the World Bank, for example, the BRICS Nations with the Russians and the Chinese.
- 8.152** If we take all of these together, where are we now? If we are to know ourselves, where are we now and how can this affect the severity of the next crisis, given that the US dollar might not reign supreme for the next 20 years—10?
- 8.153** Great question. I'm so glad you asked it because I didn't have time to extend this in that direction, and I was thinking about it. Let's first just review where we are globally. First, I would say there are a lot of things that don't seem to be changing. The incentives for the two gorillas are pretty reliable. In some autocracies, the incentives to support your grupo, chaebol, allies—pretty reliable. Those tools that we learned here are going to be of continuing relevance in individual countries.
- 8.154** What are some things that are fairly new? I would point to two things. First of all, China is in a long-term growth decline that reflects its own inefficient allocation of capital in its economy. That means that China is going to become increasingly, and I would say as it already has been, belligerent internationally in geopolitics, and that there's going to be a lot of dislocations as China is having problems, both domestically, politically dislocations, throwing different allies under the bus, as, of course, we've seen Mr. Xi has been doing. But then also more belligerence militarily, more difficulty maintaining friendly trade relations, and I would say really difficult

problems relating to existing Chinese debt and also existing old people with insufficient pensions.

- 8.155** What this means to me is I think we're going to see a major inflation in China coming from its debt problem, and we're going to see a decline of China in its importance internationally.
- 8.156** I also think that when we look beyond China, we're seeing a buildup of debt because of the low interest rate environment for other countries—that wasn't the main influence on China; it had its own domestic reasons for building up debt—but the same disease is a fairly new development in both **emerging market** and developed economies. Both types of economies have been building up enormous debt, including the United States.
- 8.157** We're now in a position where this thing called **fiscal dominance**—it means that you've issued so much debt that unless you have major reductions in expenditures or increases in taxes, you're going to be having to be pushed into a high inflation. We're now there in the US, and I would say—and I've said it in print in a recent article in the **Federal Reserve Bank of St. Louis Review**—that we are maybe four to ten years away from a major acceleration of inflation, and then the US will be tested to see if we can do big reductions in Medicare, Social Security, and Defense, which are going to be very challenging for us.
- 8.158** If not, the path of least resistance is very high inflation—and also for reasons that have to do with the way inflation gets managed—trashing the banking system, taxing the banking system as one tax of last resort through bringing back zero interest reserve requirements as a way to impose a hidden tax, because hidden taxes are great, except they destroy the banking system.
- 8.159** So just to be very specific, if we don't get pushed into a major reform, the US itself will have inflation rates certainly exceeding 10% a year, starting maybe four years from now. So all of this talk that was currently going on about the **Federal Reserve** and all that becomes irrelevant if you're in an unsustainable fiscal path. And we're not the only ones. Japan, Europe, Latin America, East Asia.
- 8.160** The low interest rate environment encouraged countries to think they could borrow hugely without putting themselves in a dangerous path because debt service costs were so low. That's changing. My analysis leads me to think that's a permanent change. So that real interest rates that were less than a half a percent, and some people believe that would continue as recently as seven or eight years ago, I think are now going to be likely somewhere above 1.5%.
- 8.161** When you're looking at—that is an absolute transformation of the global financial markets. I think that the most likely big shock that you're all going to be dealing with is for countries that borrow in dollars, that means most of **emerging markets** other than China, it's going to be big debt defaults, big sovereign defaults. For countries that don't borrow in dollars very much, it's going to be big inflation, and that's likely to be China as well as Japan, the US, Europe.

- 8.162** We're in a situation where I think it's going to be a very rocky ride unless we see major changes in the governance of our economies. Notice that neither political party in the US has a candidate who's talking about solving these problems.
- 8.163** May I make a quick follow-up?
- 8.164** Sure, please.
- 8.165** Looking at the Mexico case where there was a huge inflation and the peso at that time was back to the dollar. We saw how eventually the inflation caught up to the FX rate. Right now, the US dollar is the global currency. If we see this huge inflation take place, what does that have to say, in your opinion, about the dominance of the US dollar as the global currency?
- 8.166** Another great question. And my answer is, "Compared to what?" So yes, the dollar, if there—when Britain had its problems after World War II, the dollar emerges as the global currency because it fills a gap because it can. Who's going to replace the US dollar? Certainly not China. And not just because of their inflation problem that's looming, but because they don't have a convertible currency. You can't be the global currency if you're not convertible on capital account. I mean, it's a joke. For the Chinese government to even make the argument that China is going to be a global currency is ridiculous on its face.
- 8.167** Then what's it going to be? The Euro? I don't really see that happening for the same reason. Europe seems to be economically and financially in similar circumstances, maybe even weaker circumstances from a growth standpoint. Maybe, I don't know. Brazil and India will fill the gap. I kind of doubt it. They're also in similar, difficult circumstances. I think that the best guess is the dollar will continue to be the most important global currency, the global reserve currency, not because of dollar's strength or because of US economic strength in some absolute sense, but because there's no obvious alternative.
- 8.168** Pretty much stuck. Thank you.
- 8.169** All right. Thanks very much.